X17 Search in EAR2

Overview + Status of the July+August Run

ON BEHALF OF THE COLLABORATION

Stéphan Aune · Timote Dannely-Lamborion · Pierfrancesco Mastinu · Riccardo Mucciola
Elizabeth Musacchio Gonzalez · Dylan Neff · Thomas Papaevangelou · and all others in the n_TOF X17 collaboration

Slides and analysis built with Claude Code (Anthropic)

Outline

01 **X17**, and how we would see it at n_TOF

02 **The detector system in EAR2** — what we built this summer

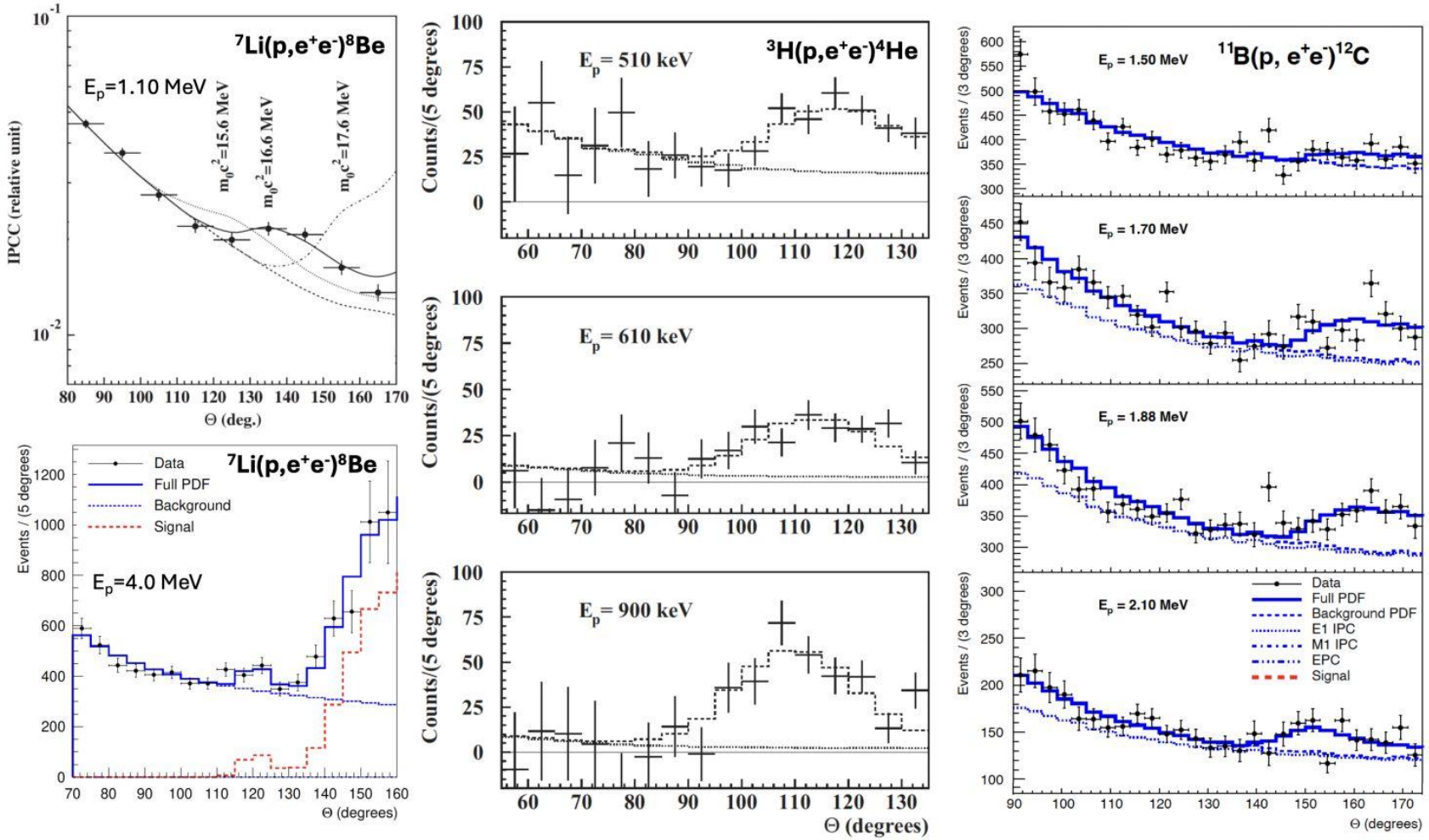
03 **Plan A: MeV neutrons** — and the γ flash

04 **Plan B: thermal neutrons** — and self-shielding

05 **Plan C: the aluminum spectrum** — capsule image, pairs to analyse

A possible new boson at 17 MeV?

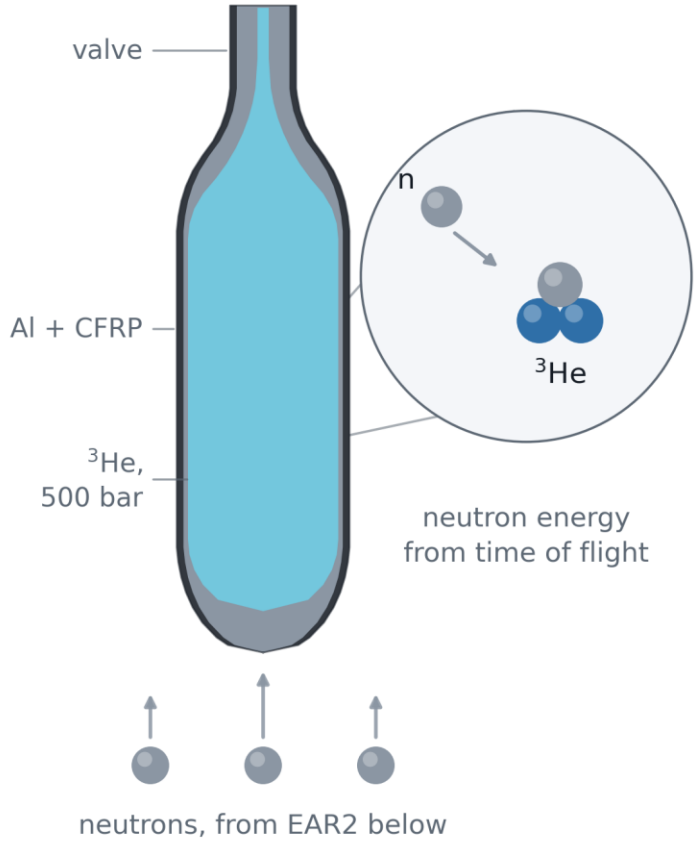
- **ATOMKI, 2016:** an excess of e^+e^- pairs at large opening angle, in three nuclear transitions
- Consistent with a new particle: **X17**, ~17 MeV, decaying to e^+e^-
- **Can we make an independent measurement?**
- The same ^{4}He* state, generated with neutrons + ^{3}He?



A.J. Krasznahorkay et al., *PRL* **116**, 042501 (2016) · *PRC* **104**, 044003 (2021) · *PRC* **106**, L061601 (2022)

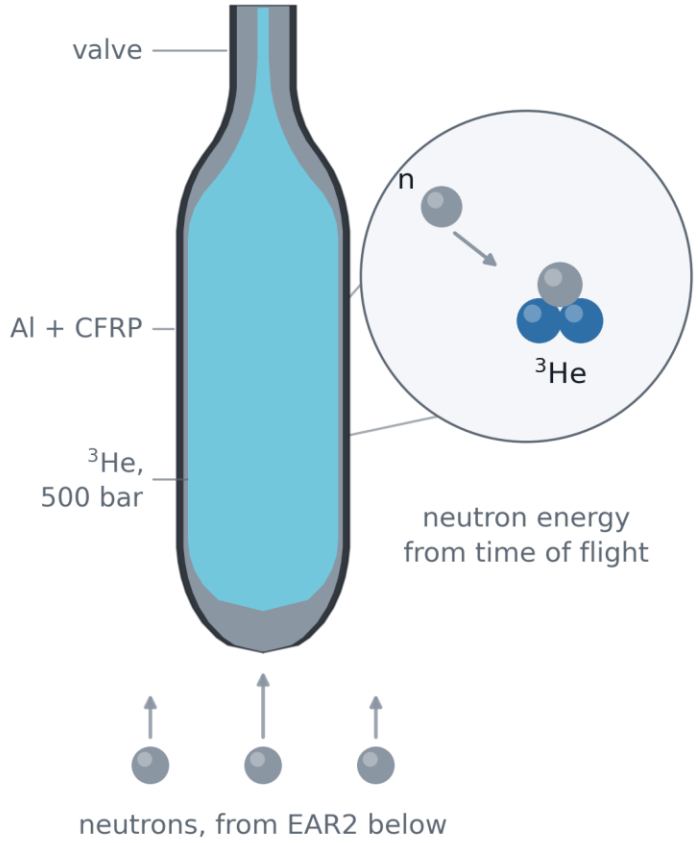
Capture on ^{3}He and how X17 could come out

1. A neutron beam on ^{3}He

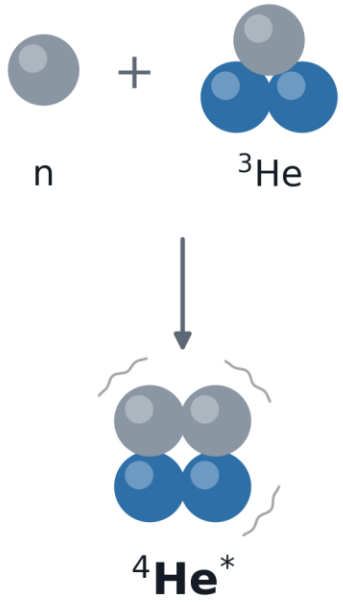


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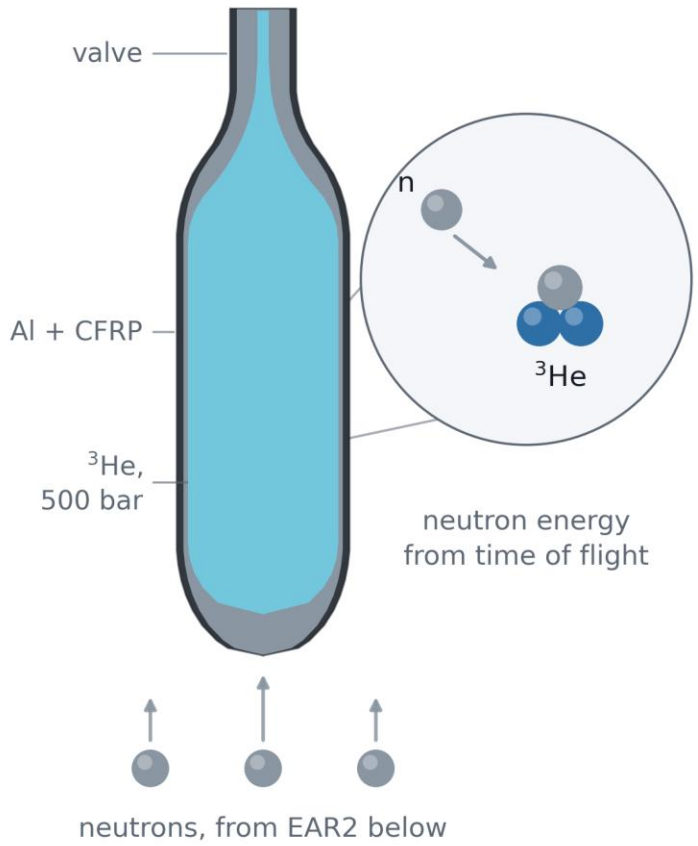
2. Capture makes ^{4}He*



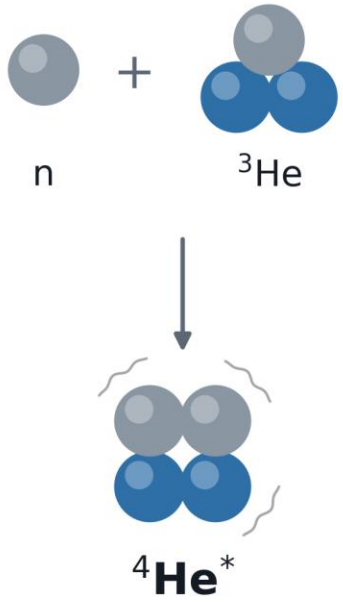
the compound nucleus is left
20.58 MeV above its ground state

Capture on ^{3}He and how X17 could come out

1. A neutron beam on ^{3}He

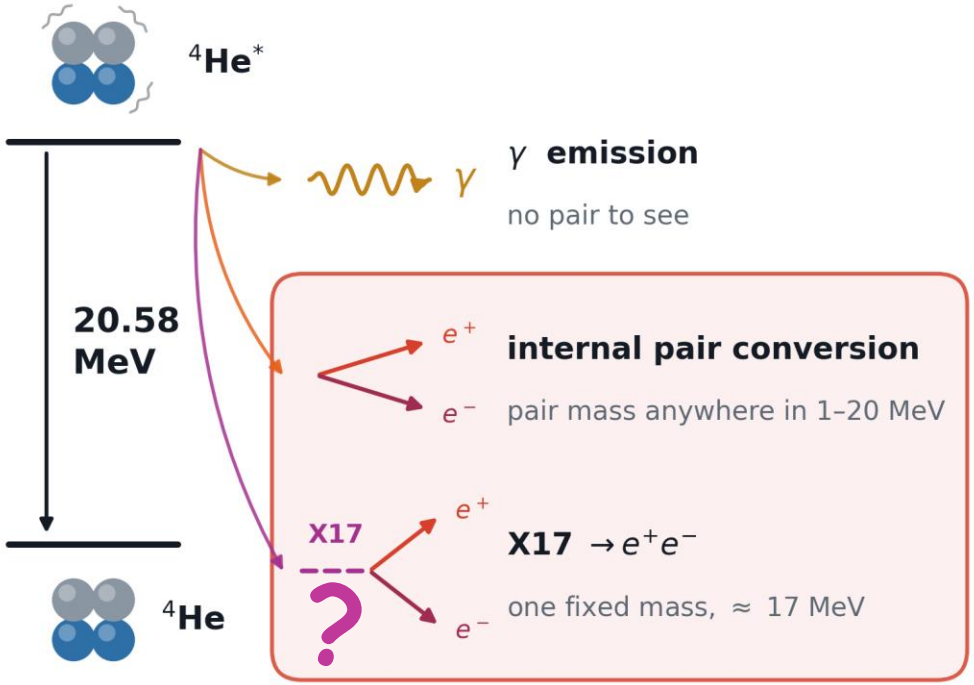


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3. Three ways to shed it

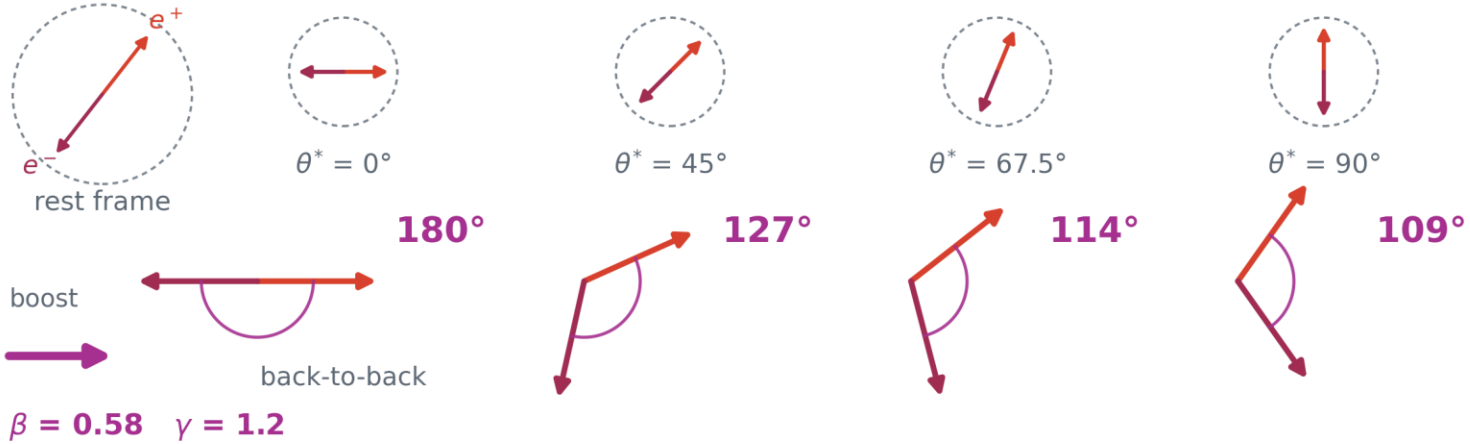


Detect the e^+e^- pair!

The opening angle observable -- X17 moves slow

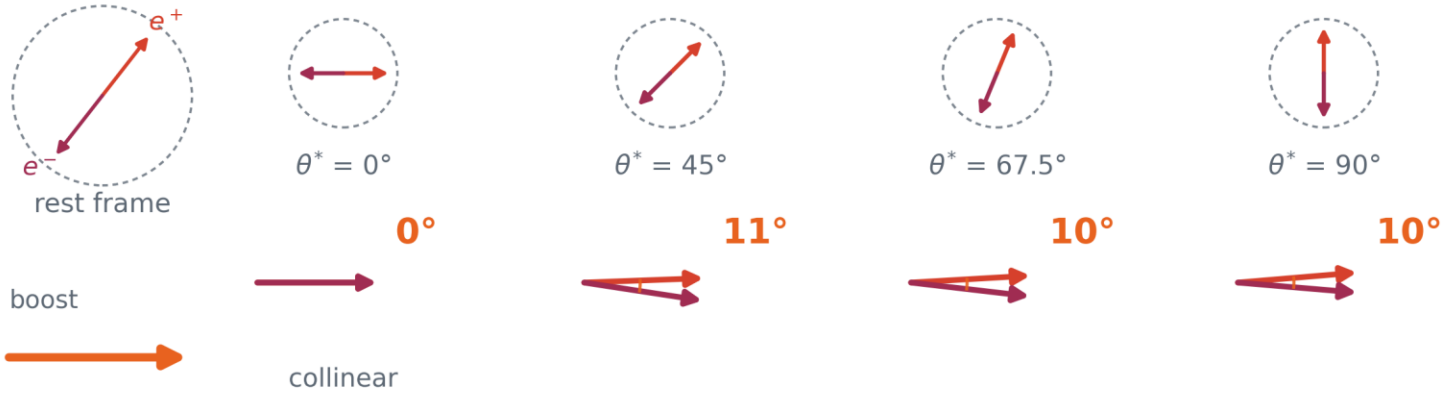
4. The boost is what makes the difference

X17 — one mass, 16.8 MeV, heavy and slow



$\beta = 0.58$ $\gamma = 1.2$

IPC — any mass, here 2 MeV, light and fast

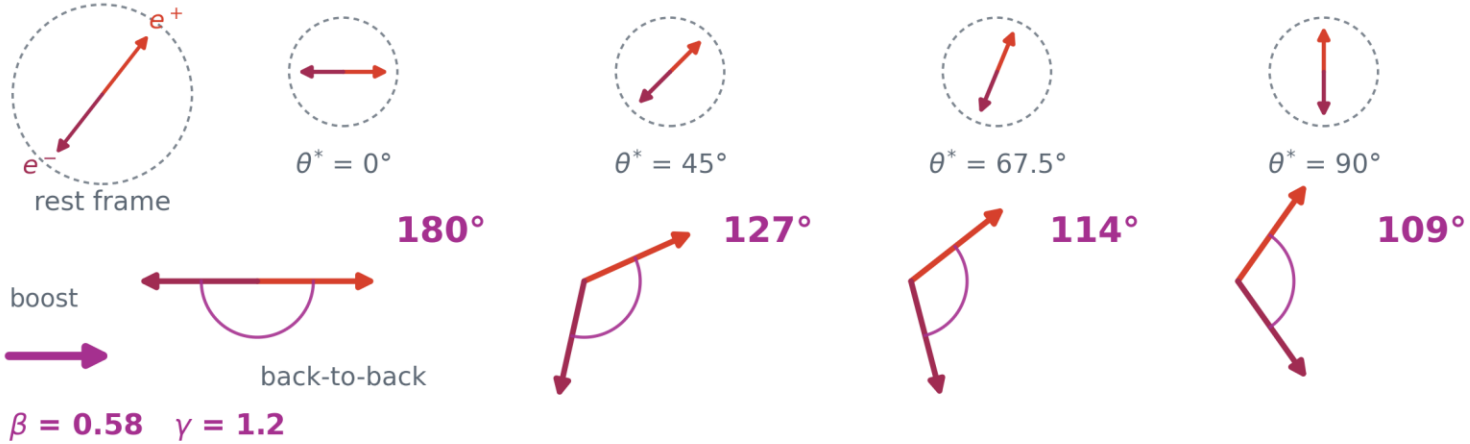


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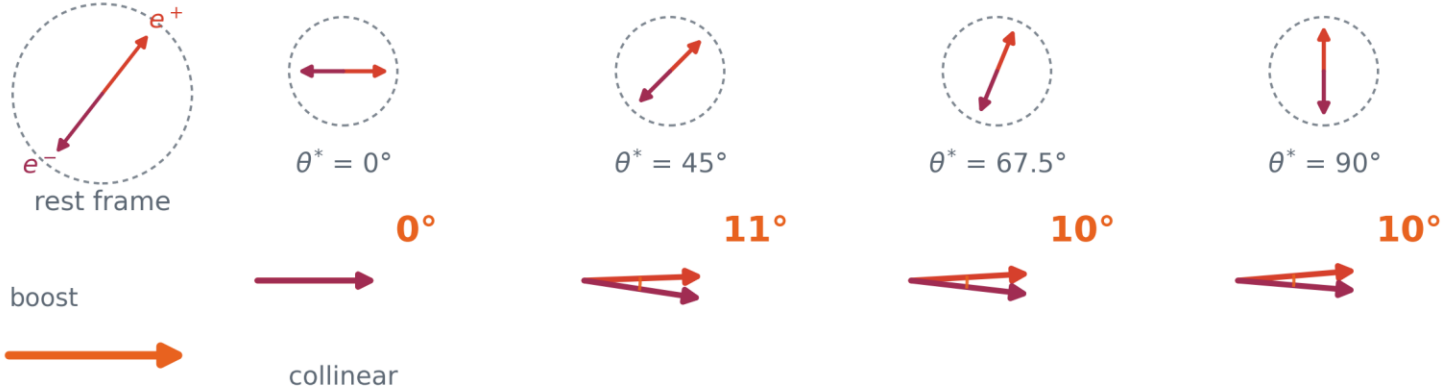
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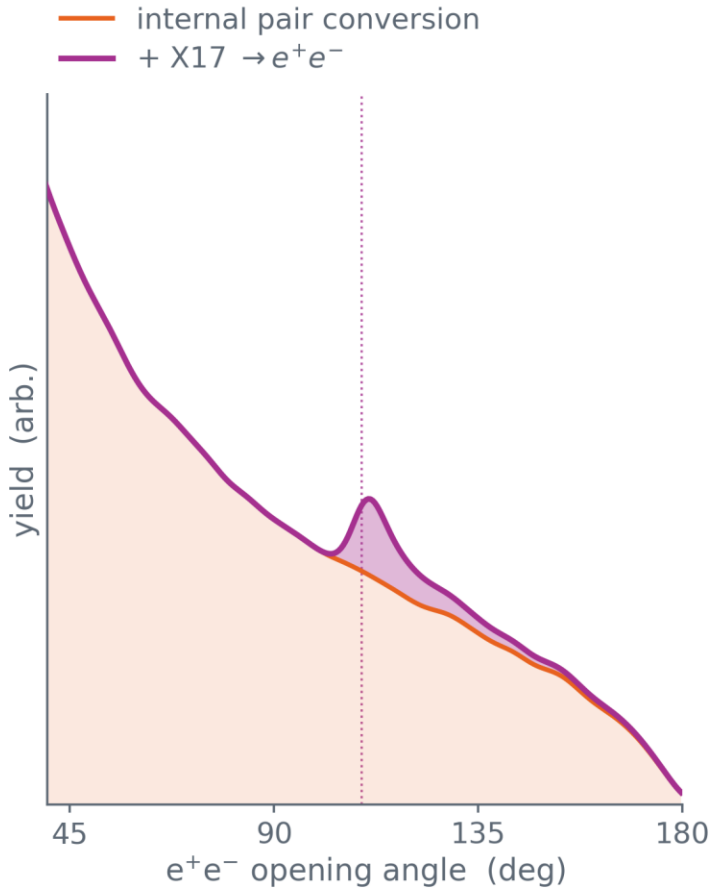
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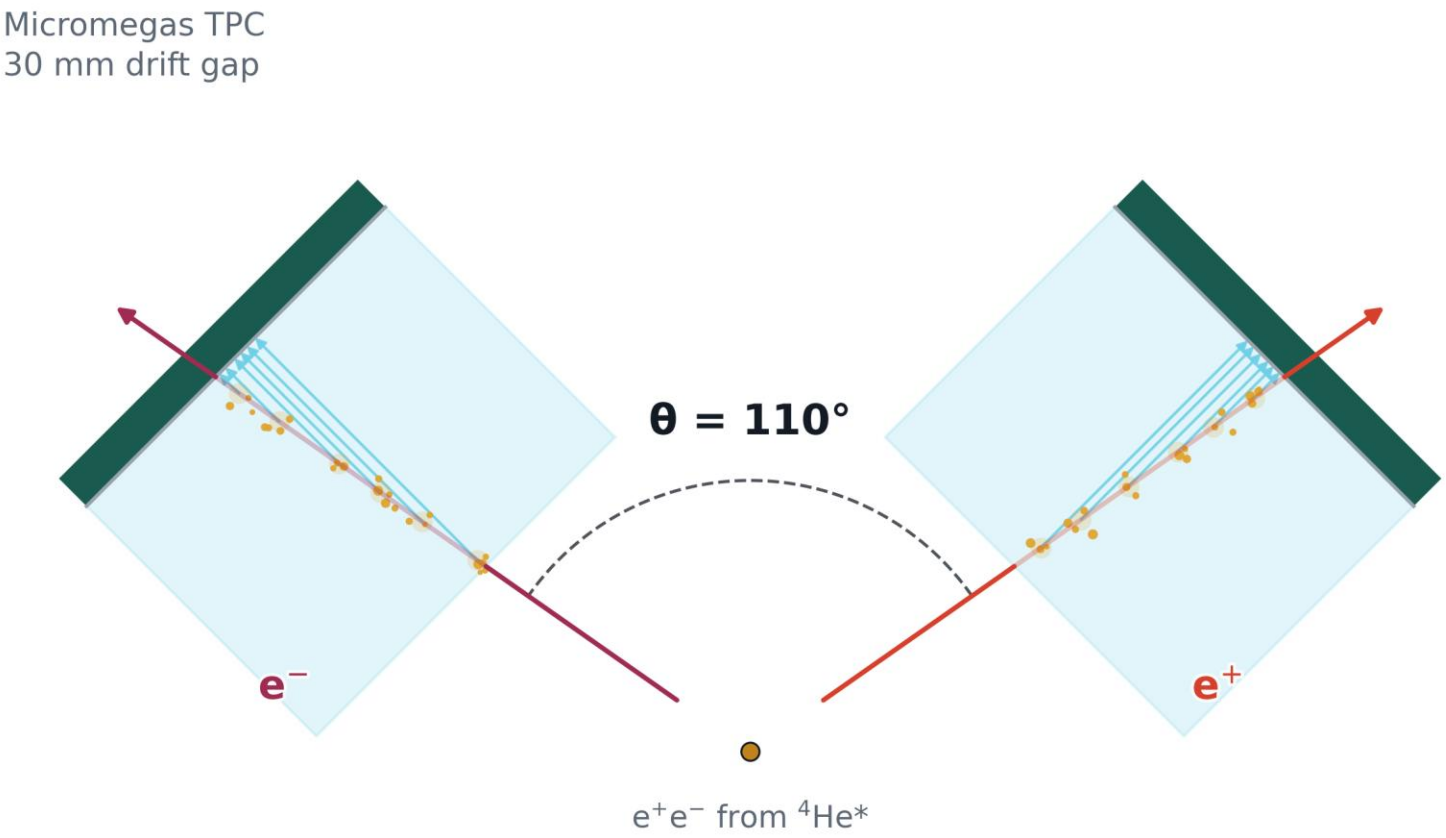
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5. So this is what we look for

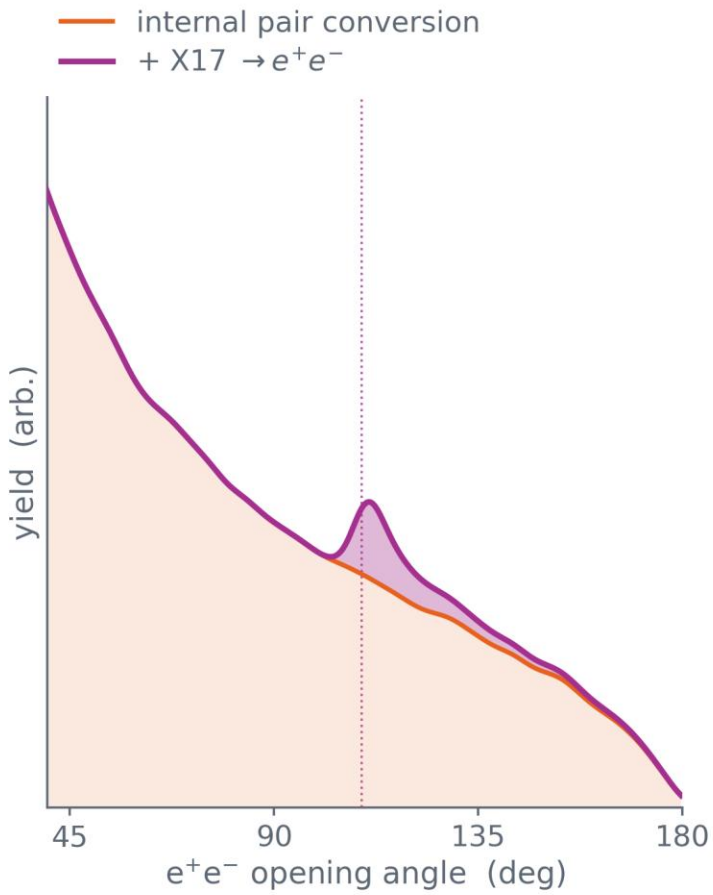


The opening angle observable -- X17 moves slow

6. Measure it with a Micromegas TPC

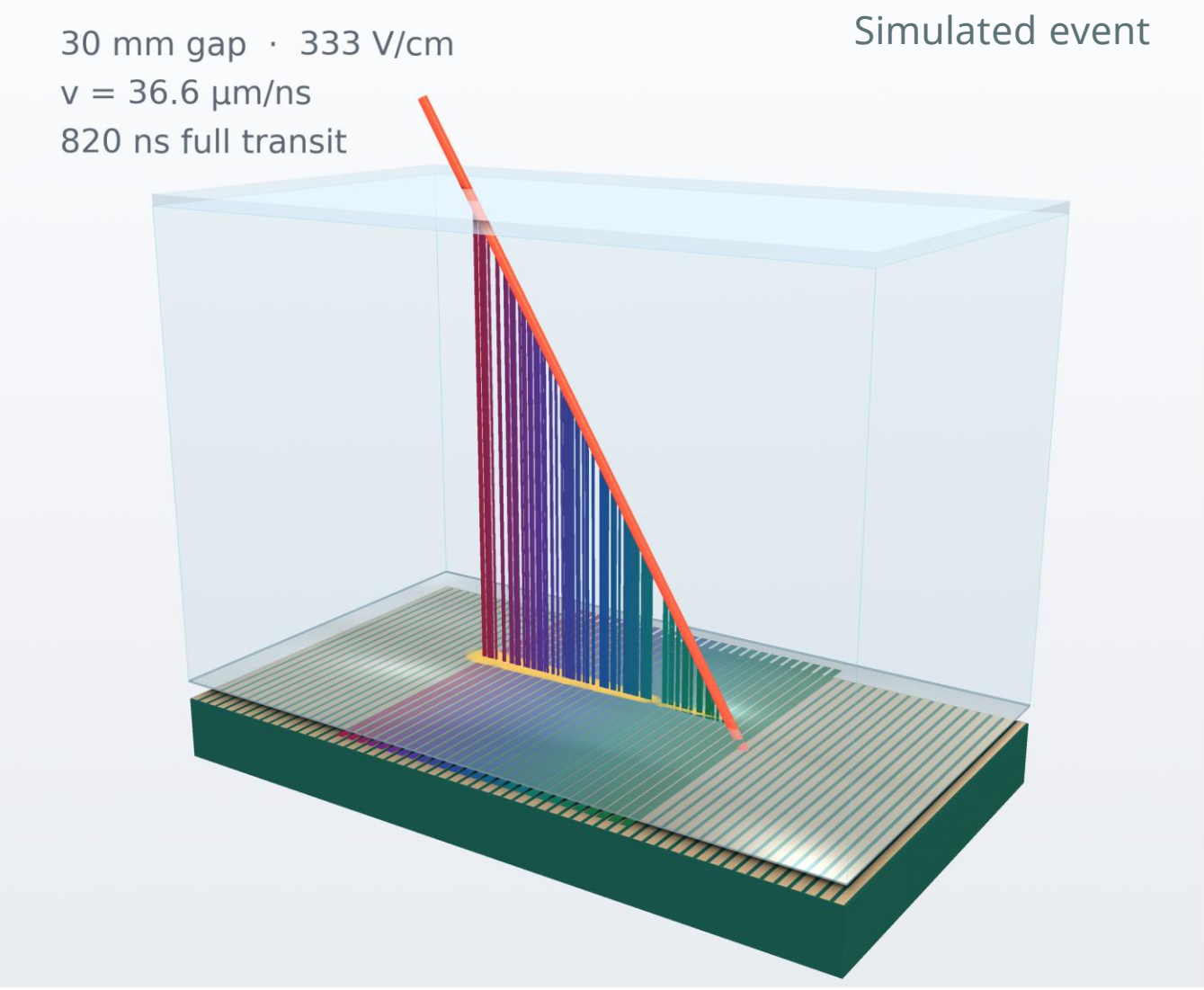


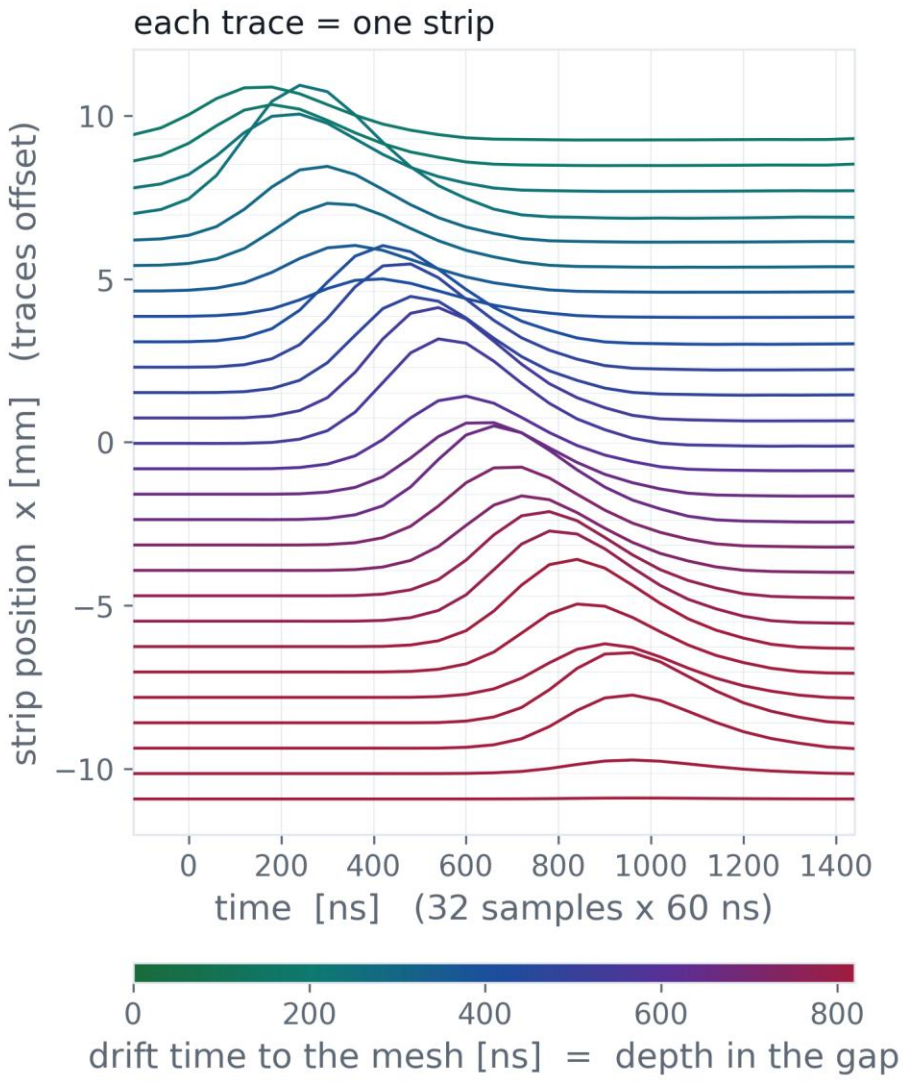
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DESIGNED & BUILT AT CEA SACLAY BY
Stéphan Aune

TPC operation

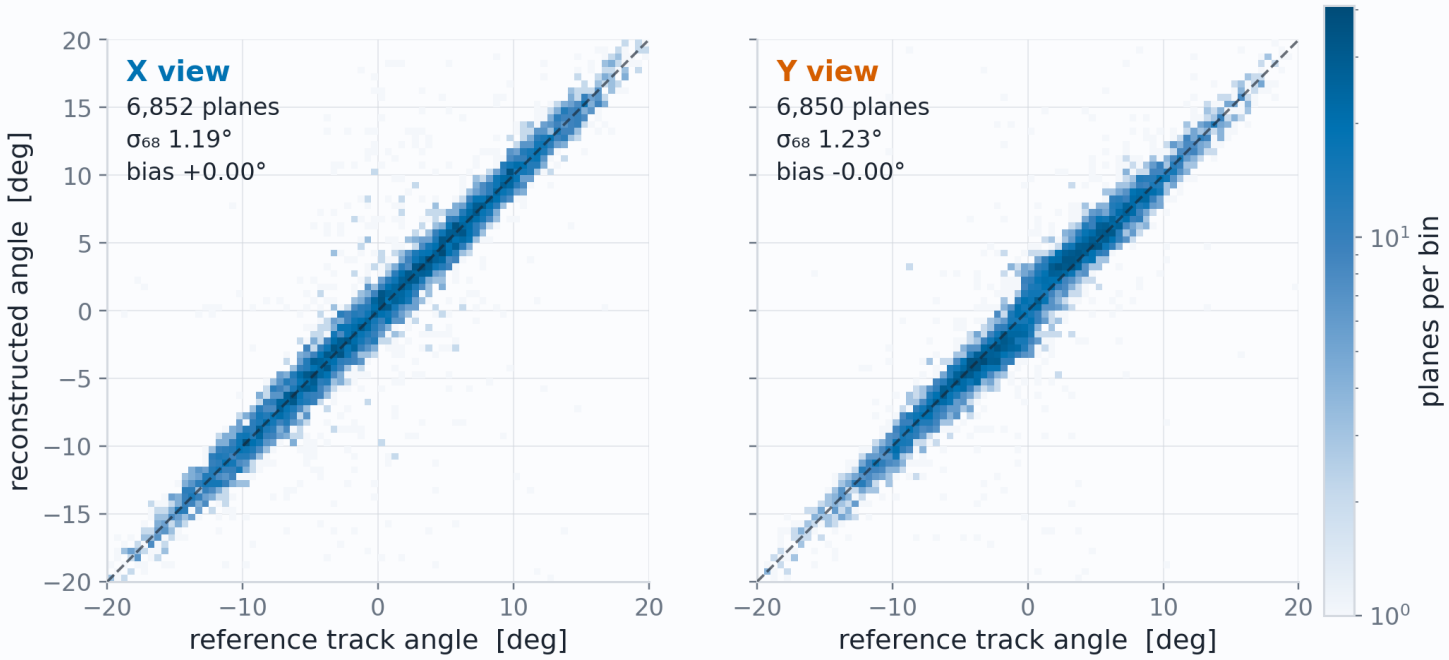




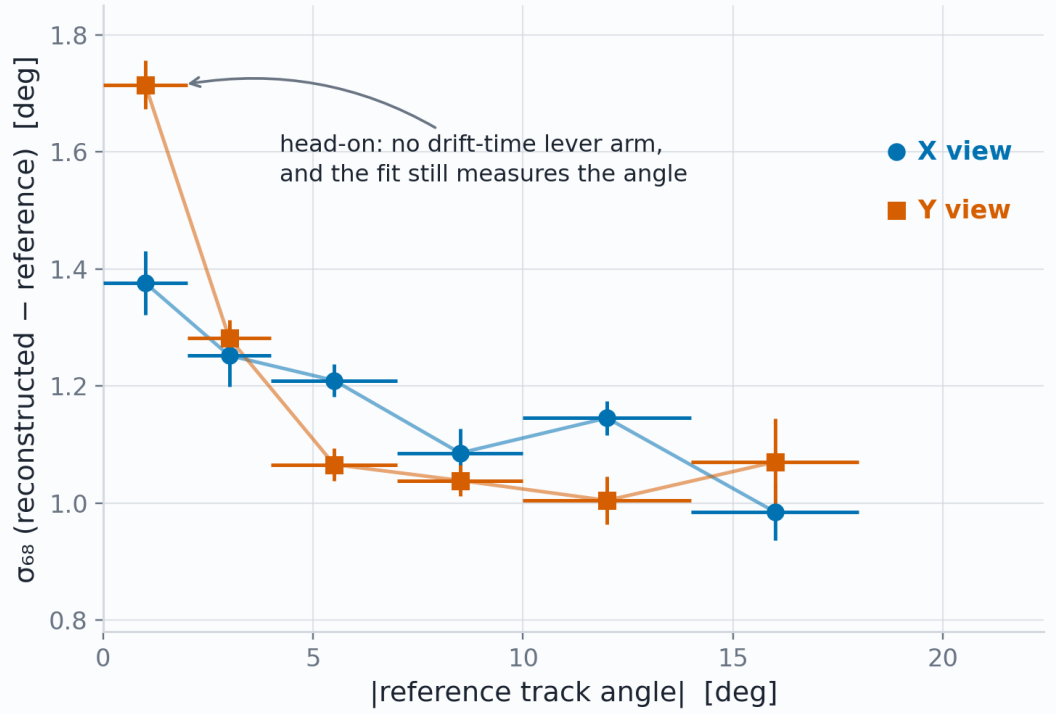
WORK IN PROGRESS

Adequate angular, spatial and temporal resolution

Reconstructed against reference, per plane



Head on tracks reasonable and angled tracks approach 1°



1.19 / 1.23°
angle per plane, X / Y (σ_{68}), bias < 0.1°

180 µm
position at normal incidence, **±0.035 mm** conservatively

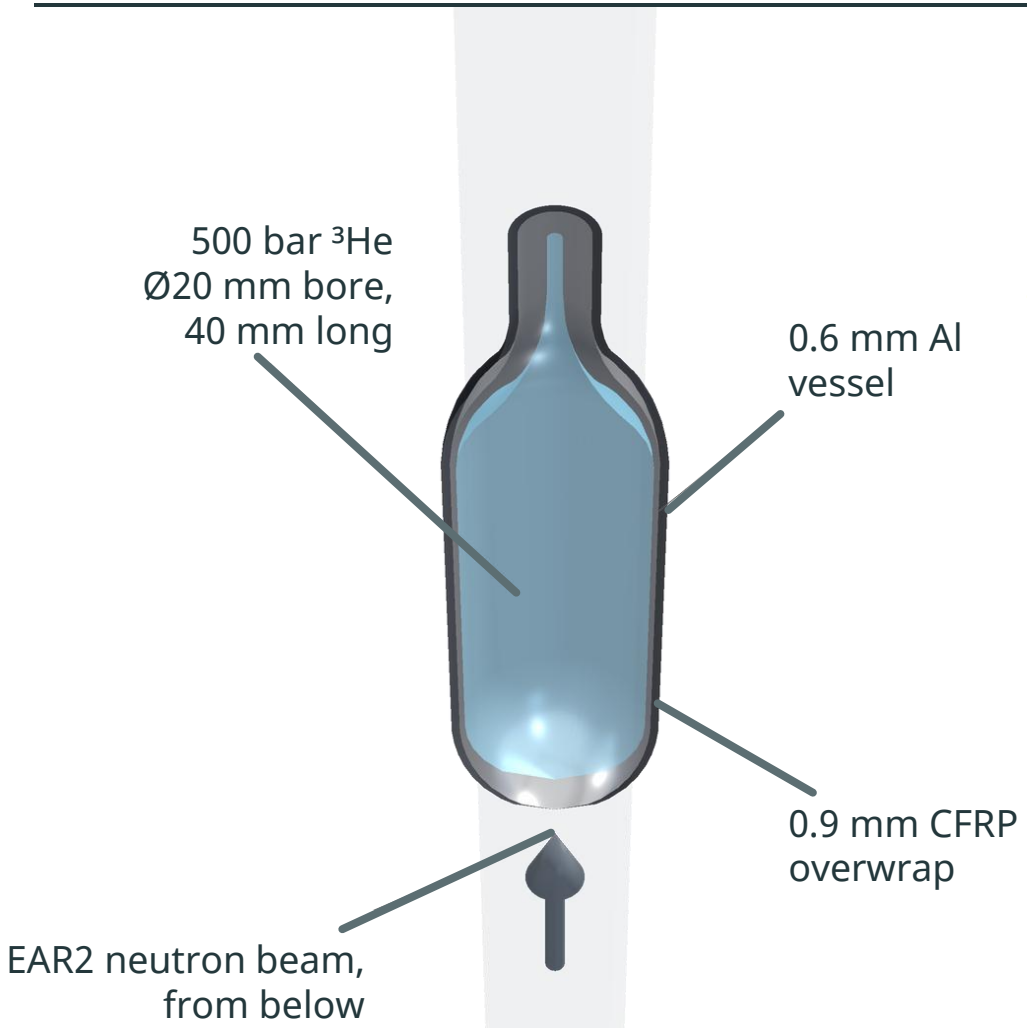
33 ns
plane-to-plane timing, ≈ 1.1 mm of drift

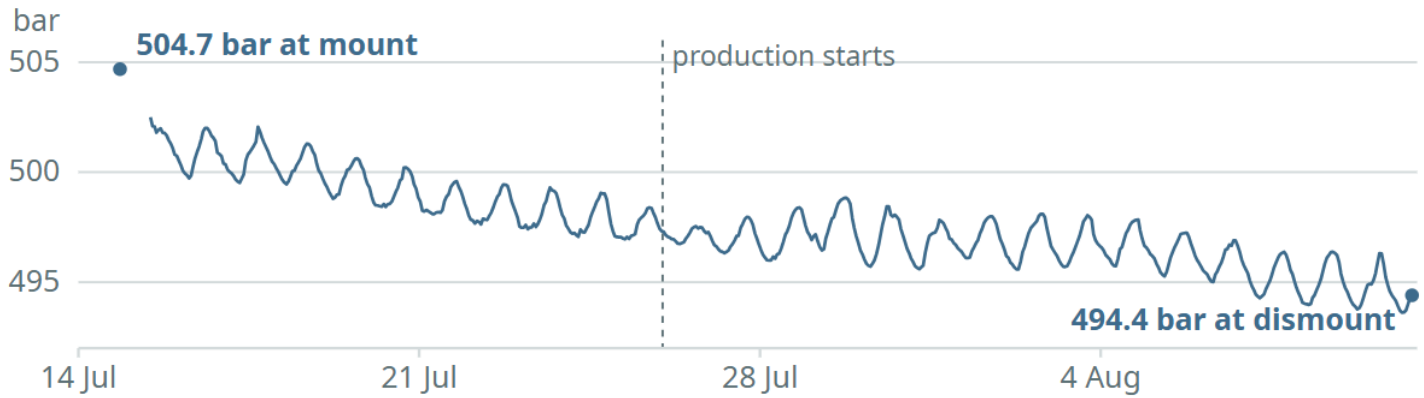
02

The EAR2 Setup

An apparatus capable of detecting X17

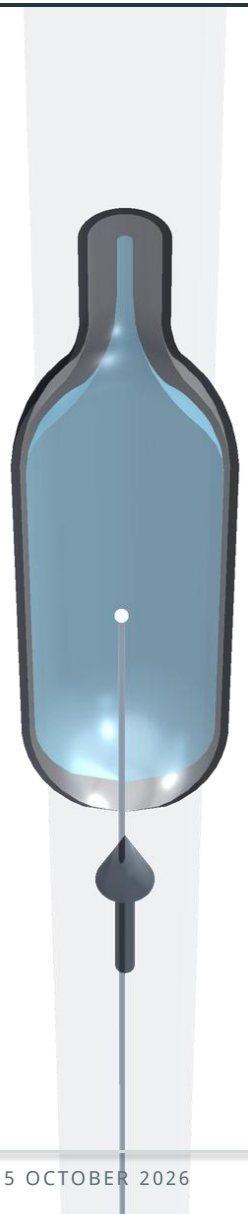
The vessel, as built: 500 bar of ^{3}He in the beam

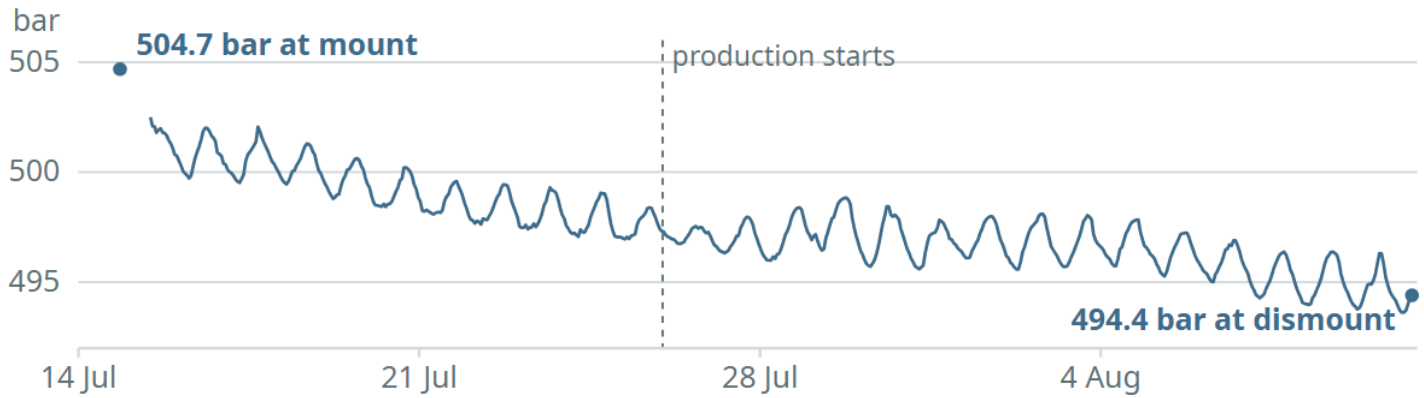




- **500 bar of ^{3}He** in as thin a capsule as we can build — least scattering on the way out, most interaction on the way in.
- Ø20 mm bore, 40 mm of gas, domed both ends — **walls of 0.6 mm Al + 0.9 mm CFRP**.

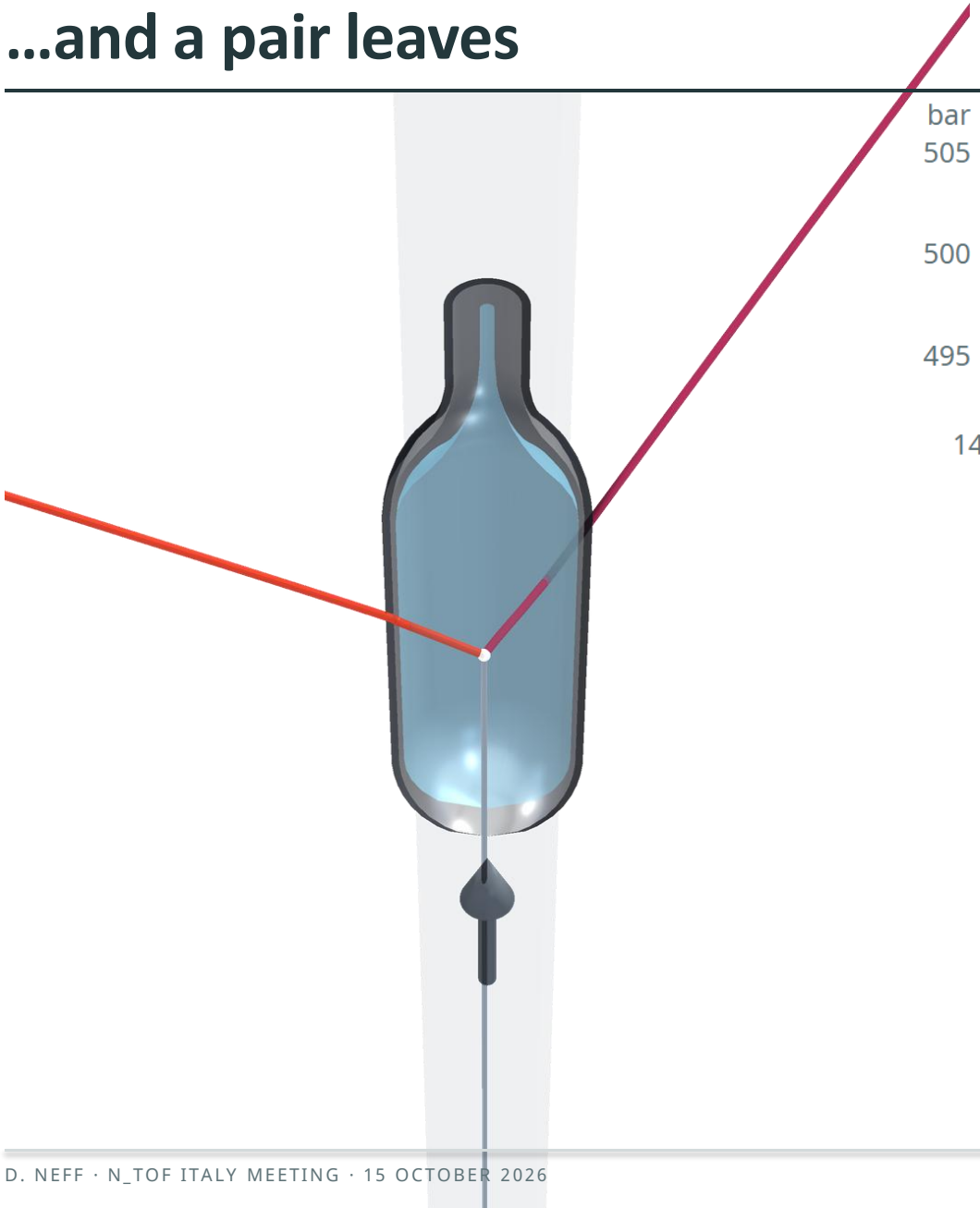
A neutron arrives

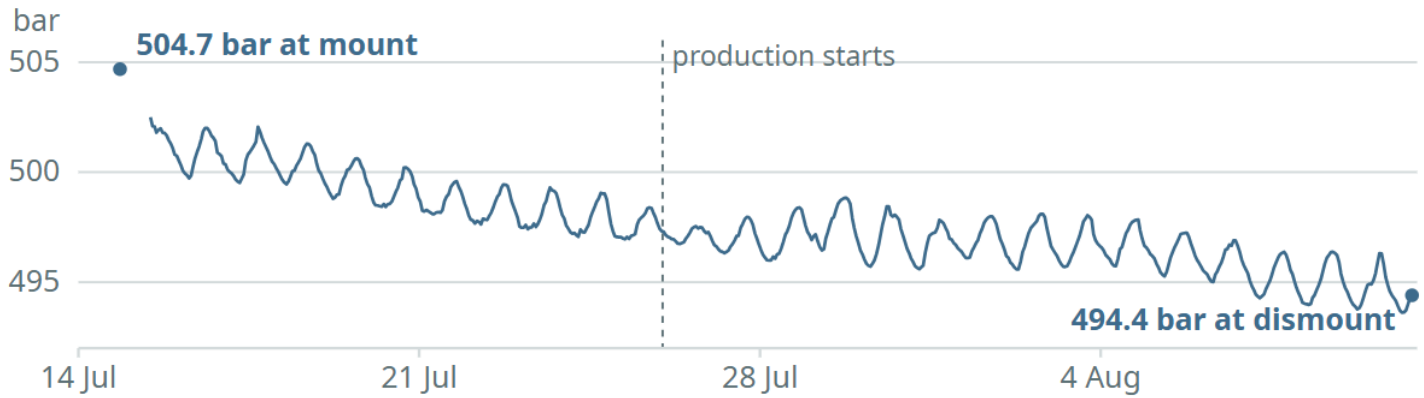




- 500 bar of ^{3}He in as thin a capsule as we can build — least scattering on the way out, most interaction on the way in.
- The **neutron beam arrives from below** and interacts in the ^{3}He gas volume.

…and a pair leaves

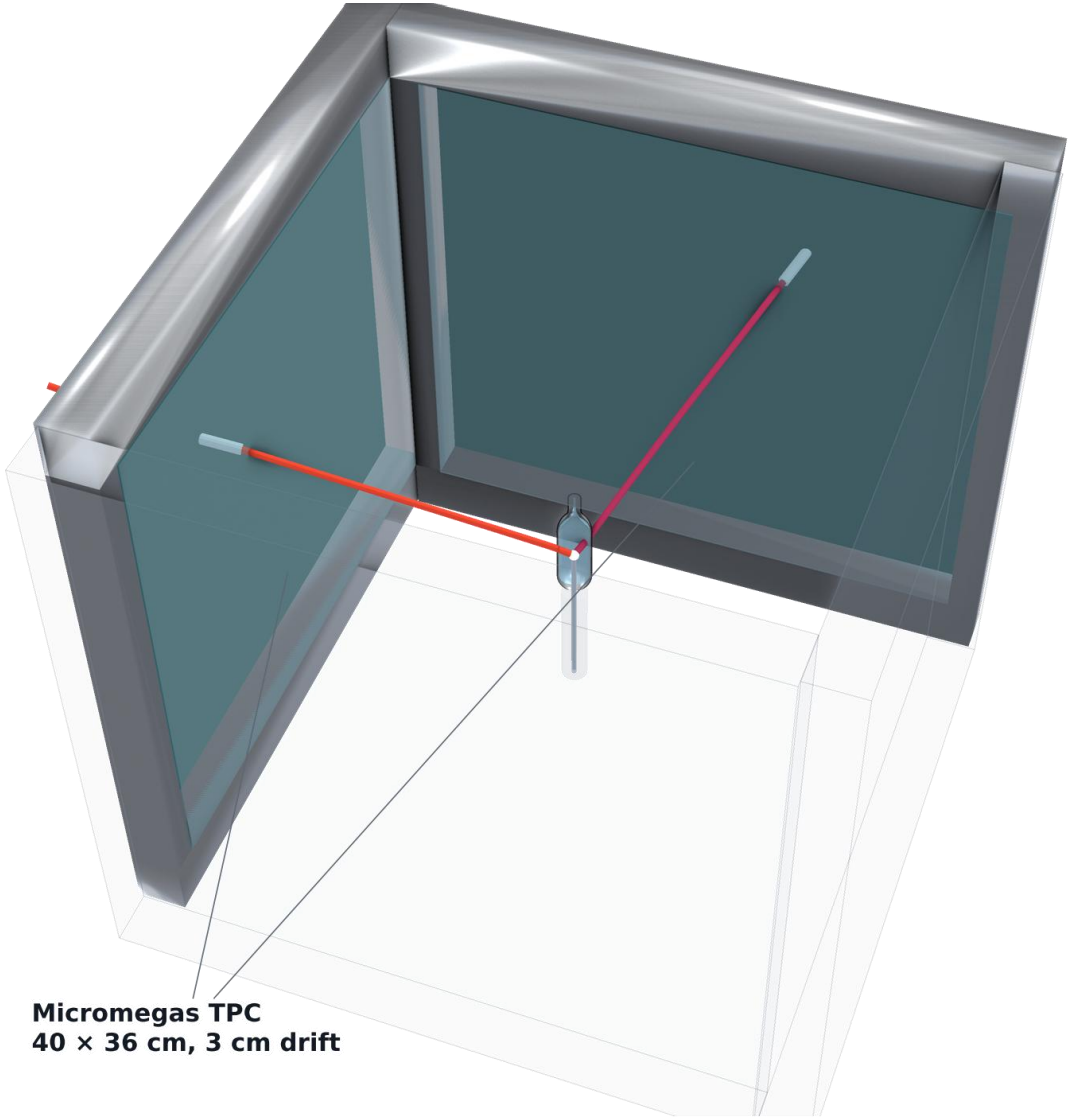




- 500 bar of ^{3}He in as thin a capsule as we can build — least scattering on the way out, most interaction on the way in.
- The neutron beam arrives from below and interacts in the ^{3}He gas volume.
- Some probability of **internal pair conversion** — or of **X17** → **e^+e^-** — leaving the capsule, with some scattering in its wall.

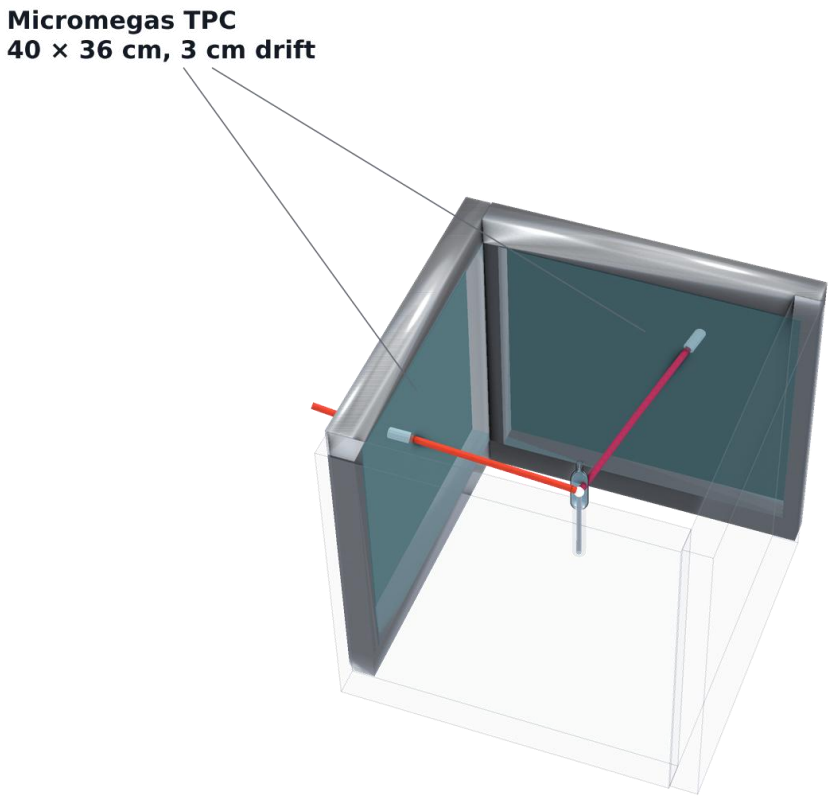
e^- e^+ neutron beam

Now build something that can catch them



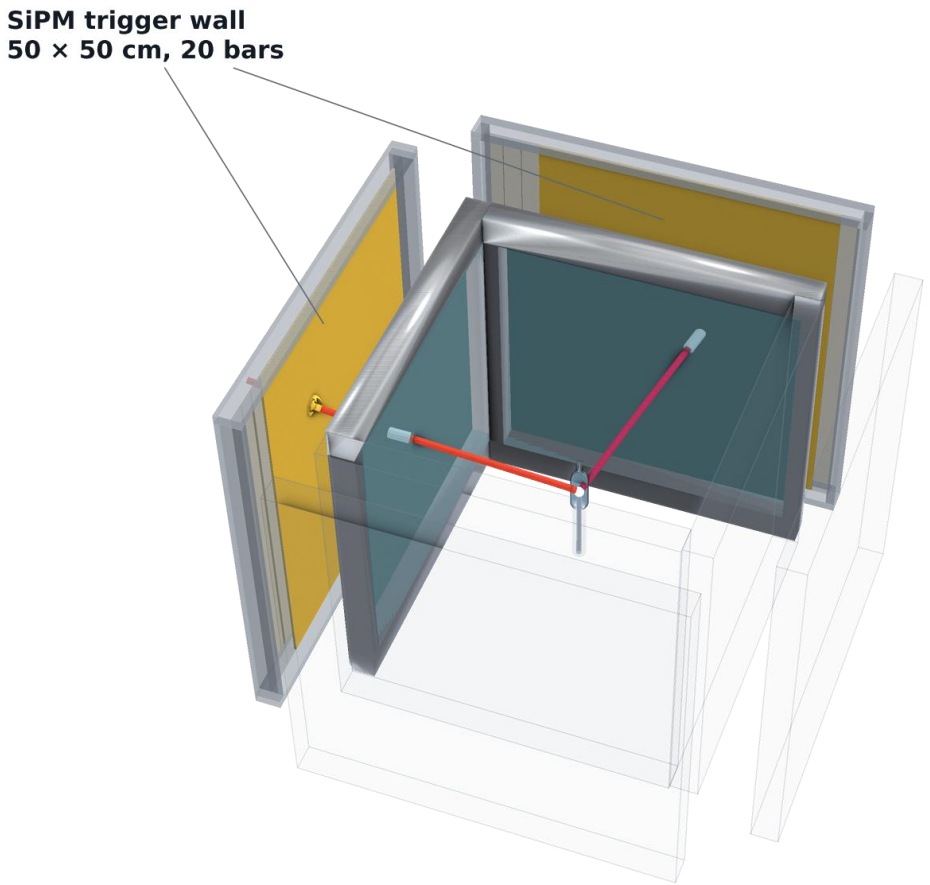
- Reduce **material budget** until the track angles have been recorded with four Micromegas TPCs

+ four Micromegas TPCs



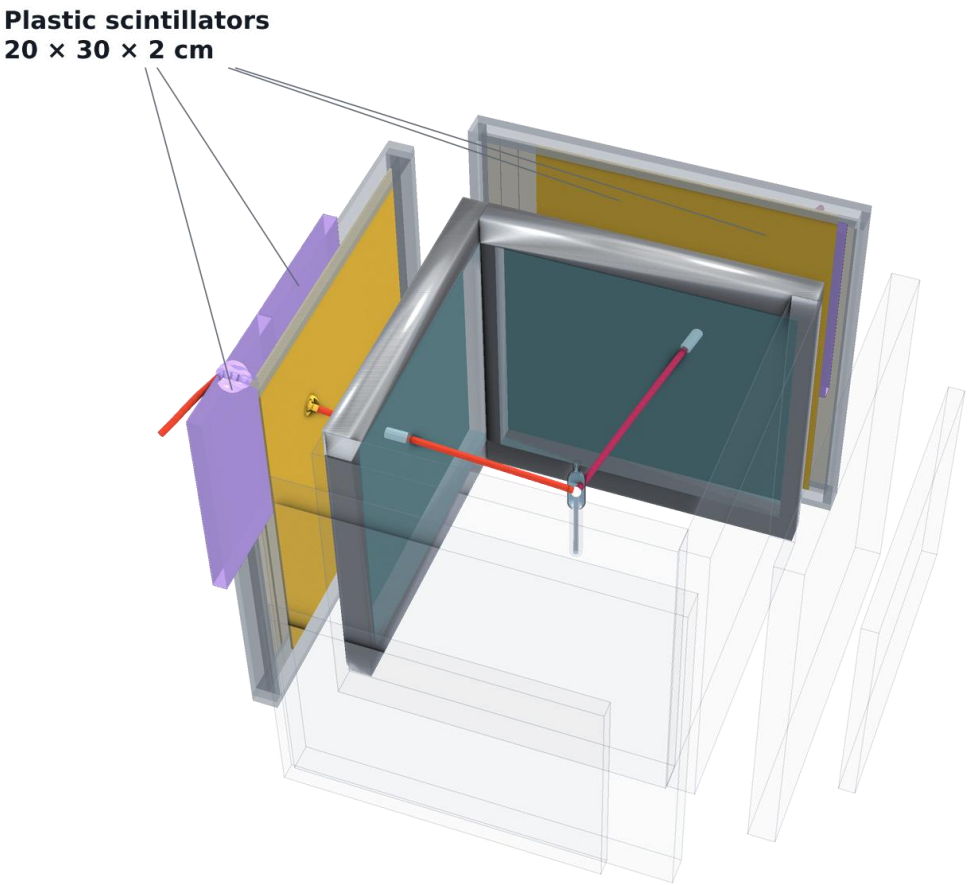
- Reduce material budget until the track angles have been recorded with four Micromegas TPCs
- Need an **external trigger** for these trackers

+ the SiPM trigger wall



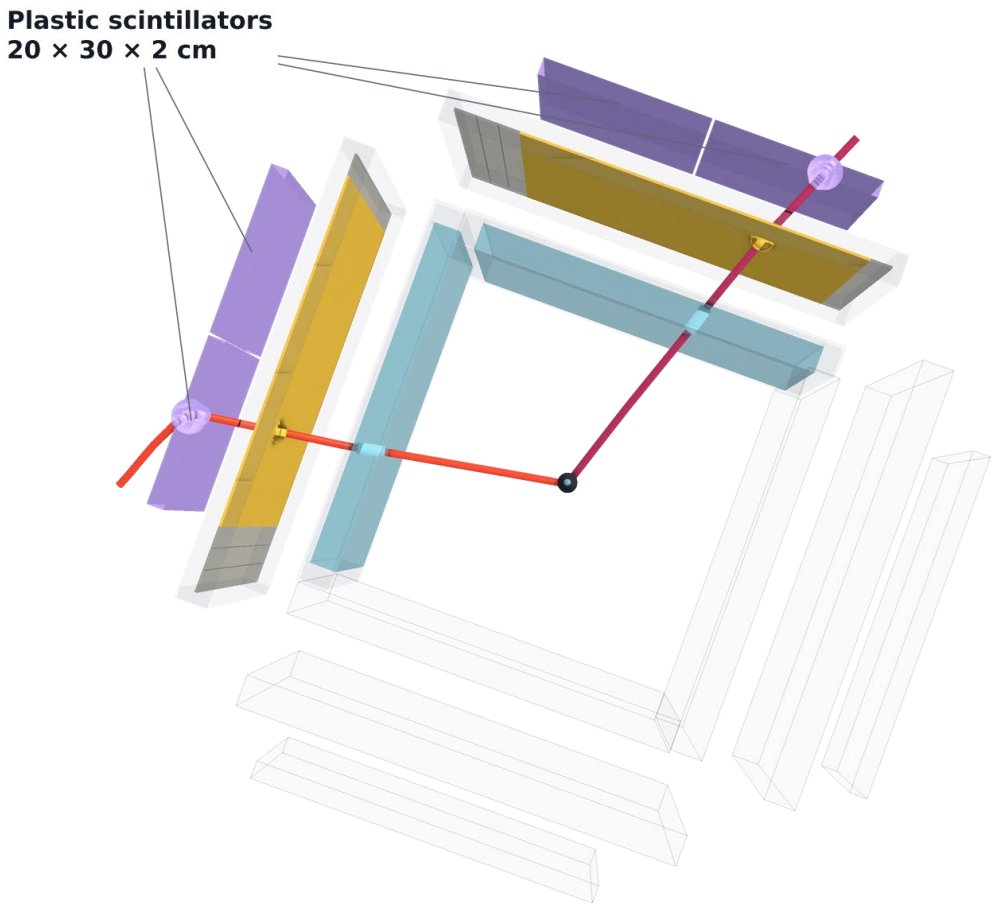
- Reduce material budget until the track angles have been recorded with four Micromegas TPCs
- Need an external trigger for these trackers
- Thin vertical scintillator bars readout by SiPMs: **register the hit** without stopping the electron or the positron. Precise timing.

+ the plastic scintillators



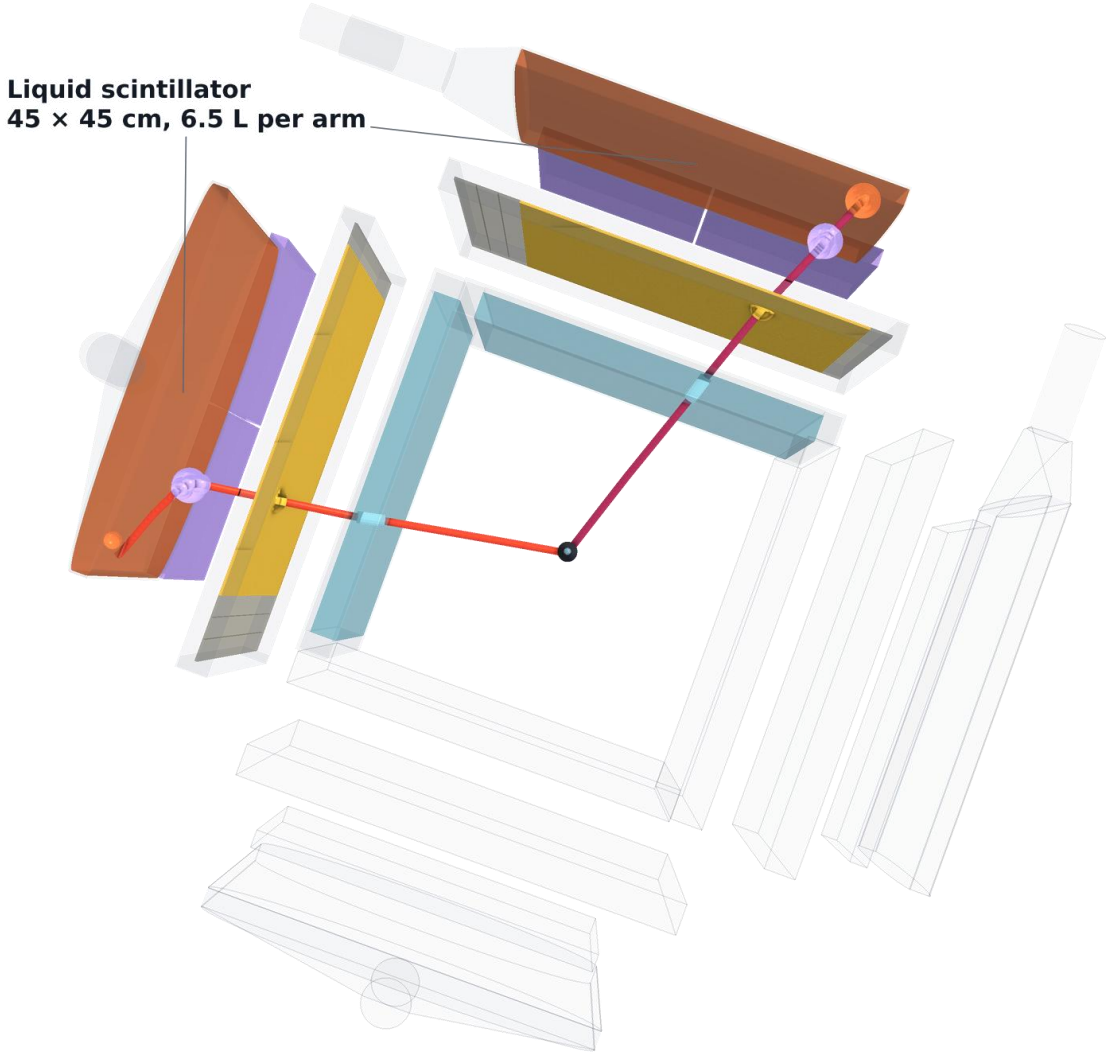
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- 2cm thick plastic scintillators **complete the coincidence** with the trigger wall.

The same event from above — where the pair goes



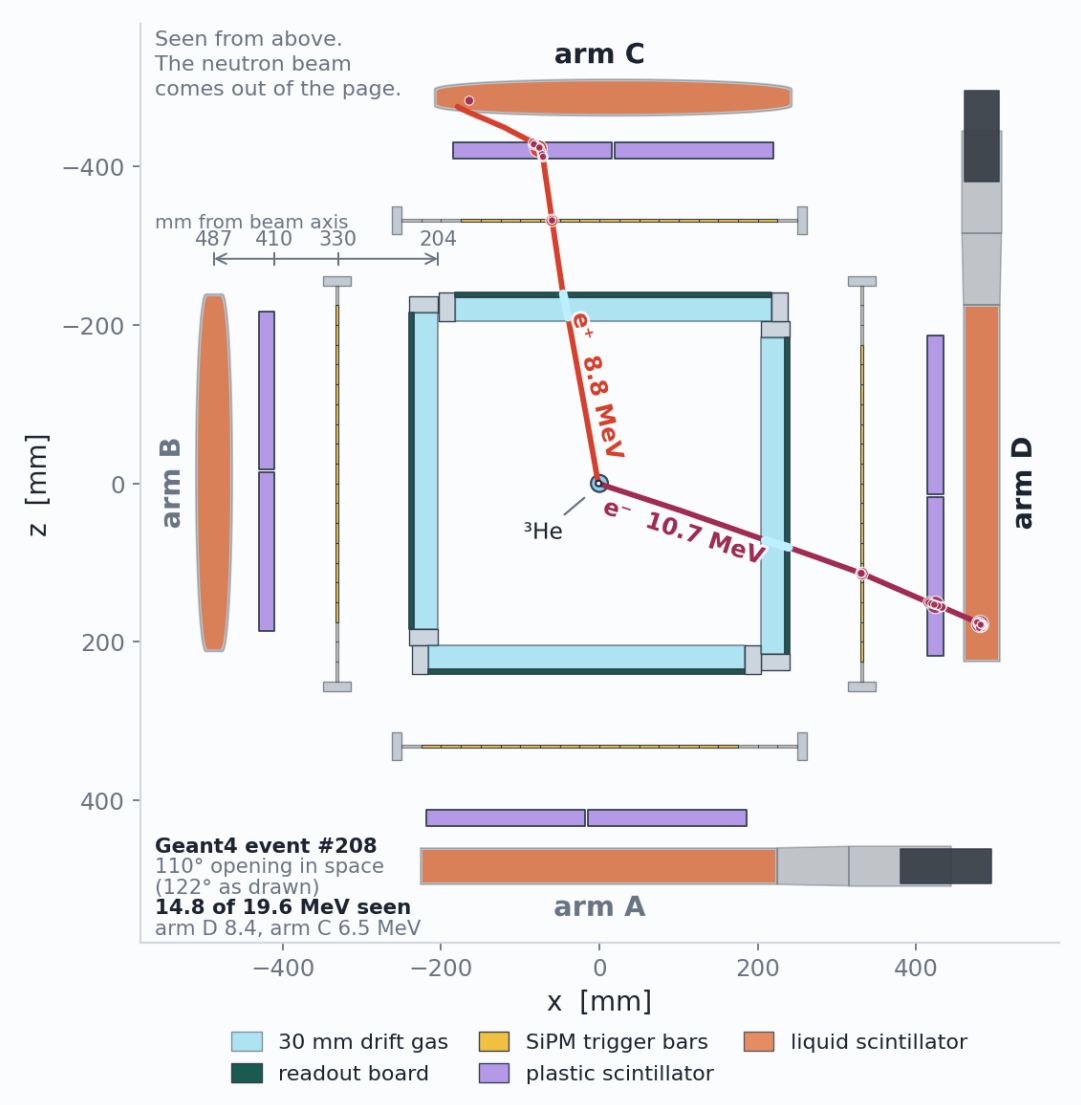
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- Thin vertical scintillator bars readout by SiPMs: register the hit without stopping the electron or the positron. Precise timing.
- 2cm thick plastic scintillators complete the coincidence with the trigger wall.
- Also provides stopping for **calorimetry**

+ liquid-scintillator calorimetry — the full setup



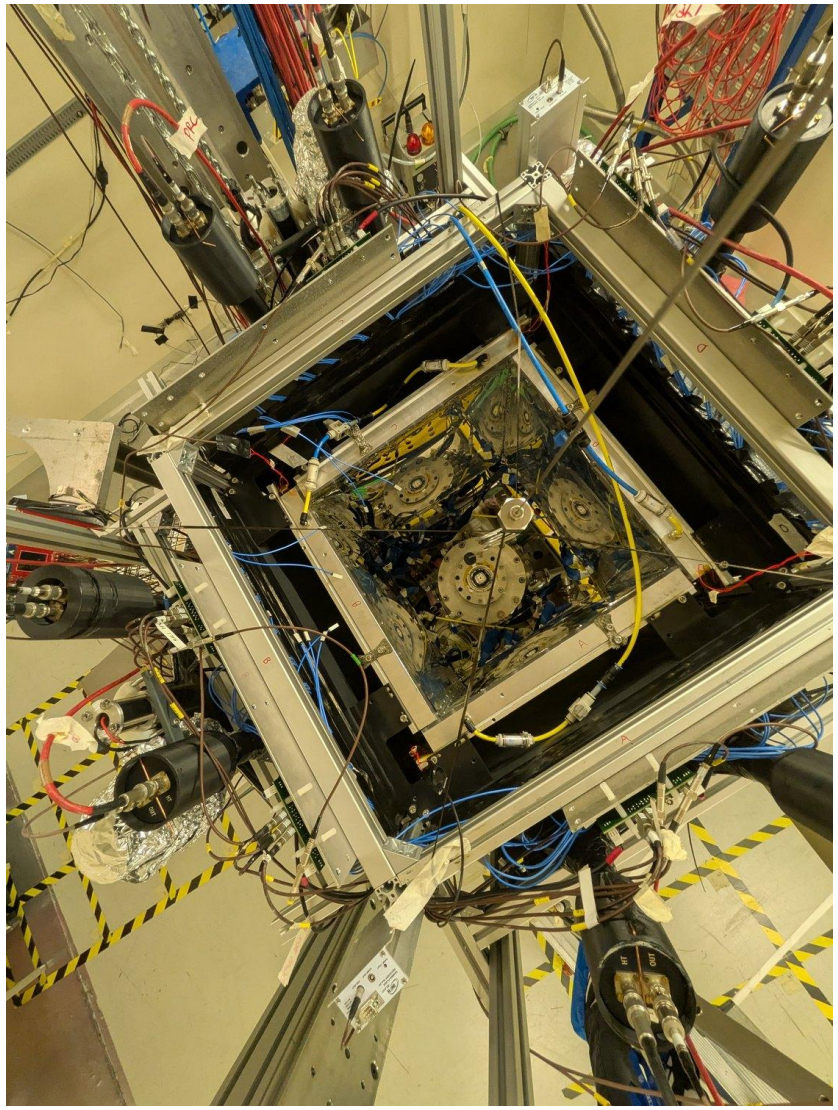
- Reduce material budget until the track angles have been recorded with four Micromegas TPCs
- Need an external trigger for these trackers
- Thin vertical scintillator bars readout by SiPMs: register the hit without stopping the electron or the positron. Precise timing.
- 2cm thick plastic scintillators complete the coincidence with the trigger wall.
- Also provides stopping for calorimetry
- Add 2cm of **liquid scintillator** to stop the pair completely

The same setup from above



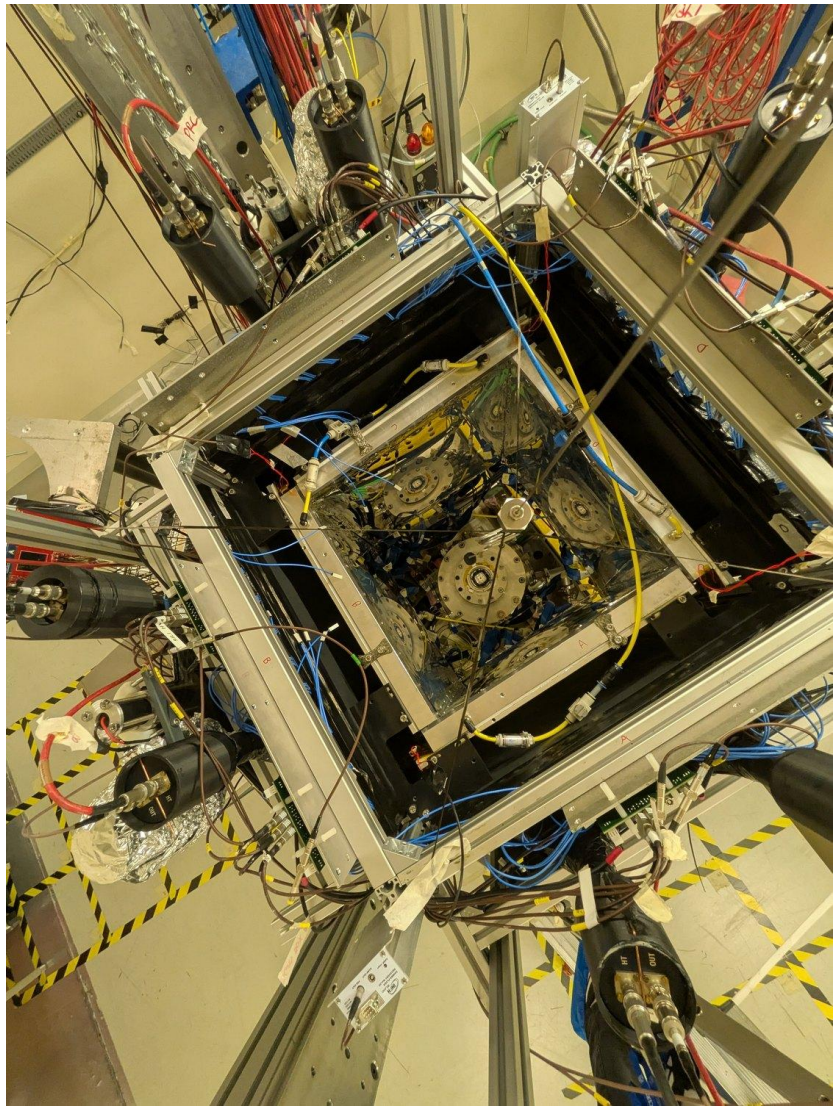
- **Opening angle** from the Micromegas, before anything else touches the pair.
- **Trigger and timing** from the wall and the plastics.
- **Calorimetry** from the full scintillator stack.

A look at the actual experiment – ran this July+August

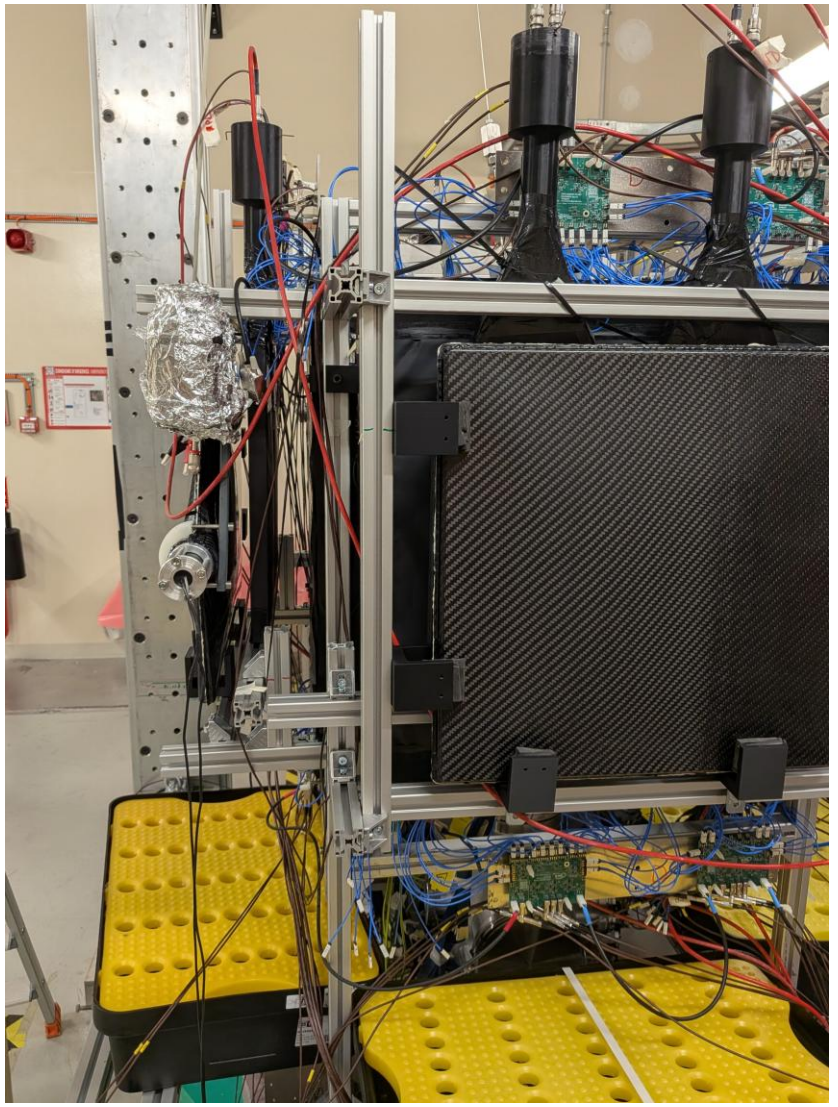


Looking down into the station · n_TOF EAR2, 10 August 2026

A look at the actual experiment – ran this July+August



Looking down into the station · n_TOF EAR2, 10 August 2026



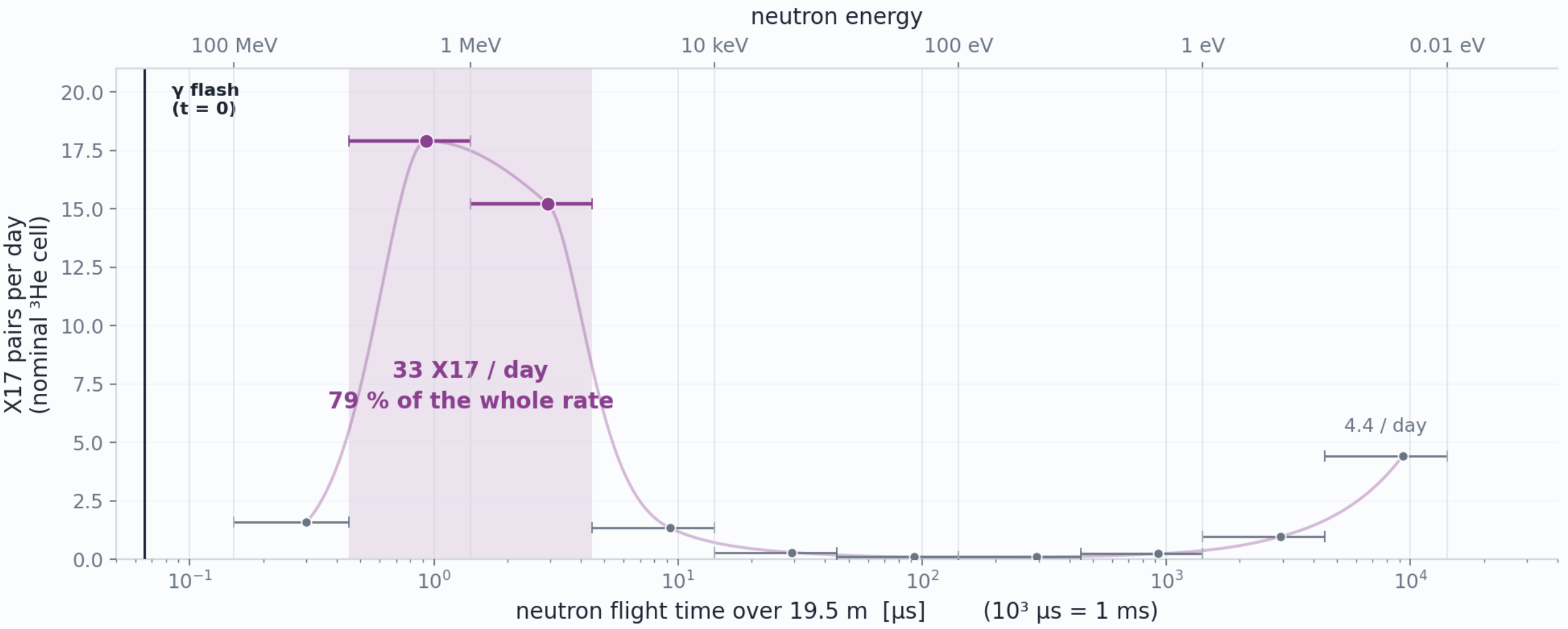
One arm, from outside · n_TOF EAR2, 10 August 2026

03

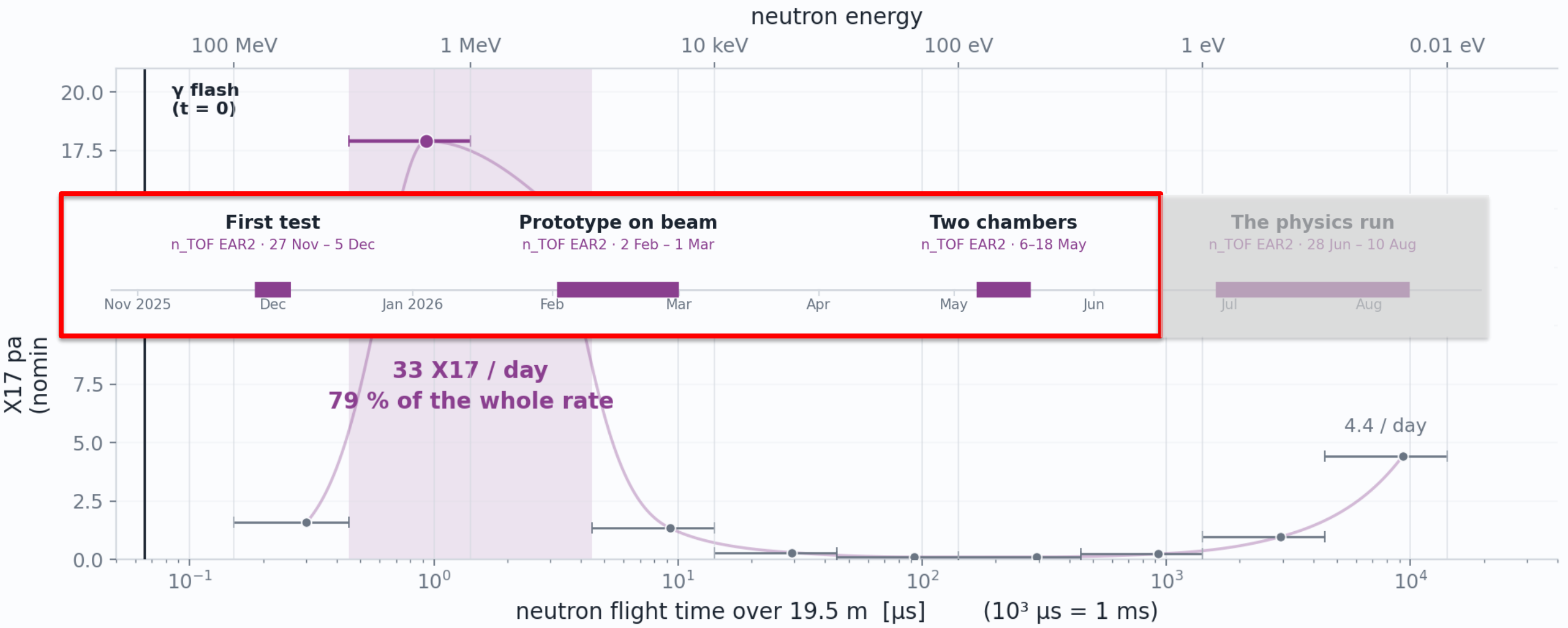
Plan A: the MeV neutrons

Where the X17 rate is, and why we could not reach it

Almost all of the X17 rate is in the MeV



Almost all of the X17 rate is in the MeV



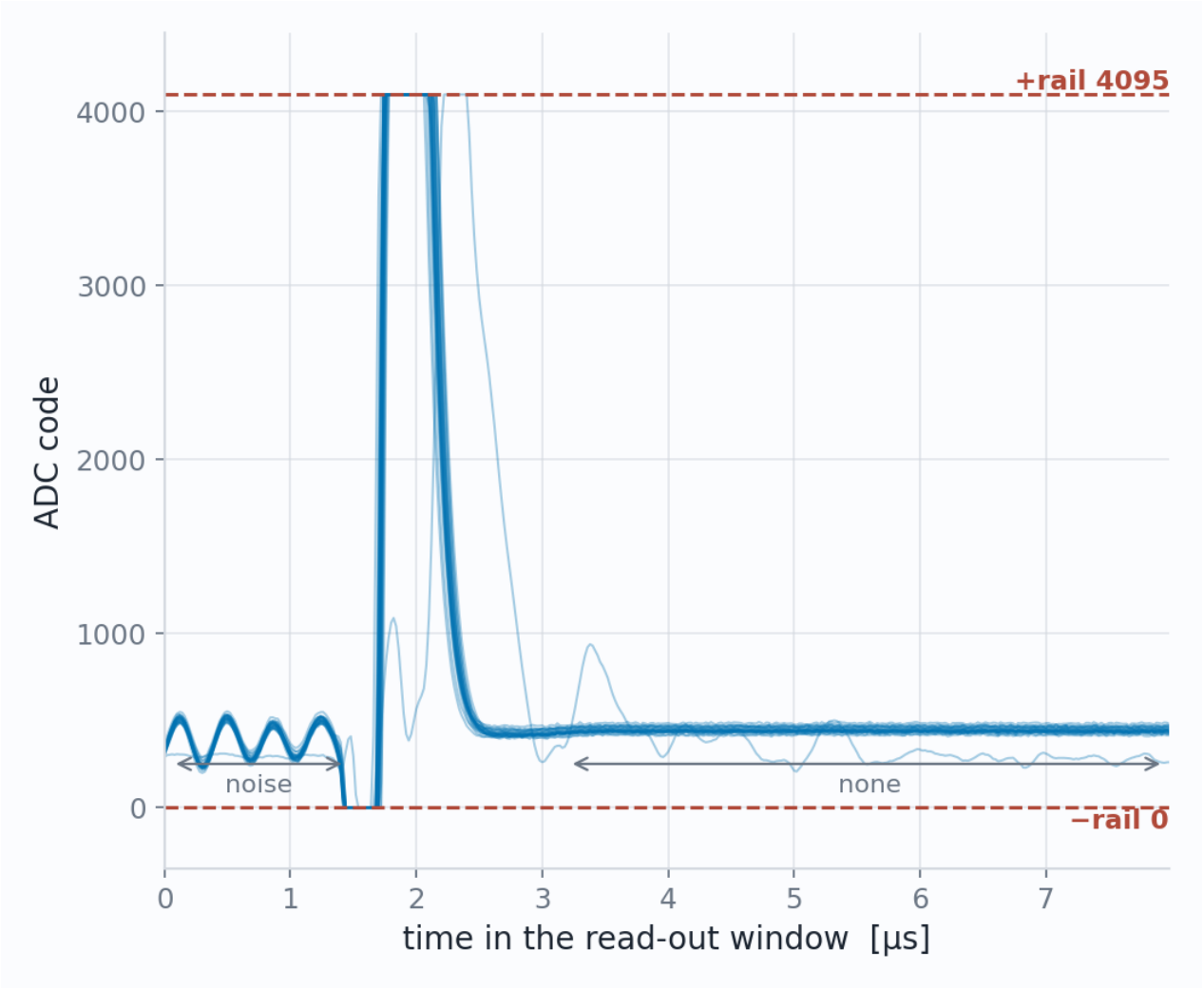
Problem: DREAM DAQ Can’t Handle Gamma Flash

- **DREAM** — *Dead-timeless Read-out Electronics ASIC for Micromegas*, developed at **CEA Saclay** and built for the **CLAS12** Micromegas vertex tracker.
- **512 channels per front-end unit** (8 ASICs × 64); **two units per chamber**, 1,024 channels for each MX17.
- Every channel: a **charge-sensitive preamplifier and shaper**, full scale switchable **50–600 fC**, into a switched-capacitor analogue memory.
- Sampled here at **20 ns**, digitized on an external trigger.

Problem: DREAM DAQ Can’t Handle Gamma Flash

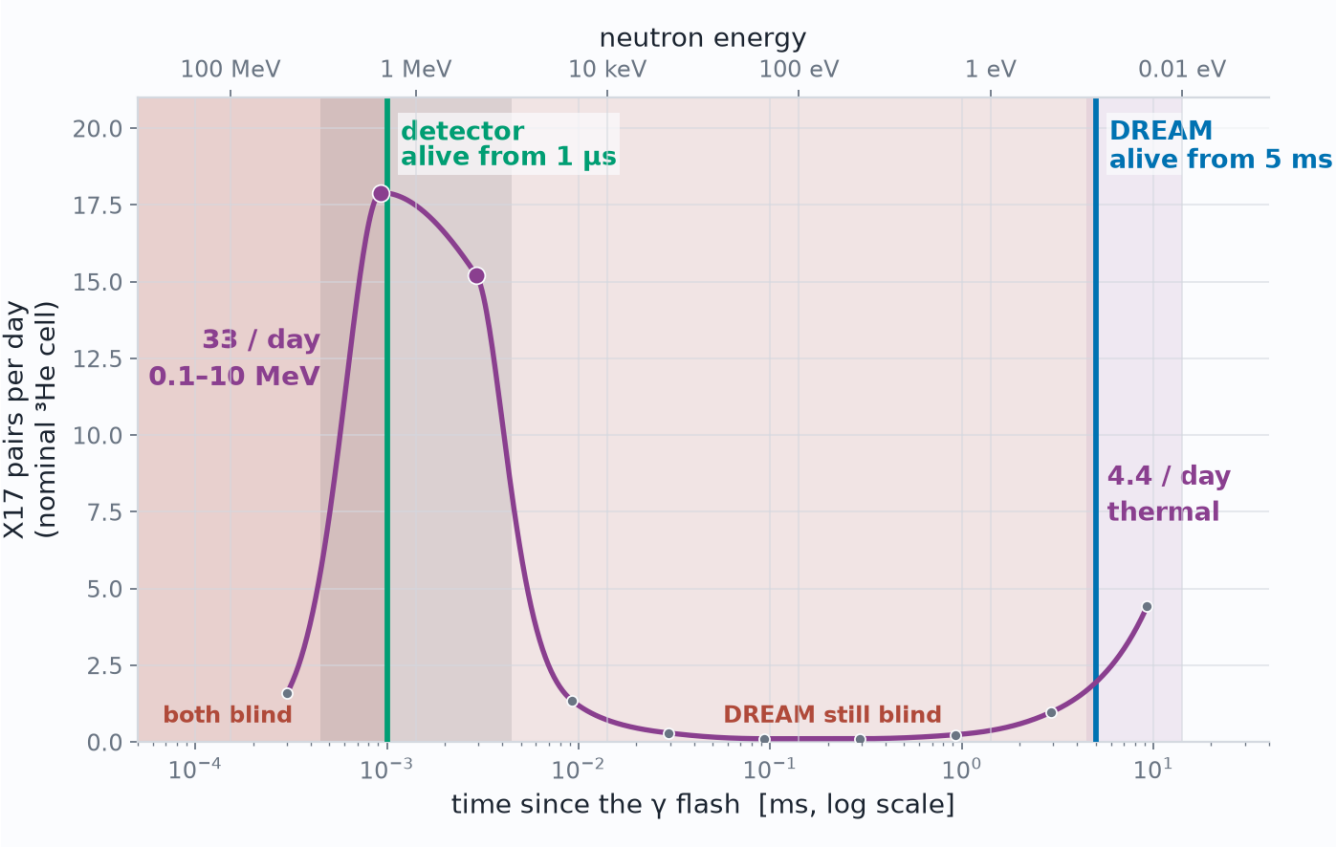
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- Every channel: a charge-sensitive preamplifier and shaper, full scale switchable 50–600 fC, into a switched-capacitor analogue memory.
- Sampled here at 20 ns, digitized on an external trigger.
- **February 2026, the first prototype on beam:** saturated — and saturated in a way that hides itself. The ADC rails to **+4095**, drops to **0**, and settles onto a flat baseline within **~3 µs**.

One flash, the 24 hottest strips



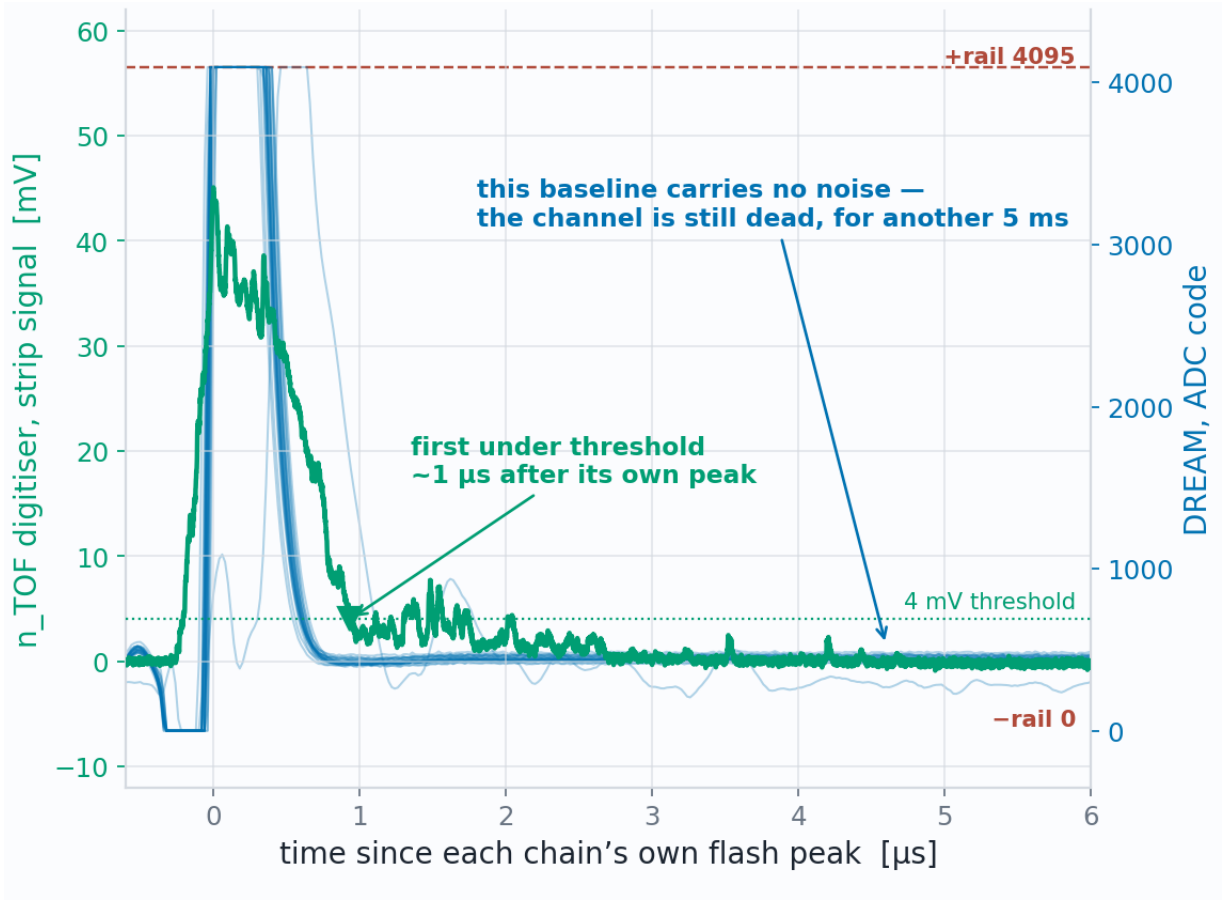
The detector is fine. The DREAM DAQ is not.

When each DAQ is usable again

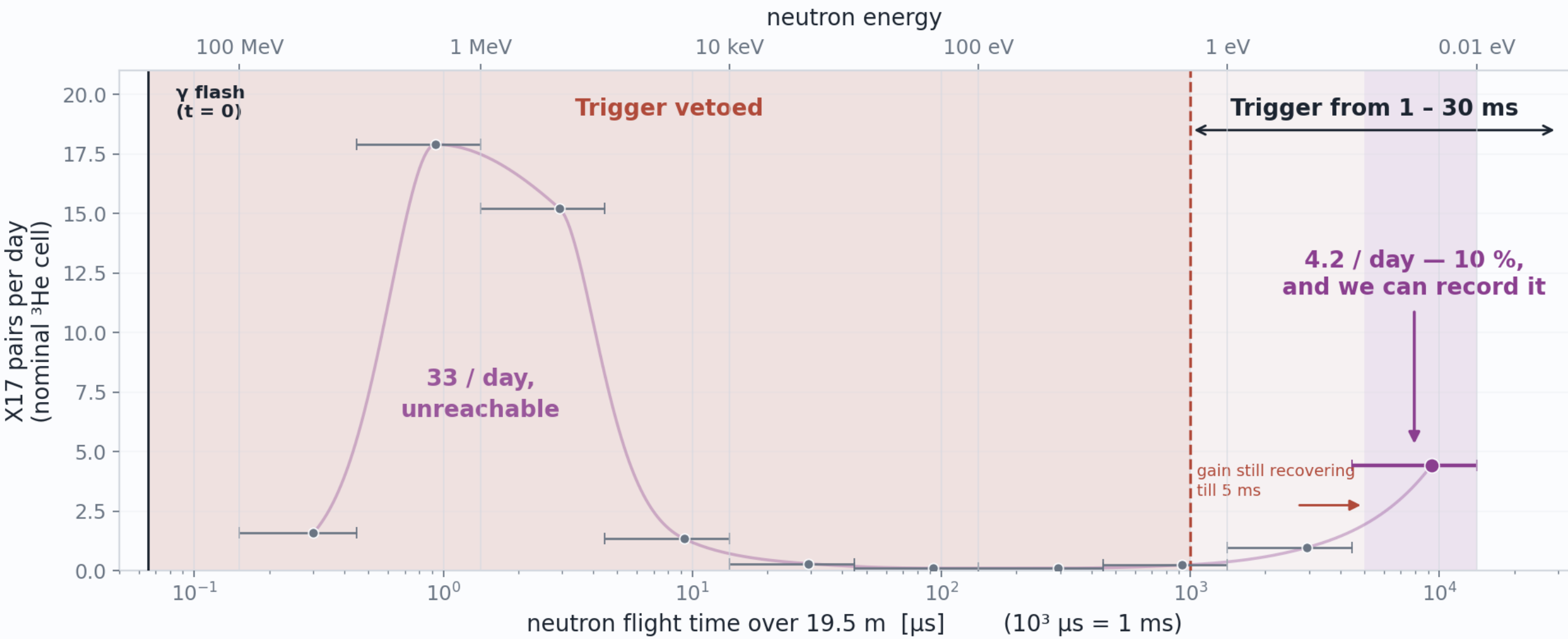


The same flash, seen by both DAQ

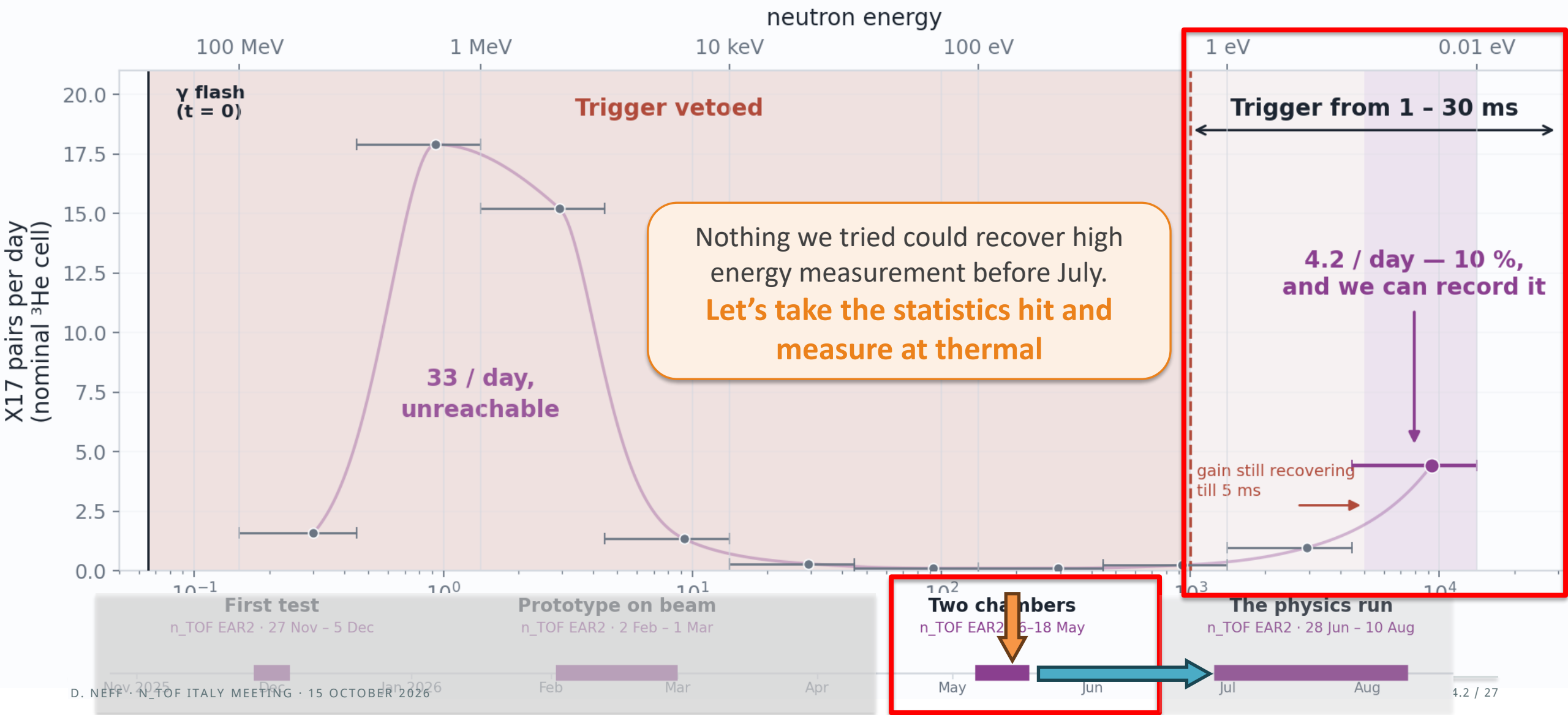
aligned on their own peaks



Plan A → Plan B : Make the measurement at thermal



Plan A → Plan B : Make the measurement at thermal

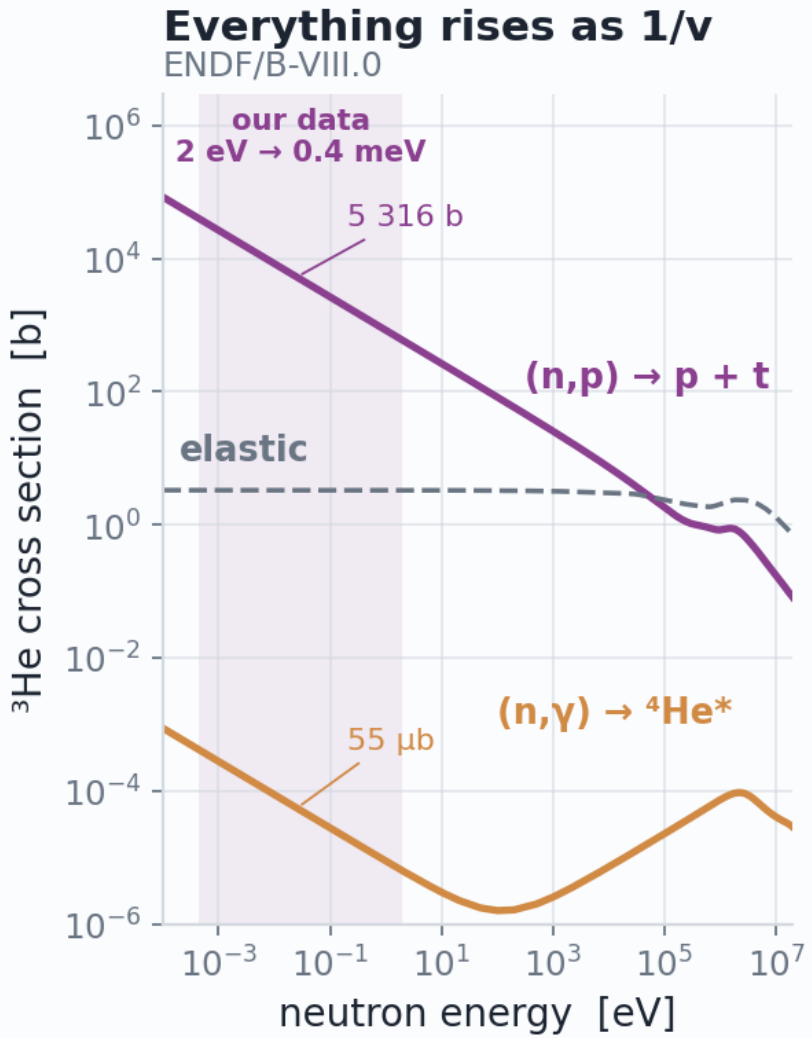


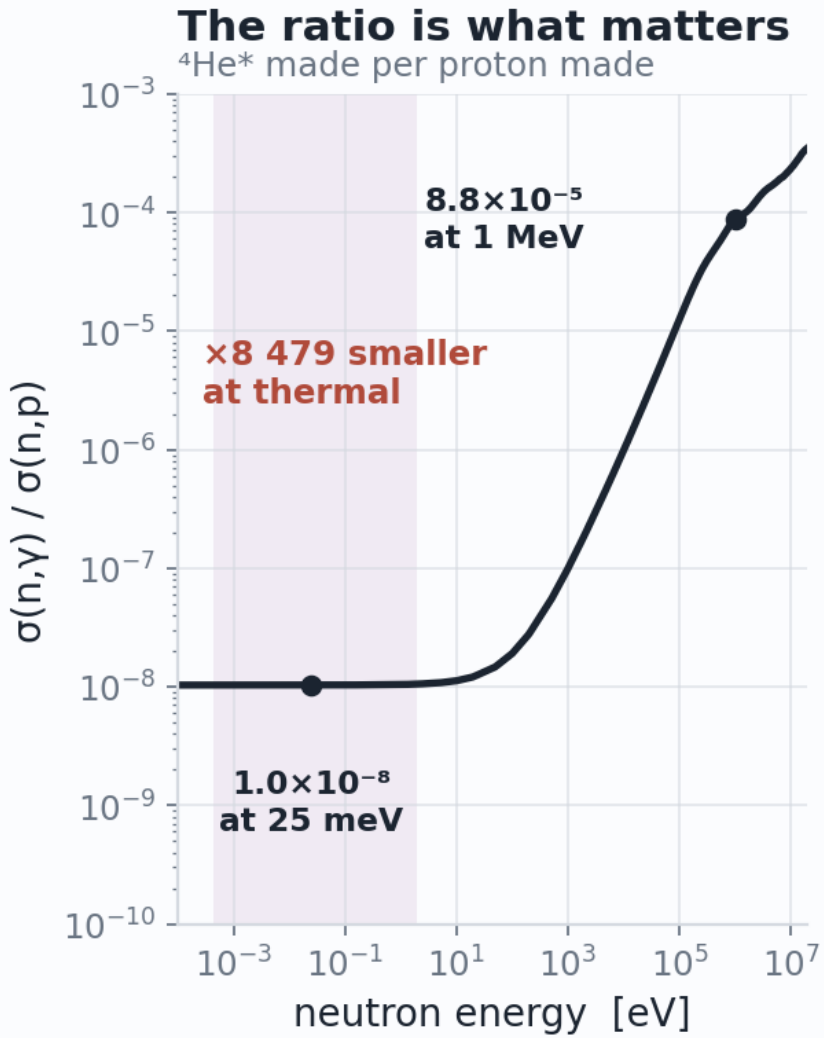
04

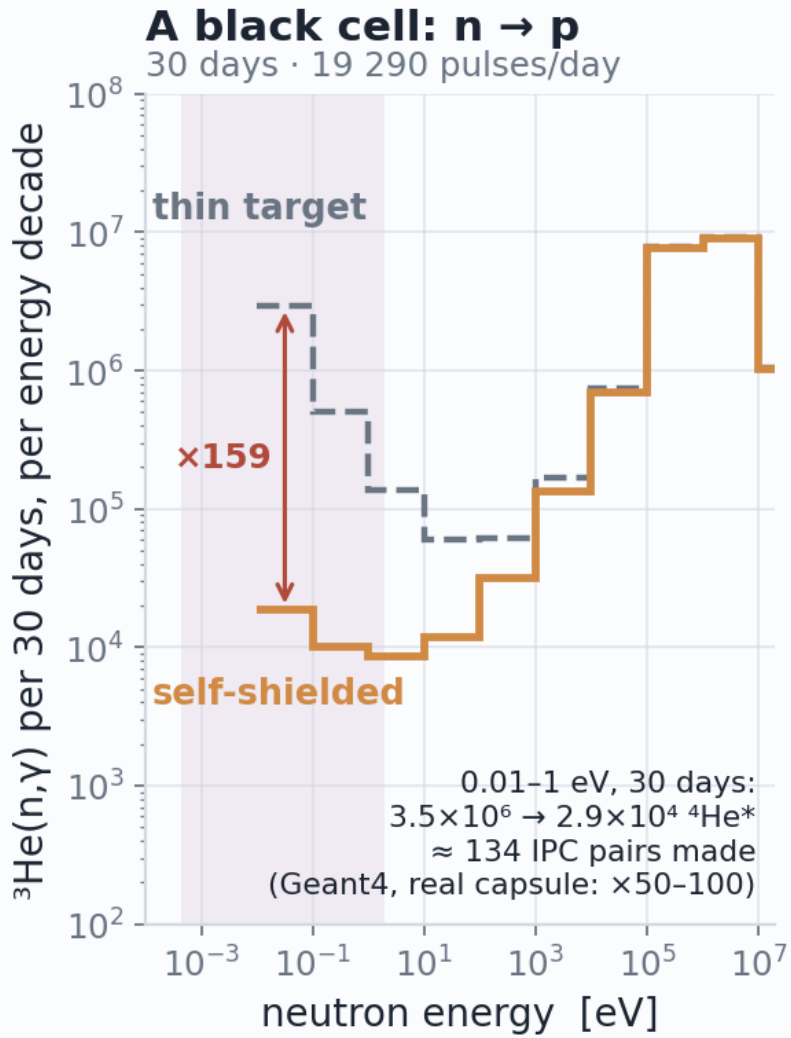
Plan B: the Problem with Thermals

What we learned from Geant4 before and during the run

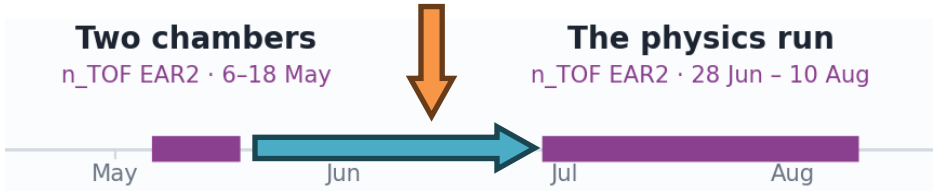
Problem: (n, p) dominates!







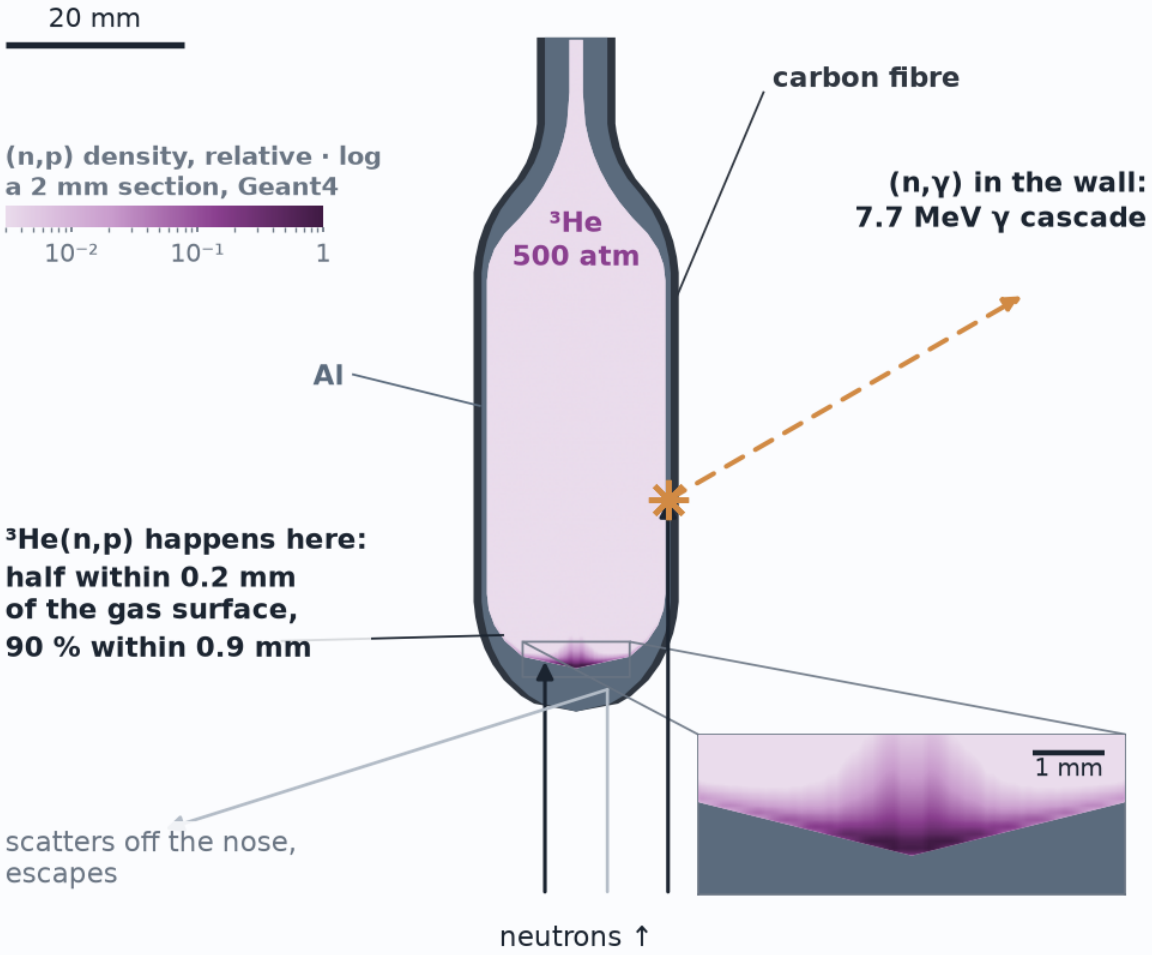
Self-shielded: (n, p) takes the neutrons



Plan B also fails

Plan C: measure what we can

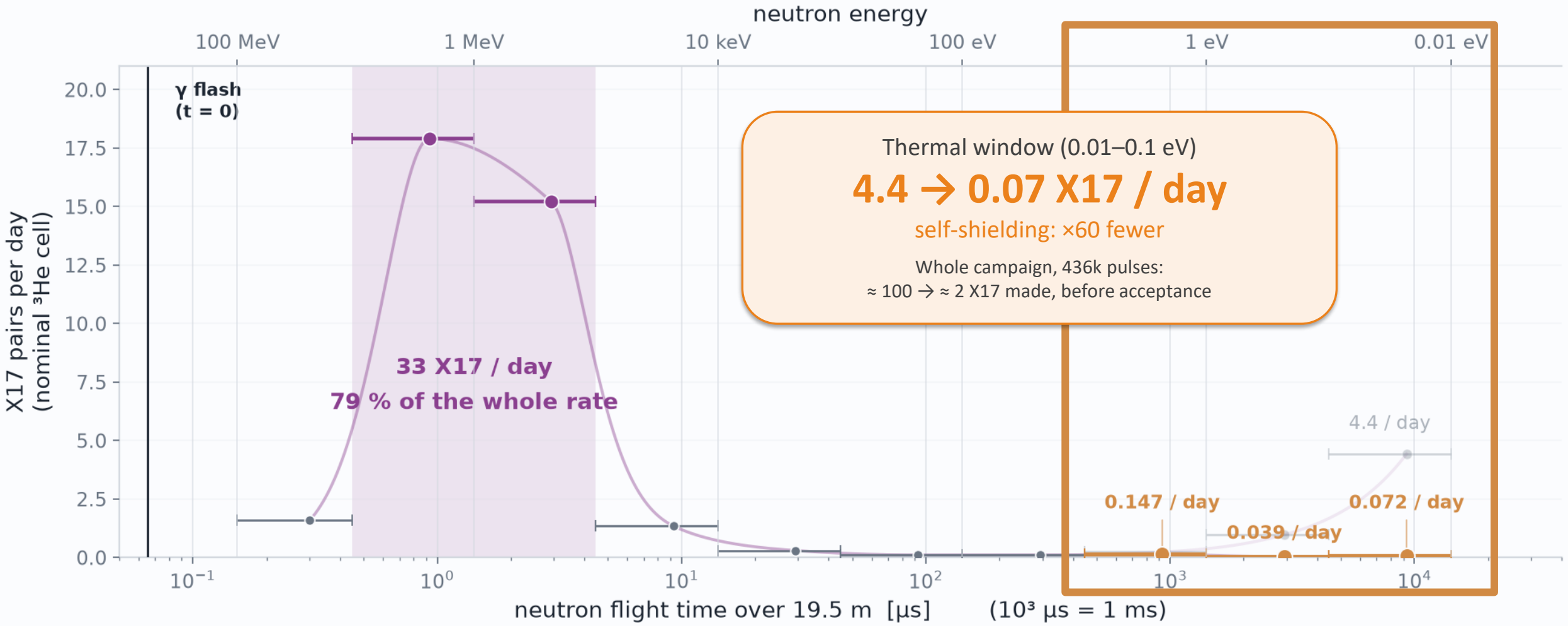
Geant4, 10^9 EAR2 neutrons below 2 eV · per neutron entering the capsule · counts per 30 days



What the neutron does	share	per 30 days
^{3}He(n,p) → p + t invisible to us: both stop in the gas	92.5 %	2.8×10^{12}
(n,γ) capture **→ the next figure opens this up**	**0.81 %**	**2.4×10^{10}**
scatters off the wall, escapes	6.6 %	2.0×10^{11}
other	0.13 %	3.9×10^{9}

where the (n,γ) happens	share	per 30 days
^{27}Al vessel	61.4 %	1.5×10^{10}
carbon-fibre wrap	3.2 %	7.7×10^{8}
scattered out, captured elsewhere	35.4 %	8.7×10^{9}
^{3}He(n,γ) analytic, self-shielded	1.3×10^{-4} %	3.1×10^{4}

What self-shielding leaves of the thermal X17 rate



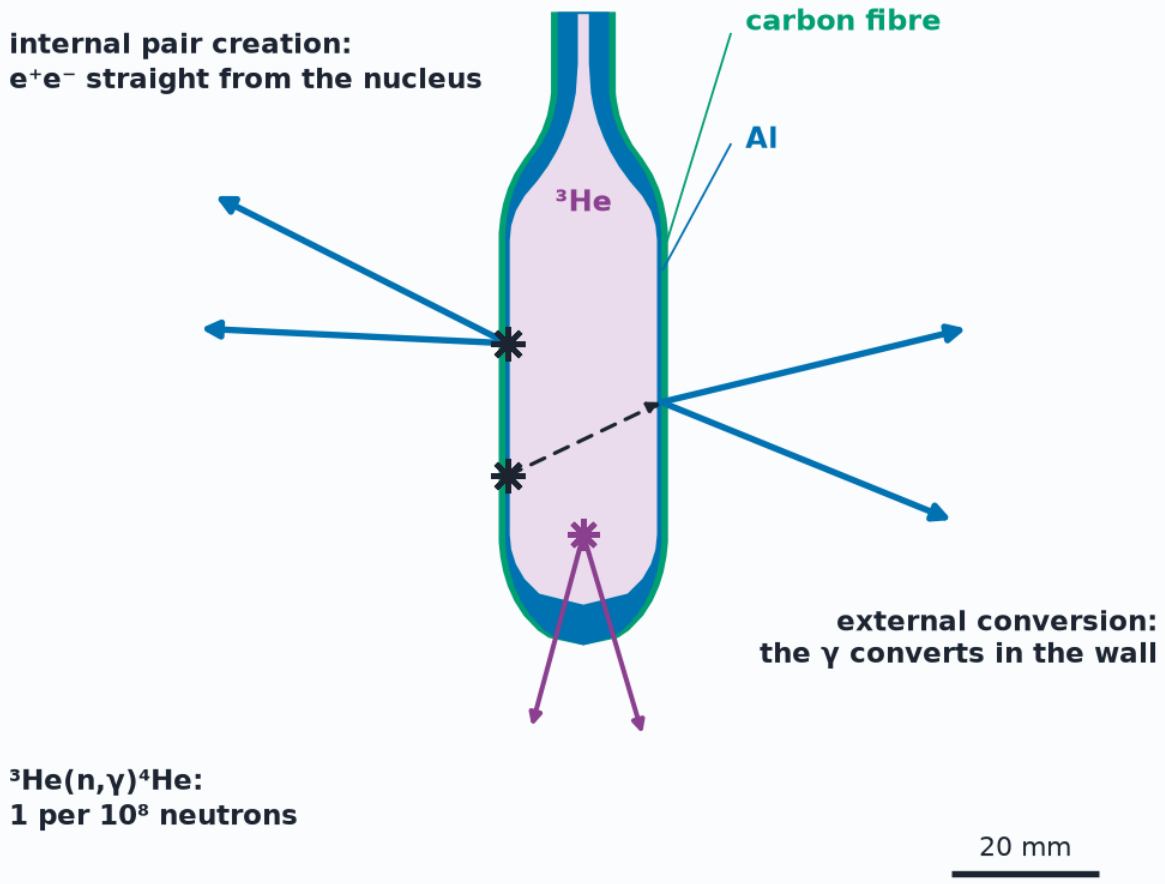
05

Plan C: the Aluminum Spectrum

Not X17, but the e^+e^- pairs the capsule does make

Plan C: the aluminum pair spectrum at thermals

capsule e^+e^- pairs with a lepton in a drift gap · Geant4 external conversion + internal pair creation added analytically



Where the capsule pairs come from	share	per 30 days
^{27}Al: γ converts in the wall external conversion · Geant4	84.5 %	7.8×10^{7}
^{27}Al: internal pair analytic: Geant4 makes no internal pairs	15.1 %	1.4×10^{7}
carbon fibre: ^{1}H + ^{12}C	0.32 %	2.9×10^{5}
γ from a capture elsewhere	0.02 %	2.0×10^{4}
^{3}He(n,γ): internal pair the signal channel · analytic, self-shielded	1.1×10^{-4} %	104

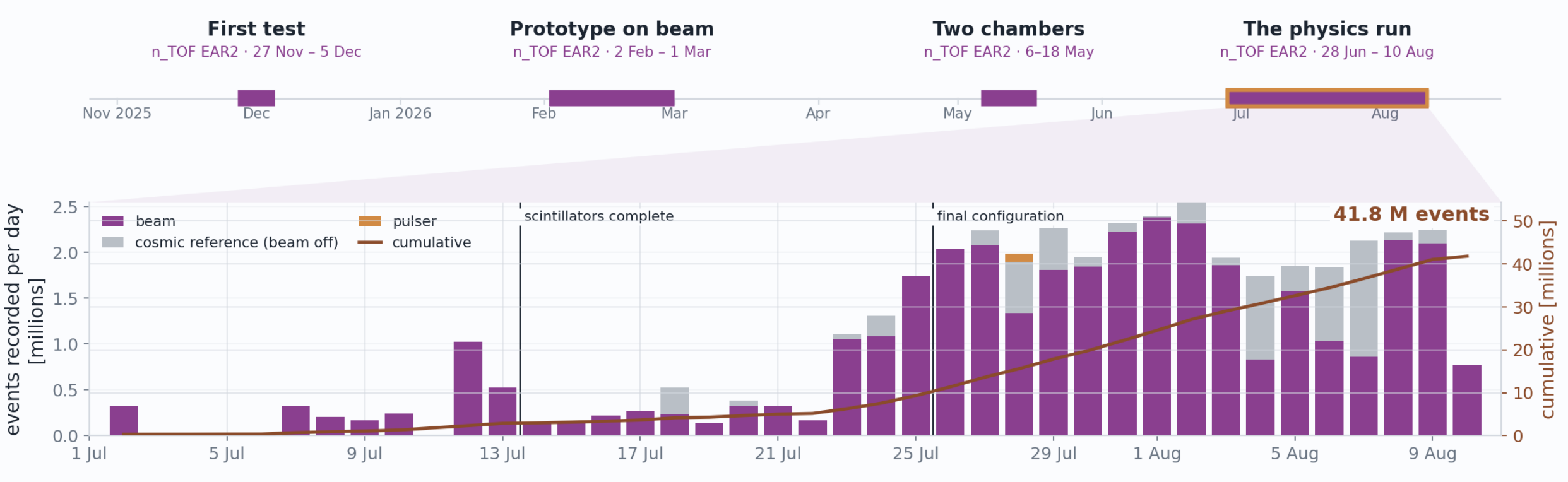
No X17 here — not even ^{3}He IPC

But ≈ 10^{8} ^{27}Al e^+e^- pairs / 30 days

A demonstrator: if we can find these pairs, the same setup should work for X17 in future

STATUS

Experiment timeline and statistics



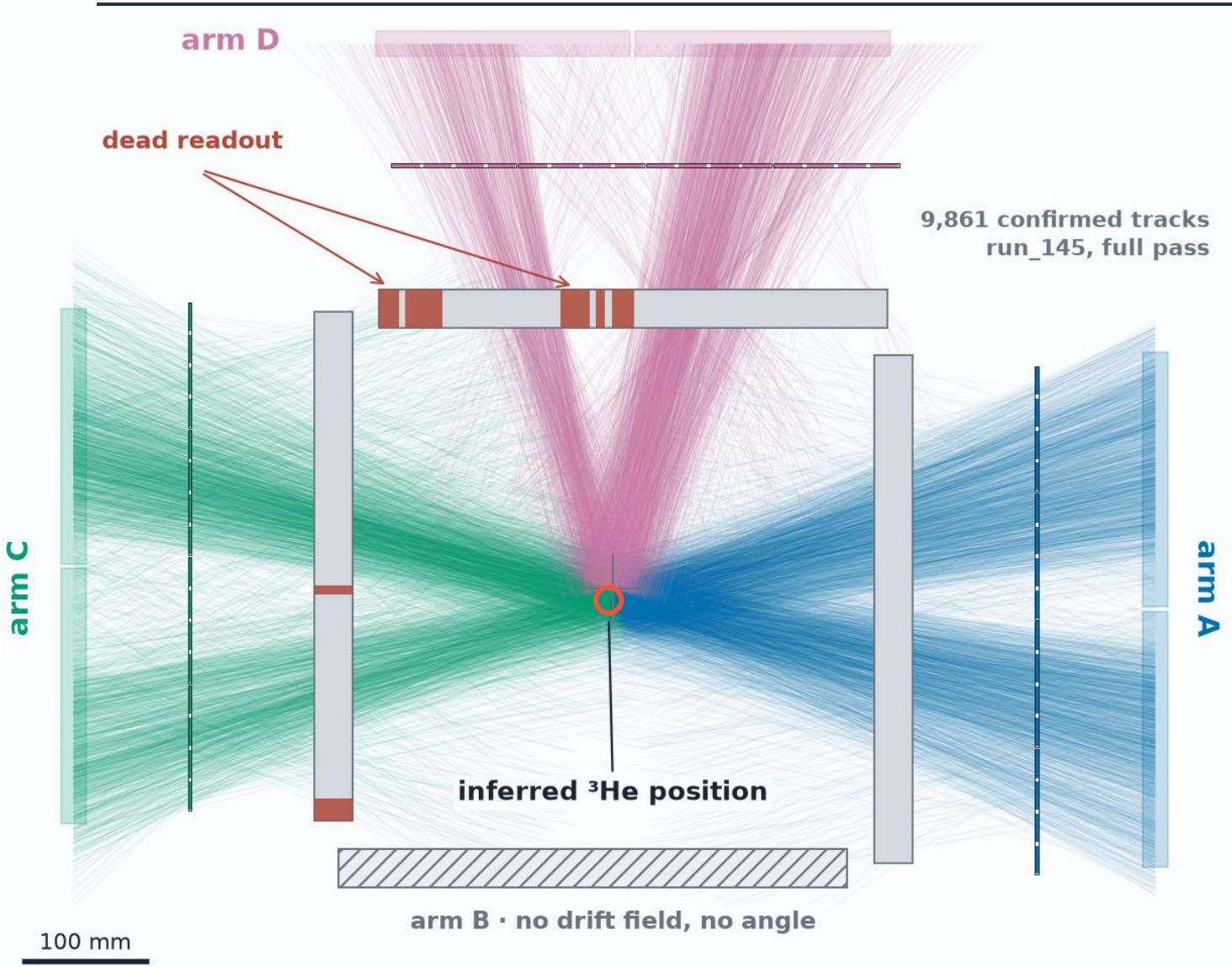
41.8 M
events recorded — 35.9 M on beam, 5.8 M cosmic reference

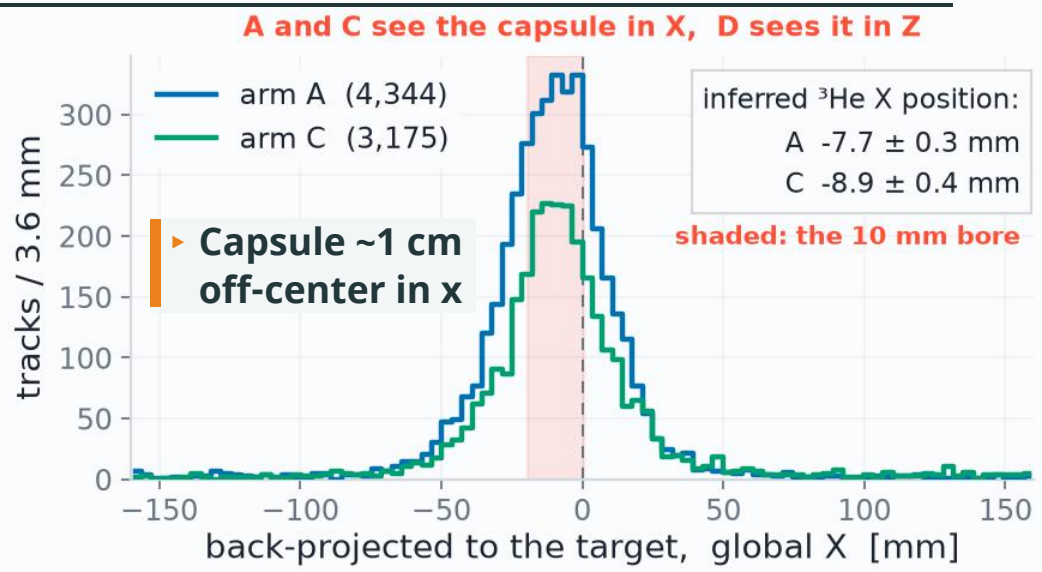
17.9 TB
on EOS, across 162 runs / 2,705 sub-runs

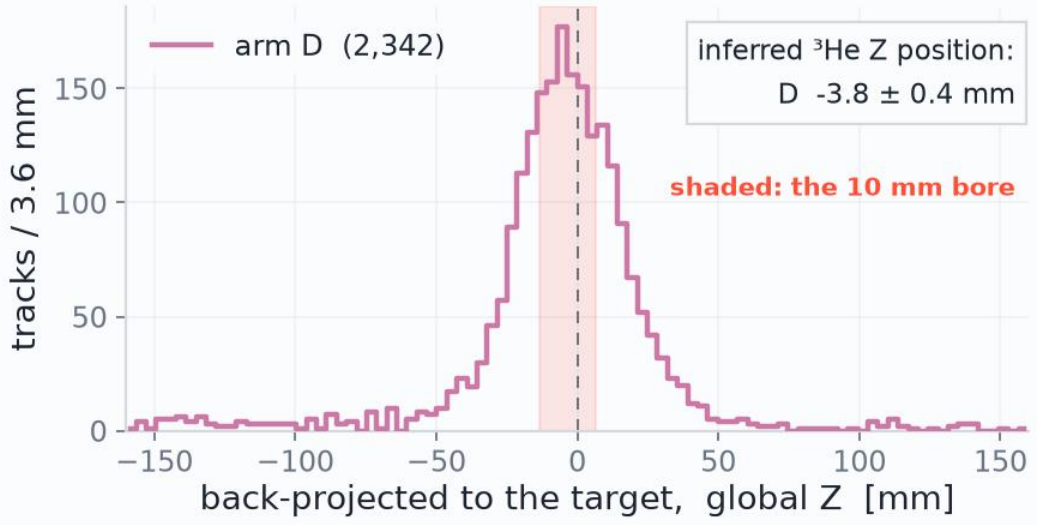
436,373
beam pulses on target over 44 days

WORK IN PROGRESS

Two opposing arms, seen from above – image of capsule

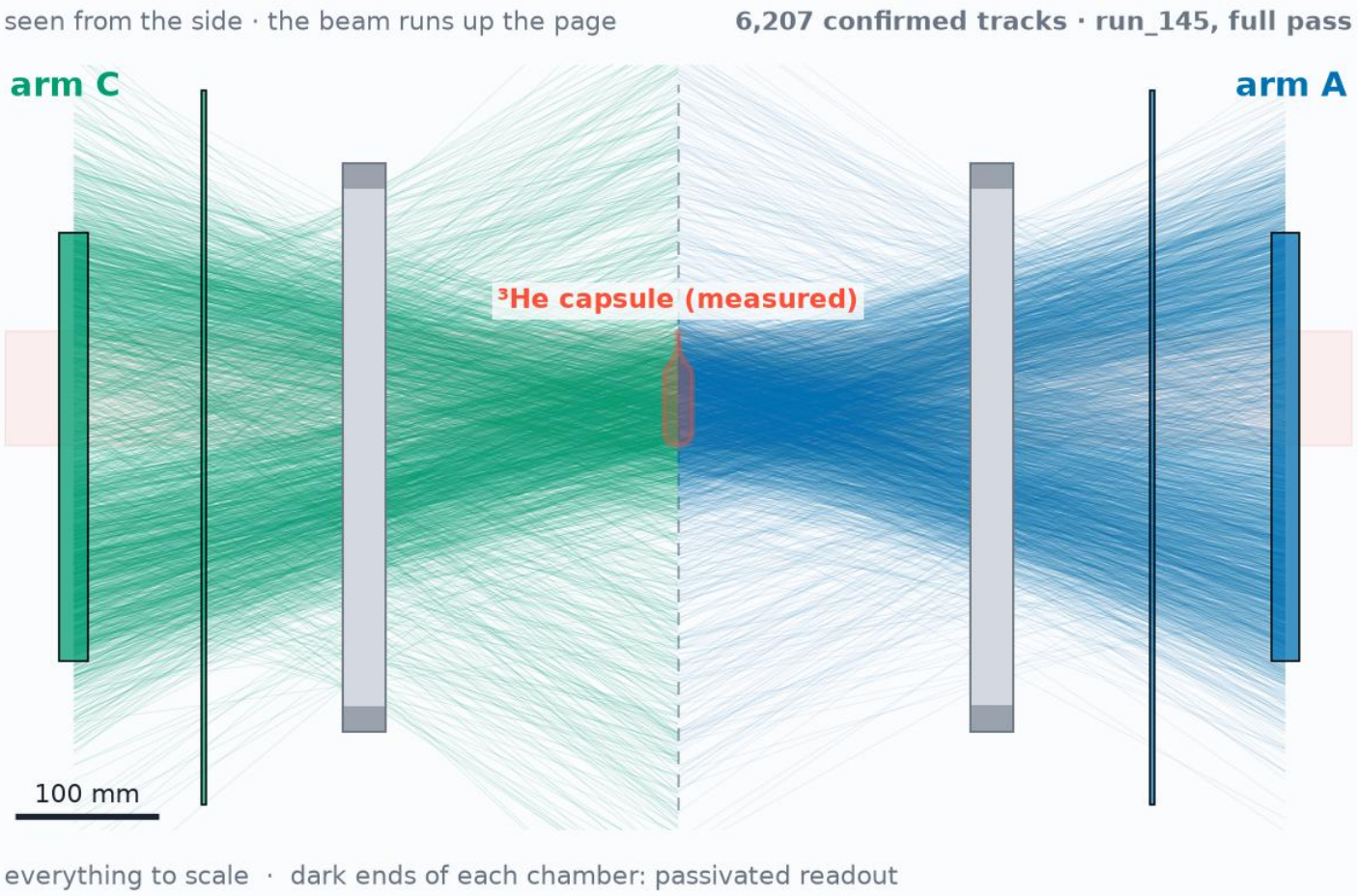


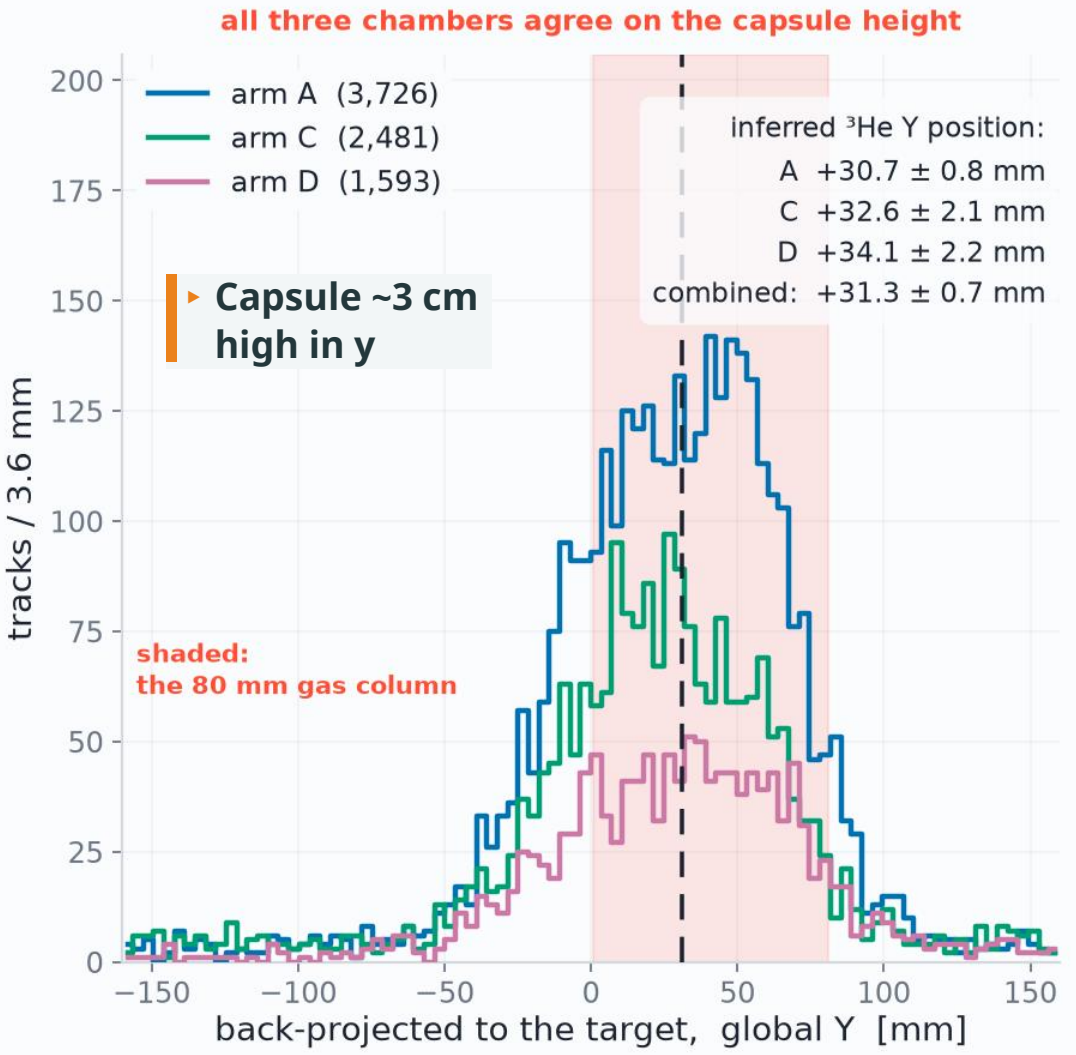




WORK IN PROGRESS

Two opposing arms, seen from above – image of capsule

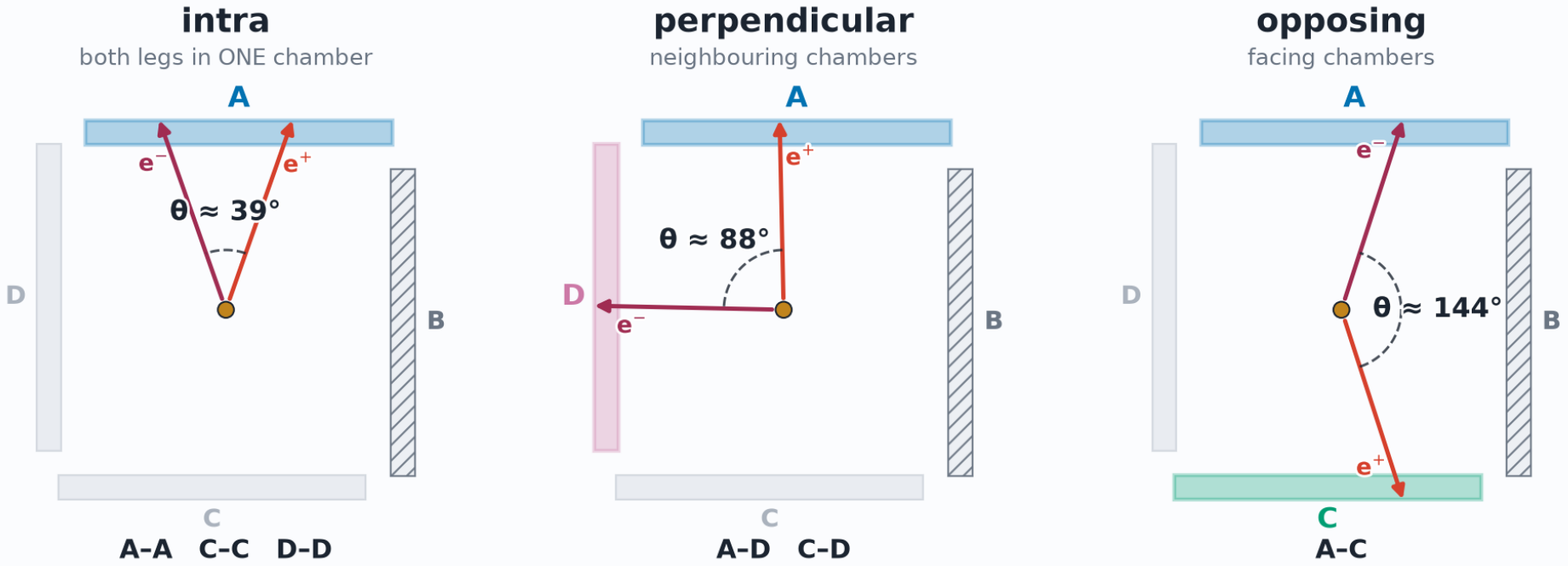


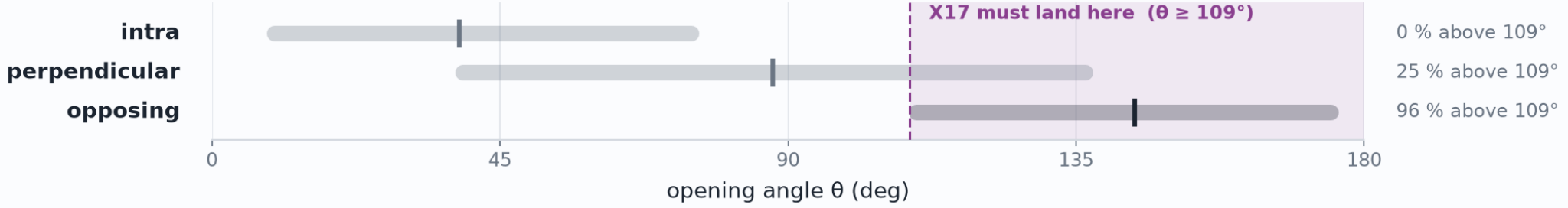


Now: sorting the pairs by where their legs land

Which two chambers a pair lands in nearly decides its opening angle

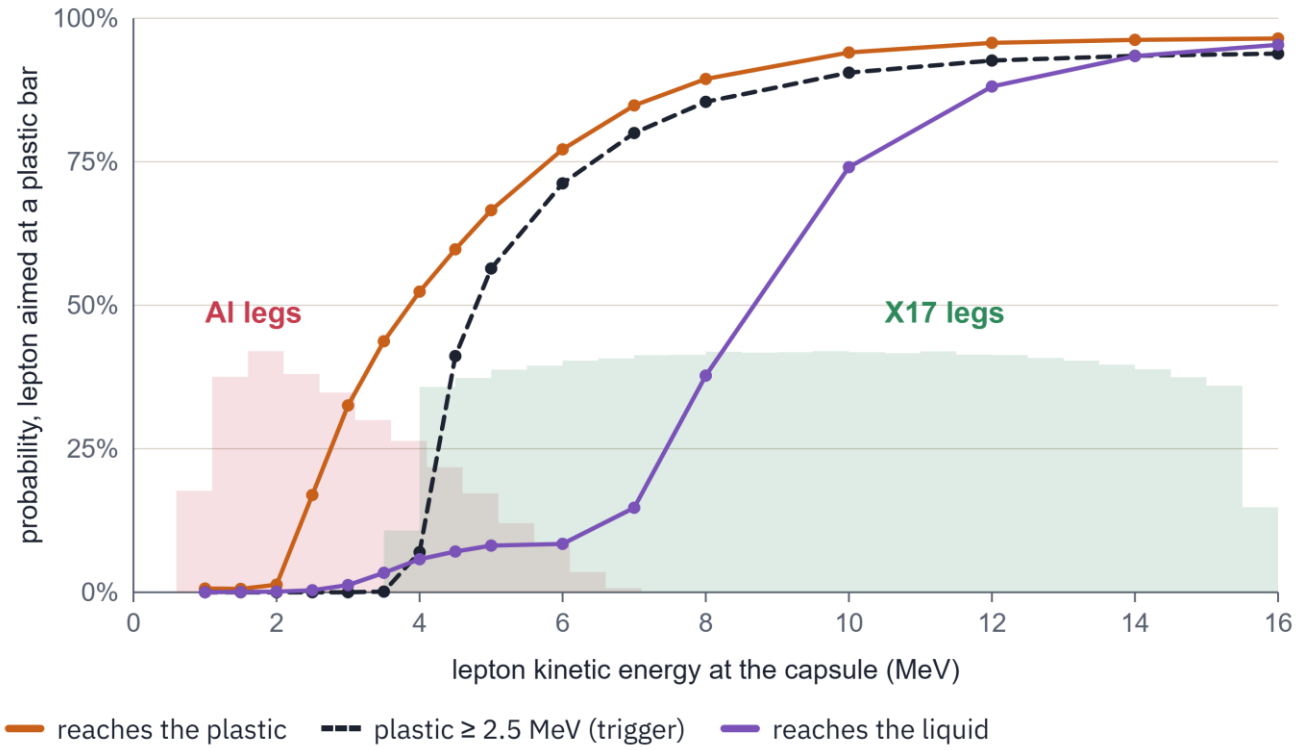
PRELIMINARY





Transverse section, true scale — each slab is one 30 mm drift gap, and the dot at the centre is the 20 mm ^{3}He capsule. θ is drawn at each class's measured median; bars span 5–95 %. Chamber B (hatched) has no field-shaping rings, so it measures no angle and enters no pairing — which is what costs the B–D opposing channel.

The Al pairs are soft: few trigger, poor pointing



Geant4: a lepton from the capsule aimed at a plastic bar, by its energy. Shaded: leg spectra. Al conversion legs have ~2 MeV at the gap; X17 legs 4–16 MeV.

Not a total loss: the double tracks are still there. A soft, close, same-chamber pair is the **hardest** two-track case, so it tests the pair reconstruction. X17 legs (~10 MeV, opposite arms) are the easy one.

11 % vs 82 %

of leptons reach the trigger level: Al vs X17

Plastic ≥ 2.5 MeV. An Al pair fires the trigger itself only 8 % of the time (X17: 45 %); most are recorded because a Compton electron of the same capture triggered.

~9° → 33 mm

scattering on the way out spoils the pointing

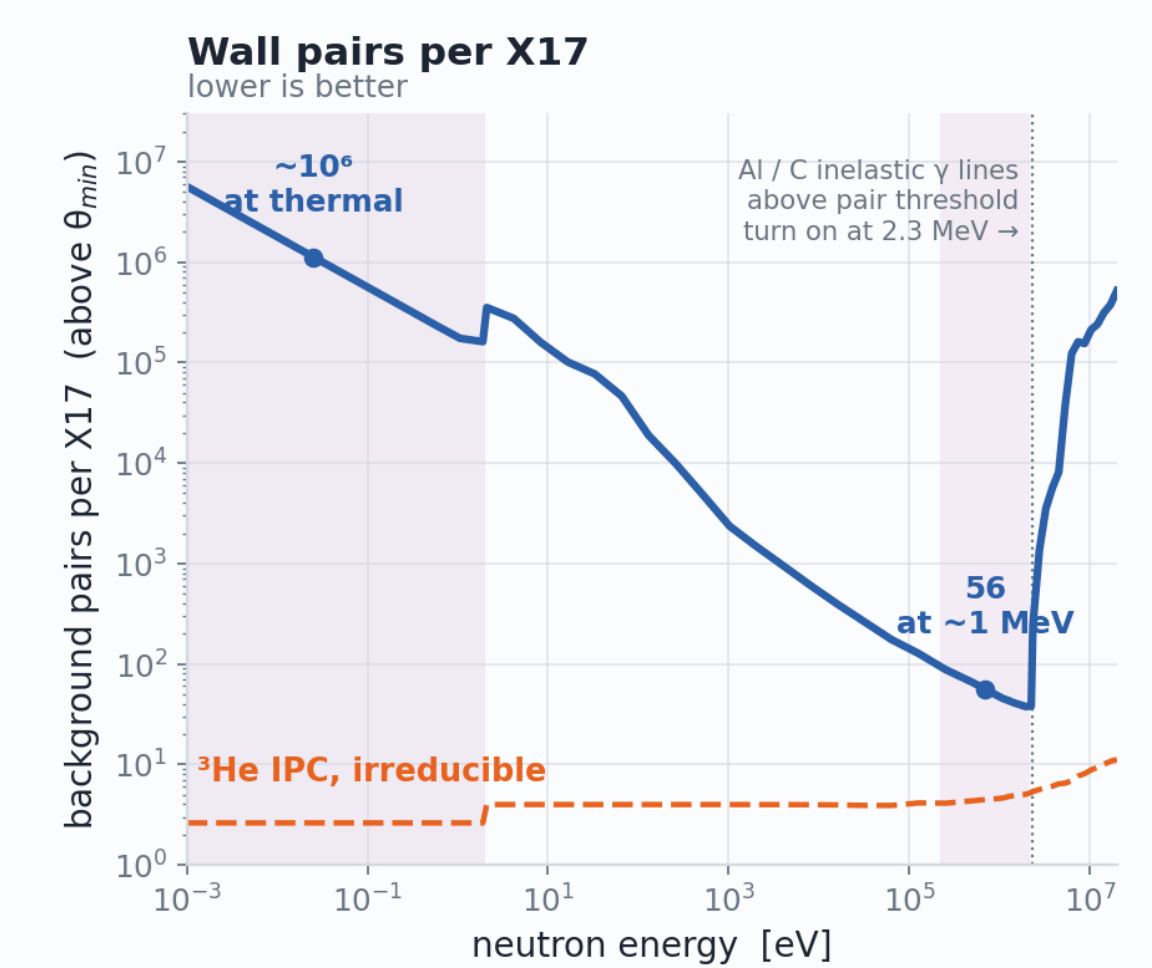
A 2 MeV leg bends ~9° leaving the capsule and its line misses the vertex by 33 mm (an X17 leg: 9 mm). Pairs pass a vertex test only at the chance rate (~2 %).

~3,700 pairs

correlated double tracks in the data

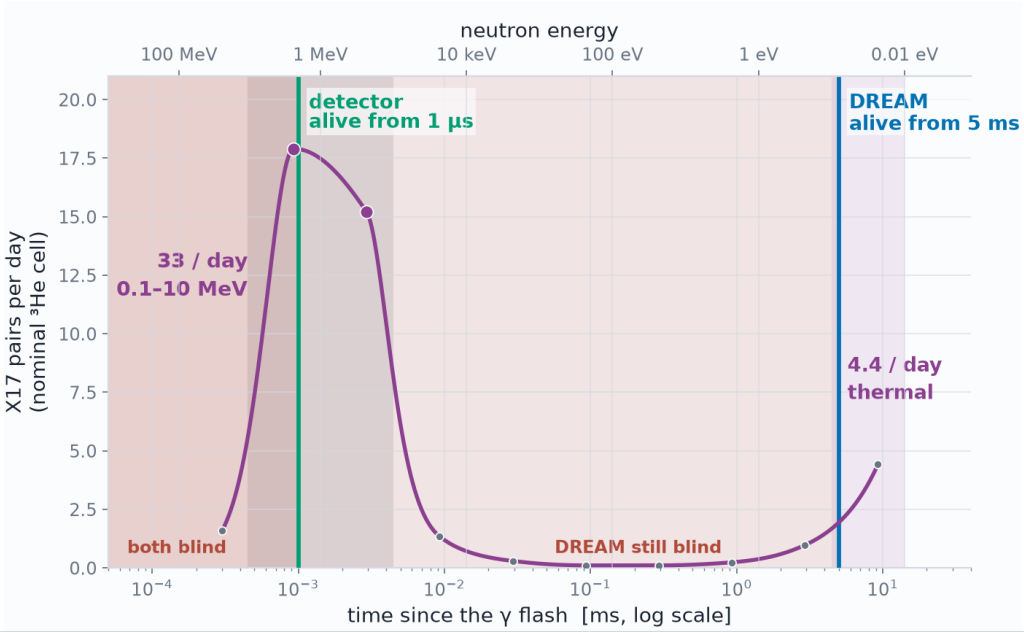
0.15 % of late events in A and C, flat while the rate falls ×3.5. Opening angle prefers Geant4 Al e^+e^-; the rate is ~⅛ of the Geant4 expectation (under study).

Plan A after LS3: still potential at MeV energies



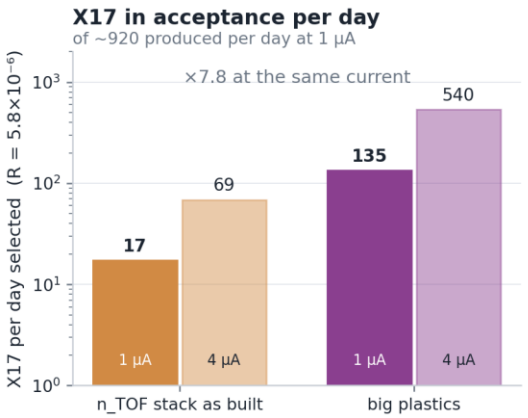
Al + C wall internal pairs only. X17/IPC = 2.5 % is the rate table's assumption; the ratio scales with it.

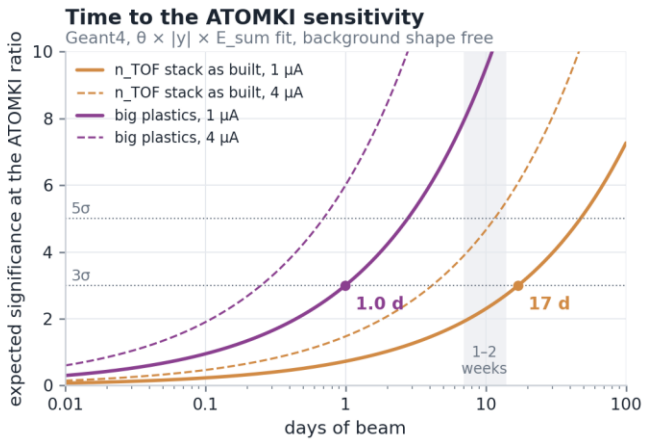
- **Find a DAQ fix** for the γ flash
- **Potential after LS3**: measure at 0.1–2 MeV
- Background per X17: **~50** at MeV vs **~10⁶** at thermal → + separate on opening angle
- Time to better **simulate the environment**

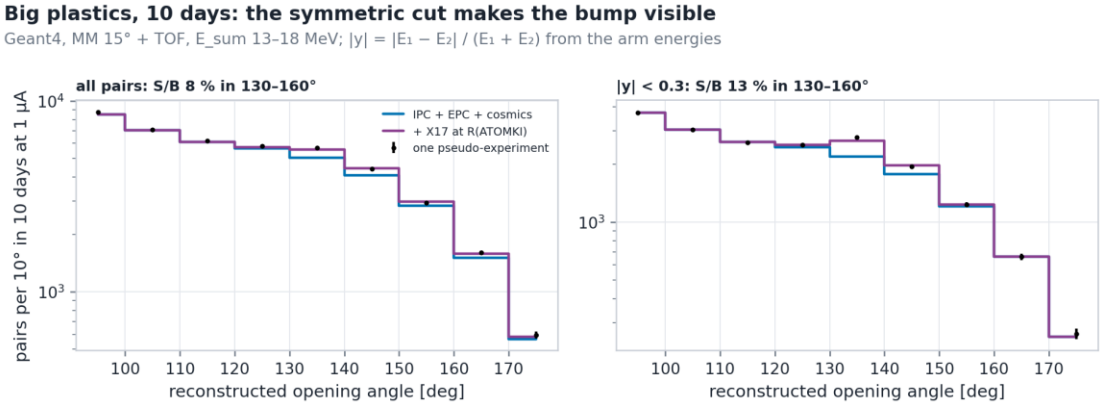


Next: repeat the ATOMKI measurement at LNL

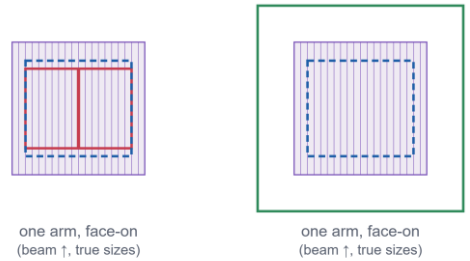
	n_TOF EAR2 2026 (ran)	LNL AN2000 / CN (proposed)
Reaction	n + ^{3}He → ^{4}He*	^{7}Li(p,e$^+$e$^-$)^{8}Be: ATOMKI's own
Precedent	none: a new channel	yes: ATOMKI 2016, Hanoi 2024
Beam	white n: the X17 edge moves 110° → 66°	DC protons ~1.1 MeV: one edge, ~135°
Target	500 bar capsule: σ_{68} 14.5°	1.5 µm Li_2O film: σ_{68} 4.6°
Background	n capture on Al/C, 10^4–10^6 × ^{3}He	the reaction's own IPC (QED) + cosmics
Statistics	no X17 expected at thermals	~135 X17 / day selected at 1 µA







Statistics are plentiful; the fight is the background. Geant4, 10 days at 1 µA, big plastics: S/B ≈ 8 %, so the background shape must be known to ~1 %. An angle × energy-split × E_sum fit measures it from the signal-free regions.



New: big plastics

four 75 × 75 × 5 cm plates replace the 20 × 30 cm bars (red) and the liquid

×7 trigger acceptance

3.3 % → 23 % of X17 pairs

3σ in ~1 day, not ~17

≈ €93k (polystyrene, budgetary)

Also studied, all with neutrons on ^{3}He: ILL PF1B (cold beam; ~7× short of ATOMKI's level in a 50-day cycle) · **GANIL-NFS** (fast n; sub-2 MeV flux unmeasured) · **NCSR Demokritos** (tandem MeV neutrons; first look).

Conclusions

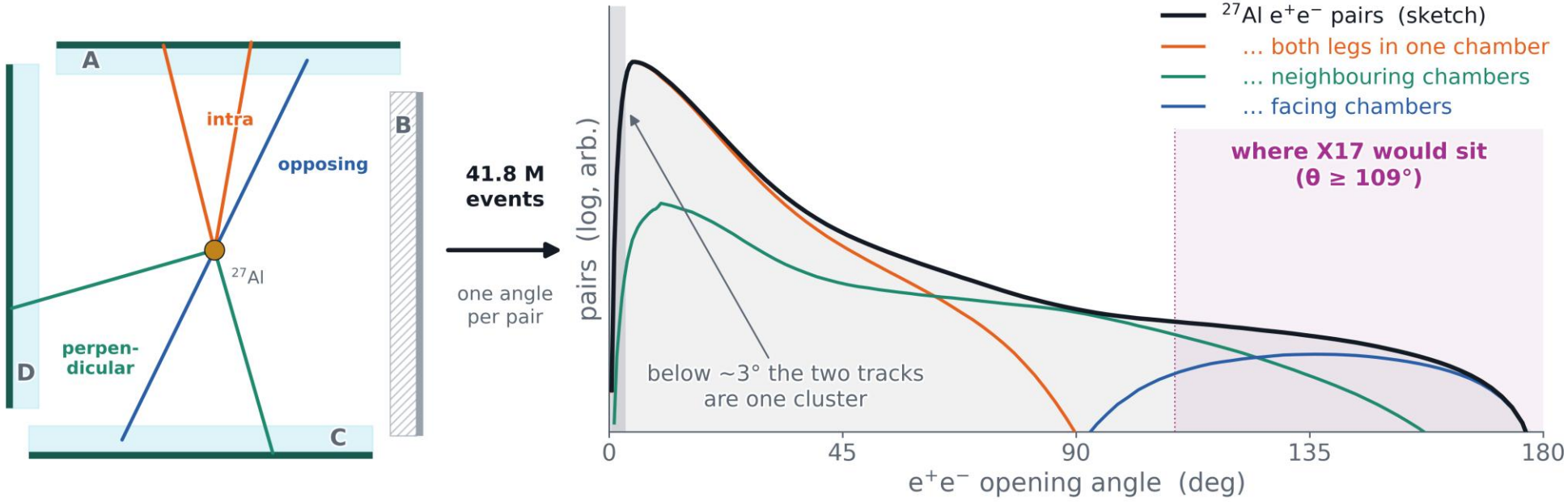
- **Plan A, MeV** (79 % of the X17 rate): blocked by the DAQ, not the detector
- **Plan B, thermal:** ^{3}He self-shields, expectation is no X17 recorded during the campaign
- **Plan C, ^{27}Al pairs:** 41.8 M events, good capsule image: a demonstrator for X17

Now: analysing the pair data

1. Find the two-track events

2. Histogram the opening angle

Illustration only — not a realistic spectrum



Thank you!

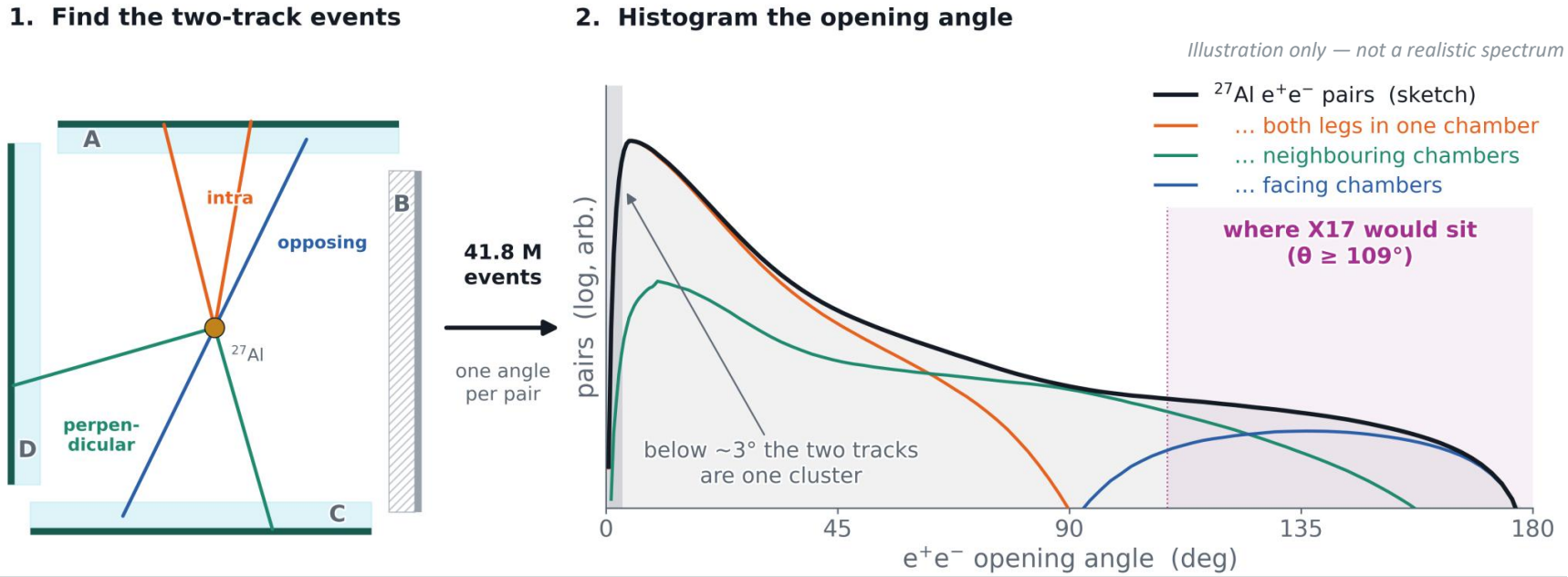
n_TOF local team — for making the measurement happen in EAR2, and the whole n_TOF collaboration

CERN GDD & MPT — for their indispensable support that enabled the measurement

EUROLABS — for the funding that made this experiment possible

CEA Saclay & Bulk Lab — for help building the detectors and supplying the DAQ

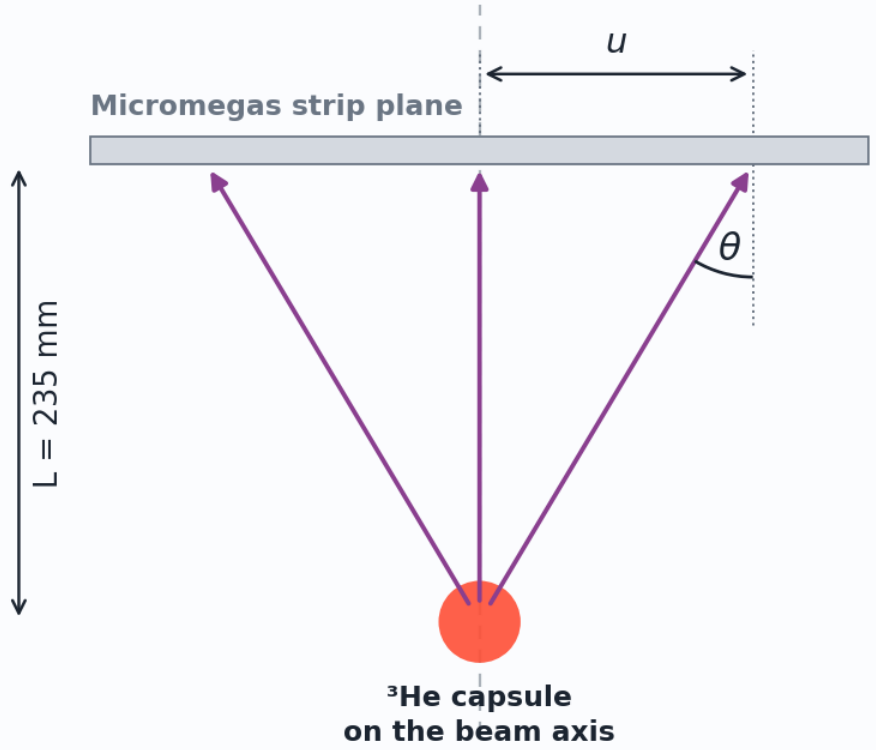
Questions?



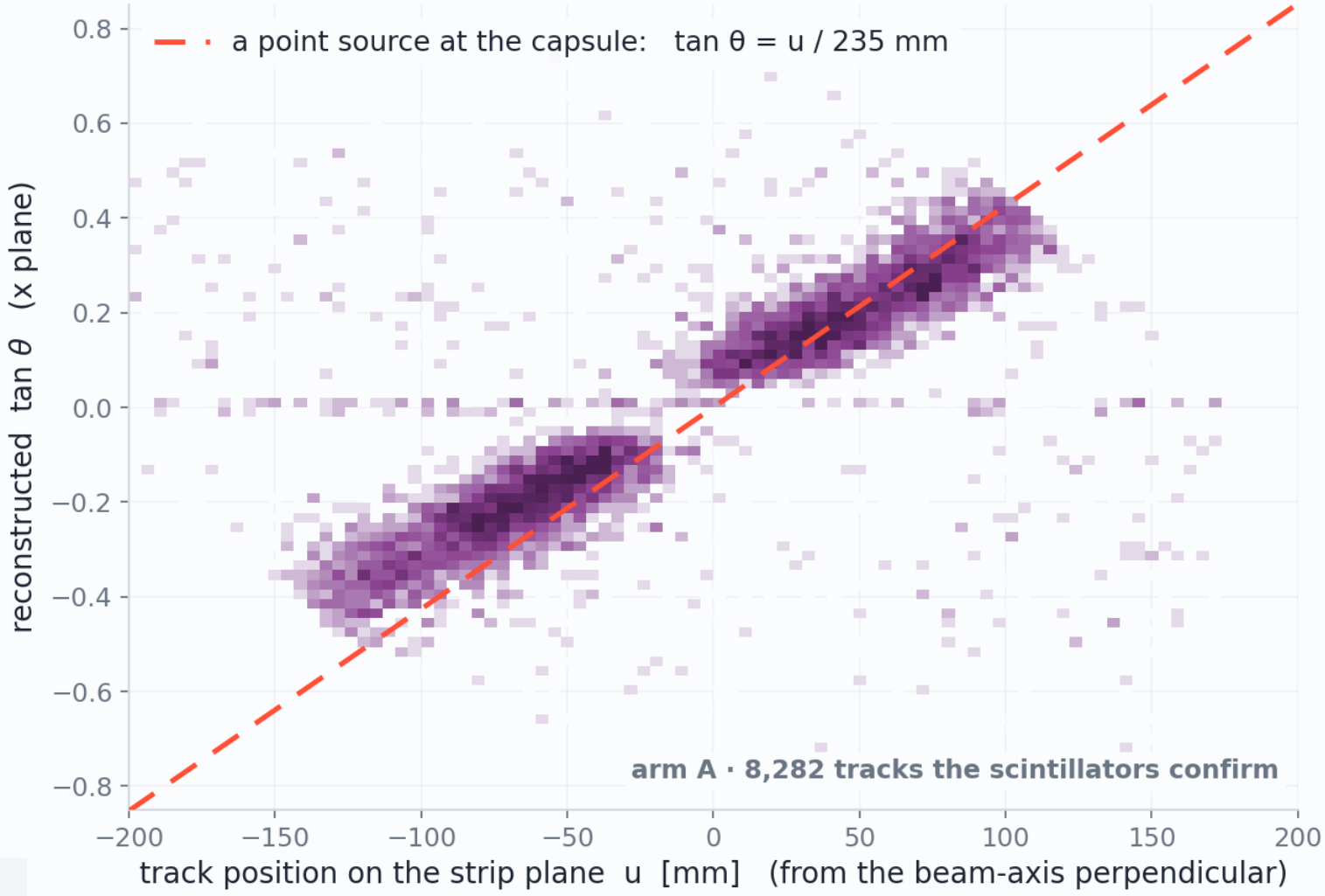
Backup

WORK IN PROGRESS

The angle a track arrives with says where it came from



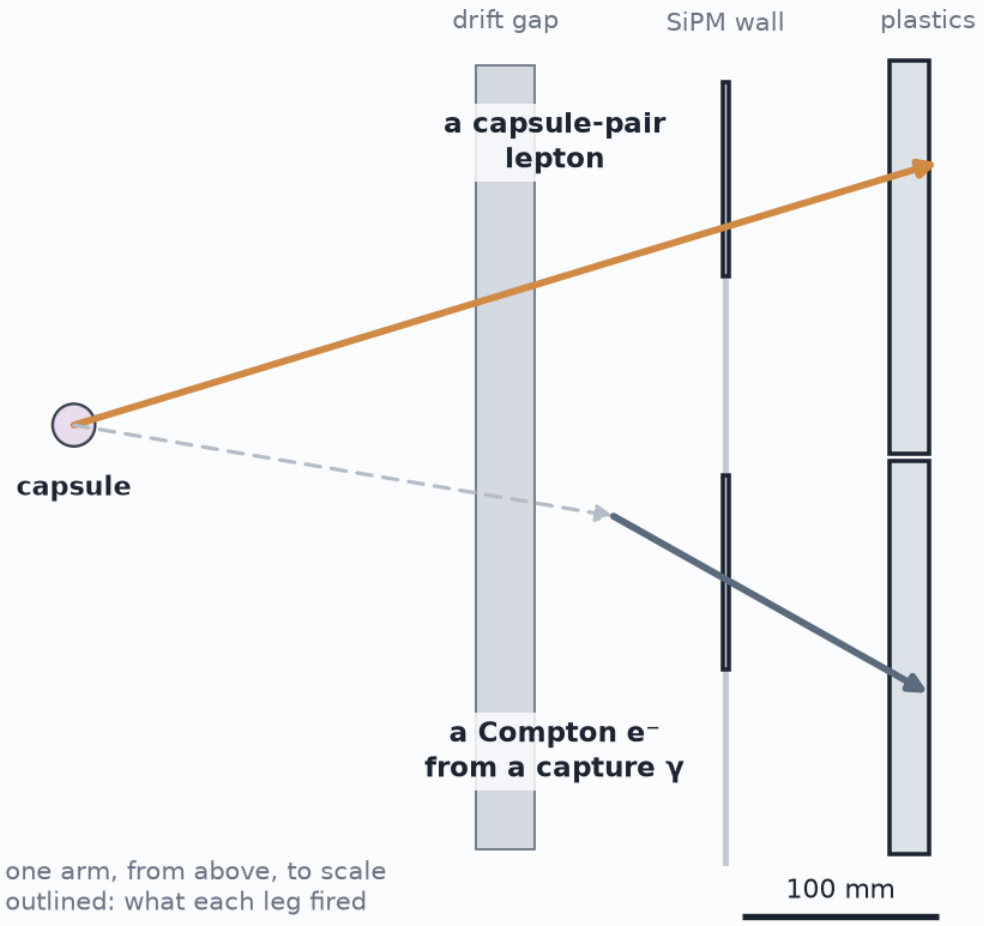
a point source can only arrive as $\tan\theta = u/L$



- Correlation implies that tracks point back to capsule

What we expect at thermals -- aluminum

Geant4 · SiPM-wall bar ∧ plastic bar ≥ 0.5 MIP within 20 ns, in one arm · the sim makes 2.1× fewer legs than the data: read the fractions



one arm, from above, to scale
outlined: what each leg fired

What made the leg

	share	per pulse
e⁺e⁻ pair made in the capsule → the next figure opens this up	**9.7 %**	**12**
anything else	90.3 %	111

“anything else”, broken down

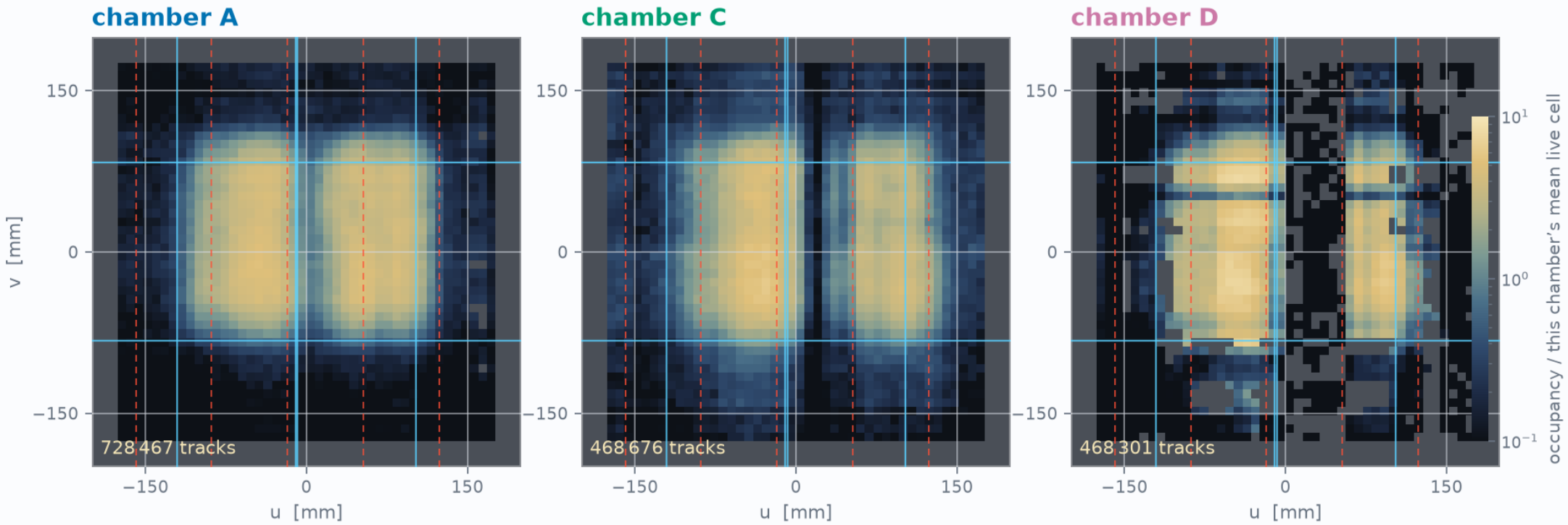
	share	per pulse
Compton from a capsule-capture γ	84.9 %	94
capsule γ converting in structure	7.5 %	8
γ from a capture outside	3.0 %	3
mixed, neutron-induced, other	4.6 %	5

Legs in two arms at once: 0.111 per pulse (26 in 10^9 neutrons). 7.7 % are one pair or one particle; the rest are two γ of one capture. One neutron per event: the sim has no accidentals.

Where a matched track lands — and where the trigger cannot look

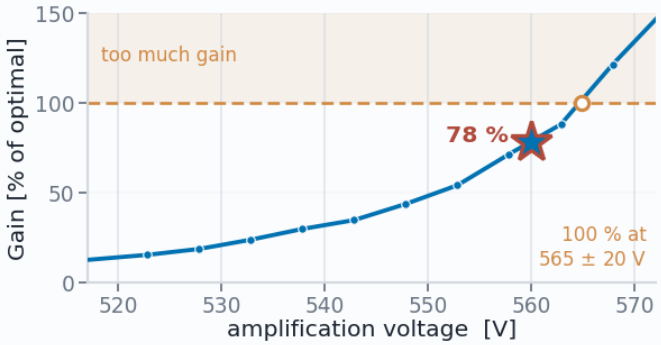
PRELIMINARY

tracks whose extrapolation hits the wall group AND the plastic bar that actually fired, in the chamber's own local frame



Red dashes: the four wall read-out groups. Blue: the two plastic bars' edges in u and their 300 mm length in v — both projected back onto the chamber through the lever arm, which is what makes them comparable with a cluster position. The bars stop at |v| = 83 mm on the chamber and the gap between them shadows u ≈ +7 mm; both are plainly in the data. Colour is occupancy relative to each chamber's own mean live cell on a shared log scale, so a hue means the same thing in all three panels. Grey: hot channels, and the fiducial edge beyond which the plane fit rails. White square: the 398.6 mm active area.

Every volt of gain costs milliseconds of beam

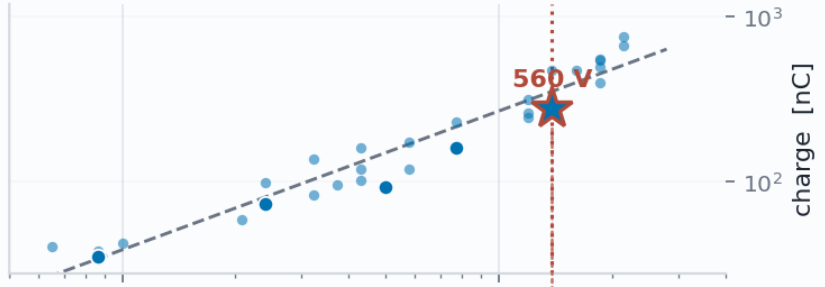


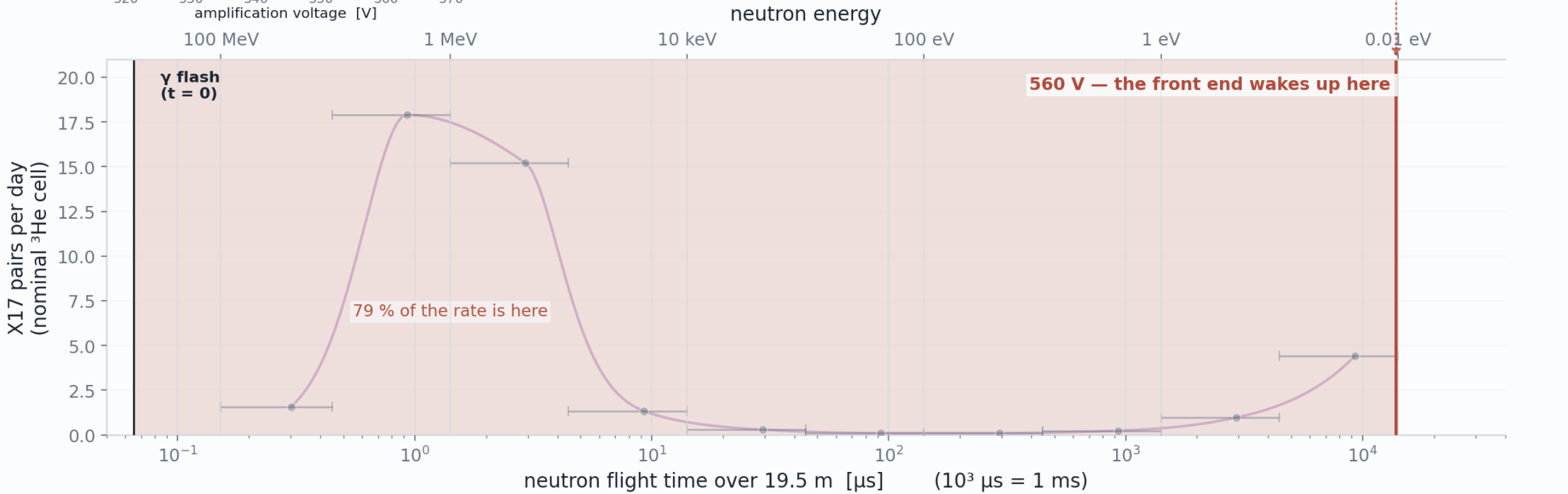
560 V where the gain wants to be

0.1 % of the X17 rate left

78 % of optimal gain (cosmic bench)

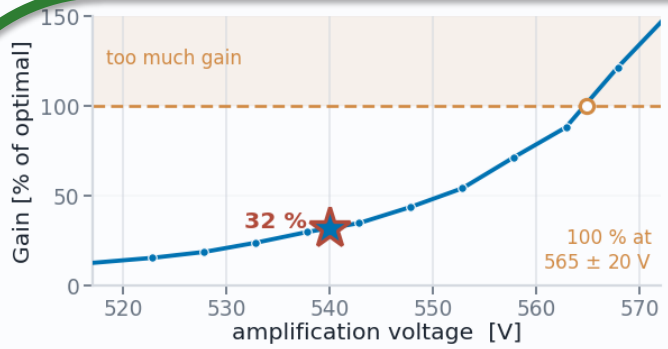
blind for 13.9 ms after every flash





WHAT WE RAN WITH

Every volt of gain costs milliseconds of beam

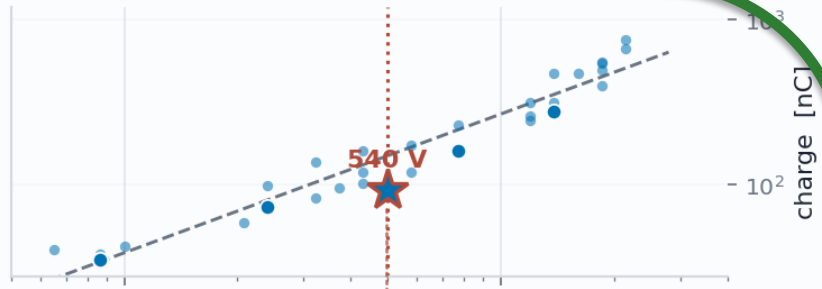


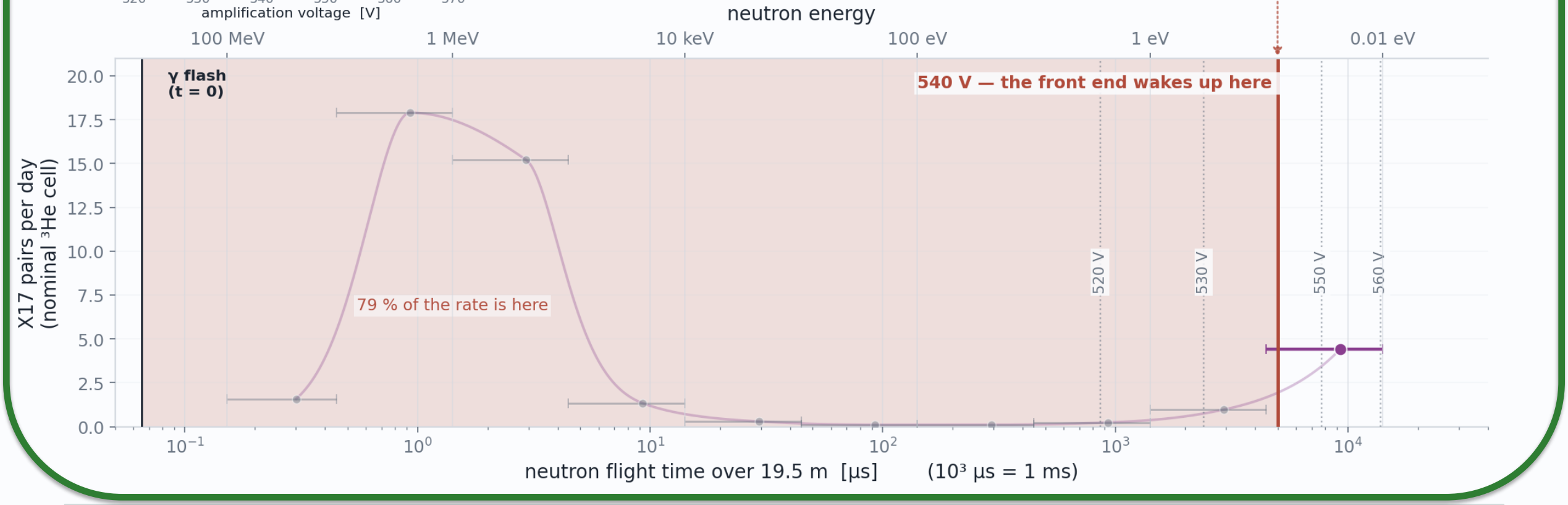
540 V where we ran

9.5 % of the X17 rate left

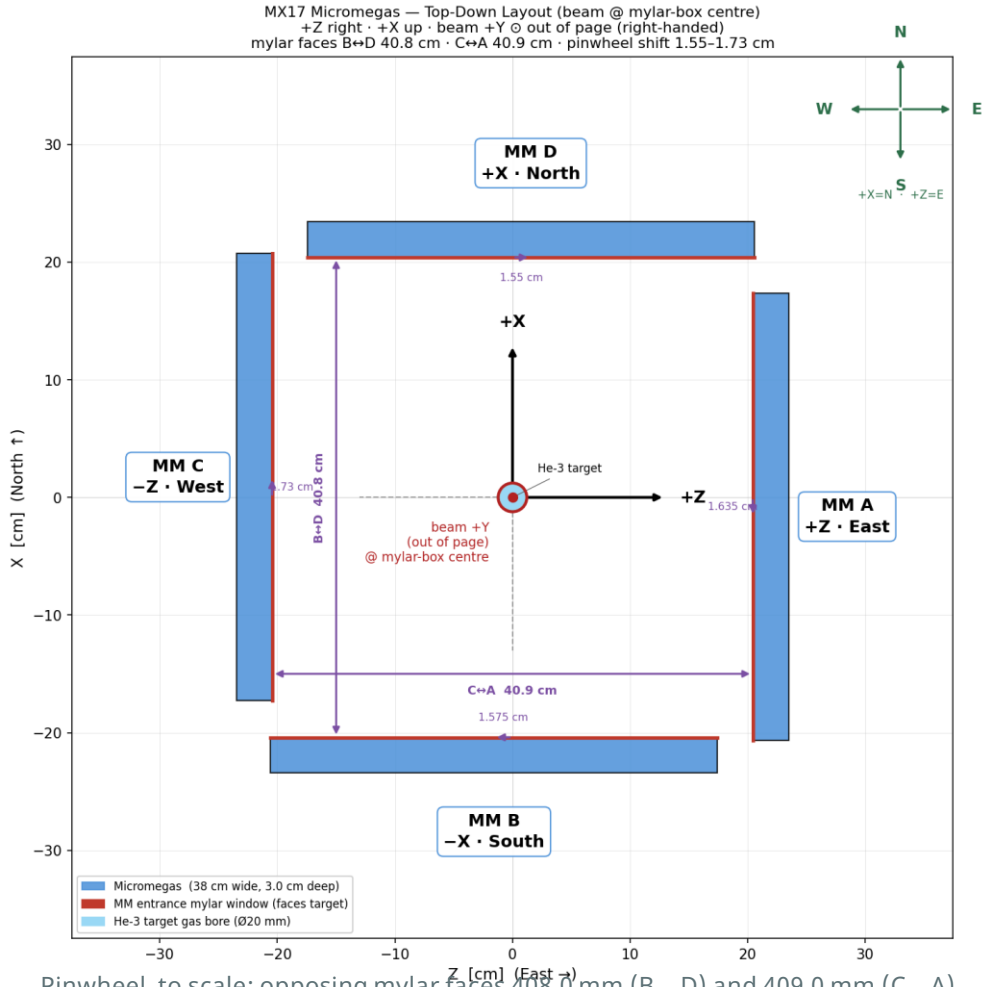
32 % of optimal gain (cosmic bench)

blind for 5.0 ms after every flash

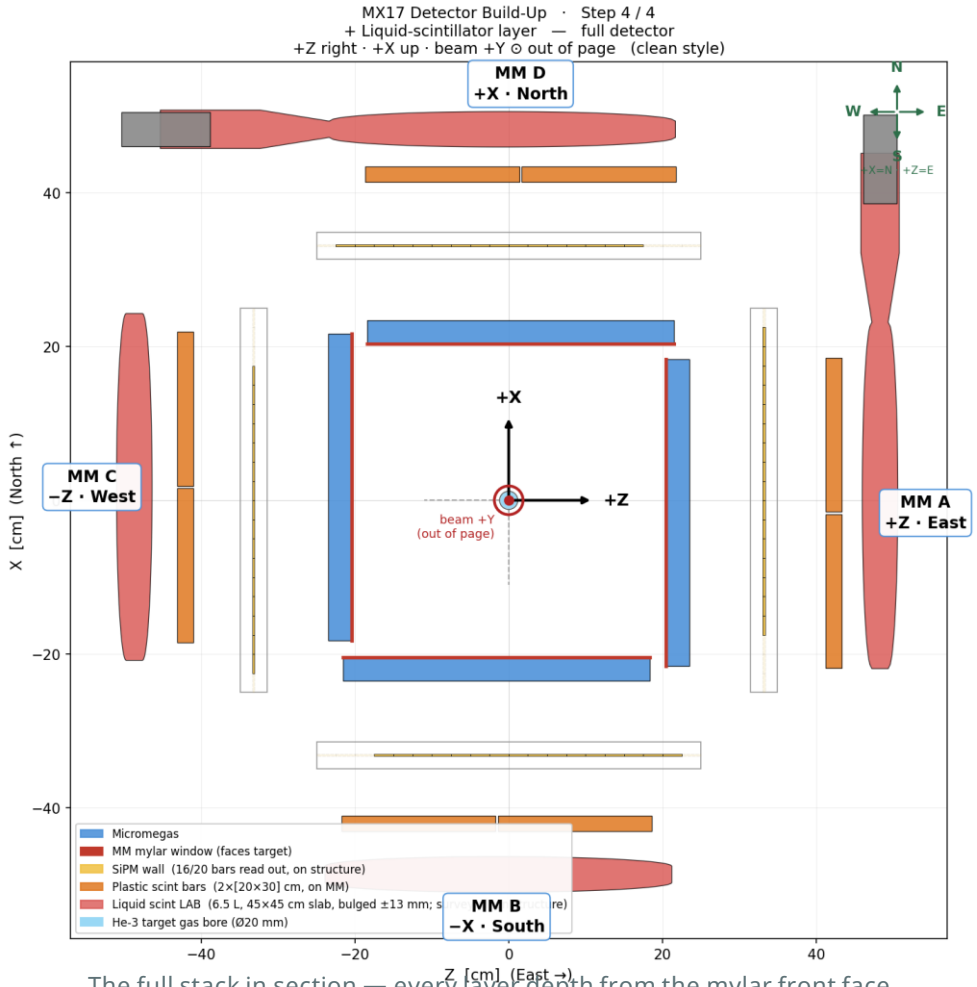




The setup to scale: top-down, and layer by layer



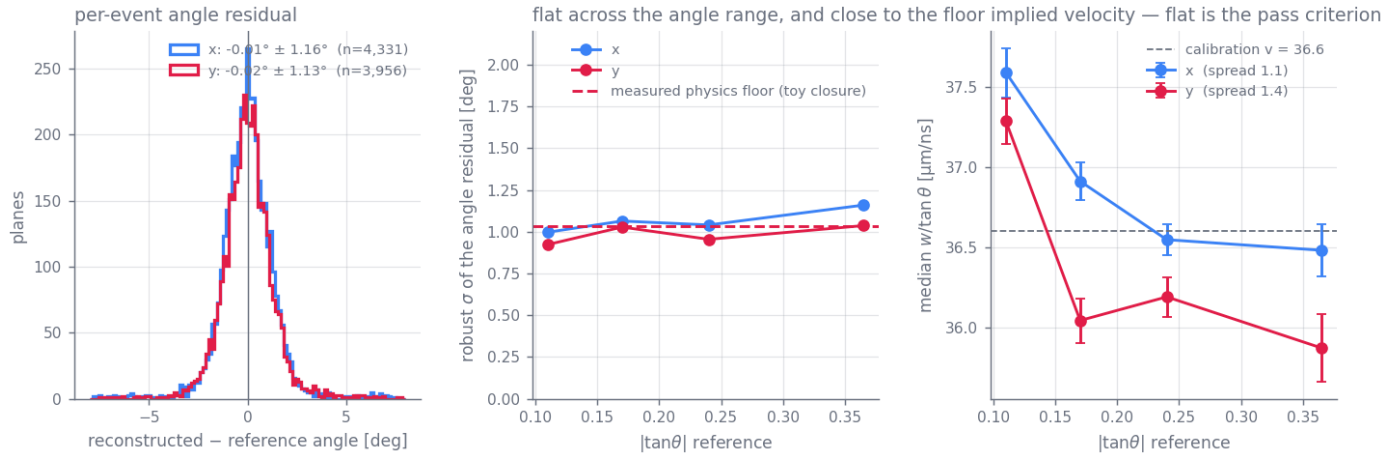
Pinwheel, to scale: opposing mylar faces 408.0 mm (B ↔ D) and 409.0 mm (C ↔ A)



The full stack in section — every layer depth from the mylar front face

Same geometry as the 3-D build-up, drawn as a measured section rather than a render: distances 2026-07-15, per-arm gaps and vessel placements 2026-07-17/18. The intermediate build states (+SiPM, +plastics) are setup_2_sipm.png / setup_3_plastic.png in assets/img/.

The ~1° per-event scatter is physics, not fitting

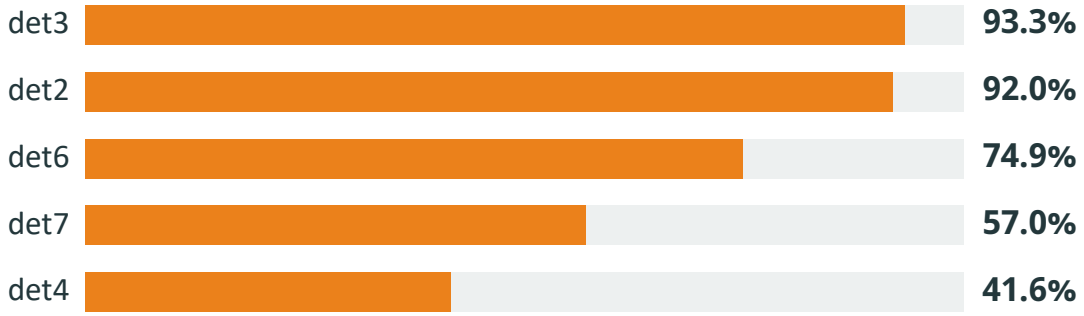


The closure test	waveforms generated from the calibrated model with **known** tracks — real geometries, real charge profiles, real noise, real clipping — then fitted back. One knob: per-depth-bin transverse jitter of the charge centroid
jitter 0	σ_θ 0.02° — **electronics noise contributes essentially nothing**
0.15 mm	0.55°
0.30 mm	**1.05°** — the scale diffusion + a few effective electrons per 2 mm of track + δ-rays actually give; **reproduces the data exactly**
0.50 mm	1.77°
errors honest?	pull width 1.16 (X) / 1.13 (Y) — ~15 % optimistic, and χ^2/dof genuinely ranks which fits to distrust

Full run, 6,852 / 6,850 reference-matched events, reference never in any fit: σ_θ = 1.16° / 1.13°, bias < 0.1°, **flat across the angle range and sitting on the measured floor** (middle, dashed); implied v flat to ~1.4 μm/ns (right)

So the per-event fit is extracting **essentially all the information the detector geometry physically has** — gains beyond ~1° come from ensemble averaging, not better fitters, which is why the reported errors carry a physics floor added in quadrature and the optimiser tolerances are deliberately loose. Ablations agree: dropping any single model refinement moves per-event σ by < 0.05°; what the refinements buy is *ensemble physicality* (implied-v flatness), the thing that cannot be tuned. Doc Part VII §41–43.

The fleet, and what limits each chamber

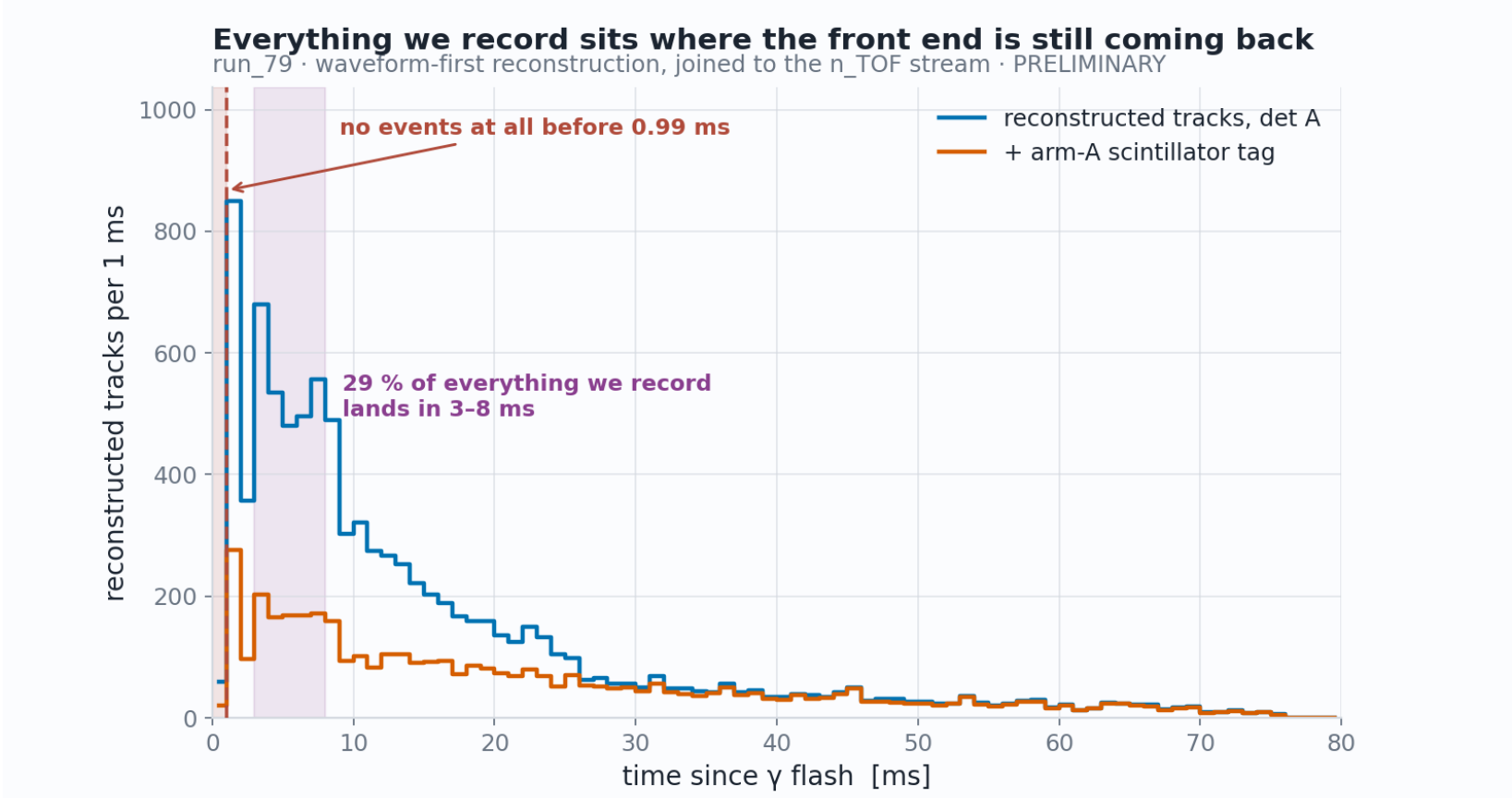


det3 / det2	Healthy. **100%** of crossings fire; the whole residue is a 2.4–3.2% discharge coincidence and a 4.0–4.3% near-miss tail.
det6 / det7	**Discharge-limited** — 22.3% / 37.4% of firing events are >50-strip discharges. Both ran at resist 495 V, *above* their measured optima (~480 / ~440 V): this is a chosen operating point, not a ceiling.
det4	**Gain-limited**, and it is not blind: it produces a signal on **95.8%** of crossings, but on 38.3% too few strips fire to form a valid X+Y point. Independently confirmed at the SPS: 62% of its area does not amplify.
Position core σ	0.43–0.46 mm on det2/det3/det6, 0.56/0.62 mm on det4/det7 (radial) — on the good chambers the M3 reference contributes only ~10–20% of the variance.

Reconstructed within 5 mm of the M3 reference track (χ^2<1.0 & NClus=4). One high-statistics sub-run per chamber, 3.7k–12.3k reference muons, waveform-first reconstruction, 2026-08-13 reprocessing.

Superseded 2026-07-14 values were 92.9 / 91.3 / 57.8 / 43.1 / 20.7%. The last three chambers gain 14–21 points from the matched-filter reprocessing of the raw waveforms — low-amplitude hits the previous analyzer never admitted — plus a further 0.4–4 points from reconstructing position from the waveforms instead of the strip hit times.

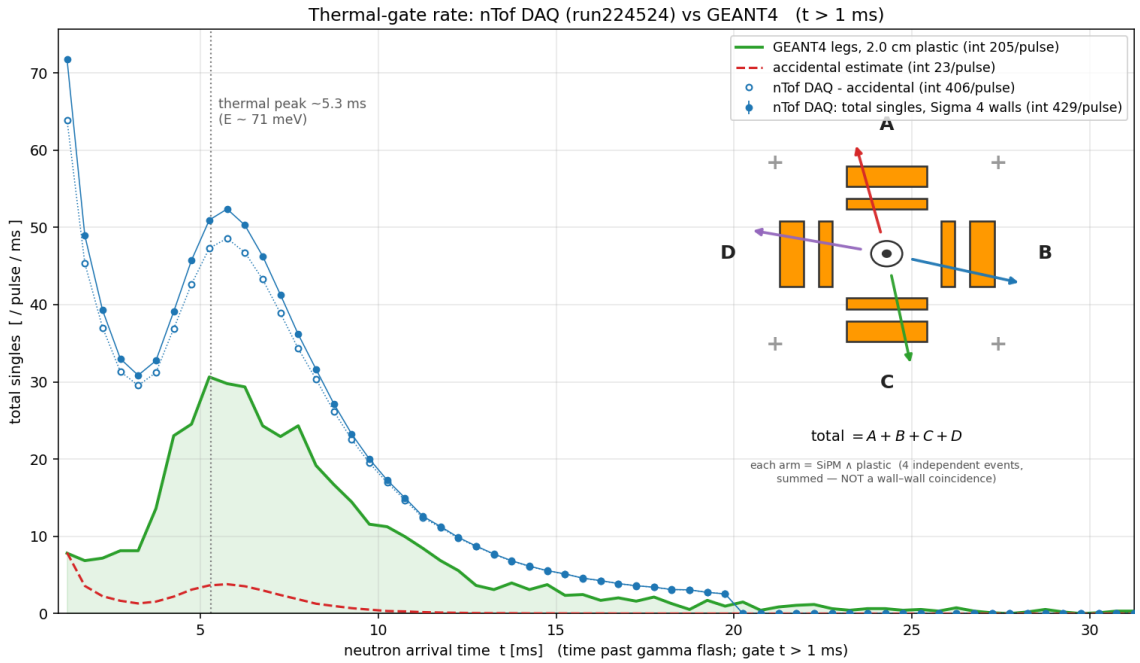
What we record, and when



Shaded purple = the thermal-neutron arrival window (Geant4 timedist_2cm, peak 5.3 ms); shaded red = before the earliest event in the run. The distribution turns on sharply at 1 ms — that is the DREAM gate, not a reconstruction effect — and its bulk sits in the first ~10 ms, which is exactly the interval the recovery map says the chambers are still blind or only partly alive in. Tagged and untagged keep the same shape, so the scintillator tag is an acceptance and not a time-dependent selection. PRELIMINARY: transferred bench calibration, no in-situ calibration (RUN79_PRELIM_2026-07-30.md).

Reconstructed tracks on beam data, joined event-by-event to the n_TOF scintillator stream. The turn-on at 1 ms is the DREAM gate, not the reconstruction; the bulk of everything we record sits in the first ~10 ms — the same interval the recovery map says the chambers are only partly alive in. That overlap *is* the experiment's central operational problem, stated in data.

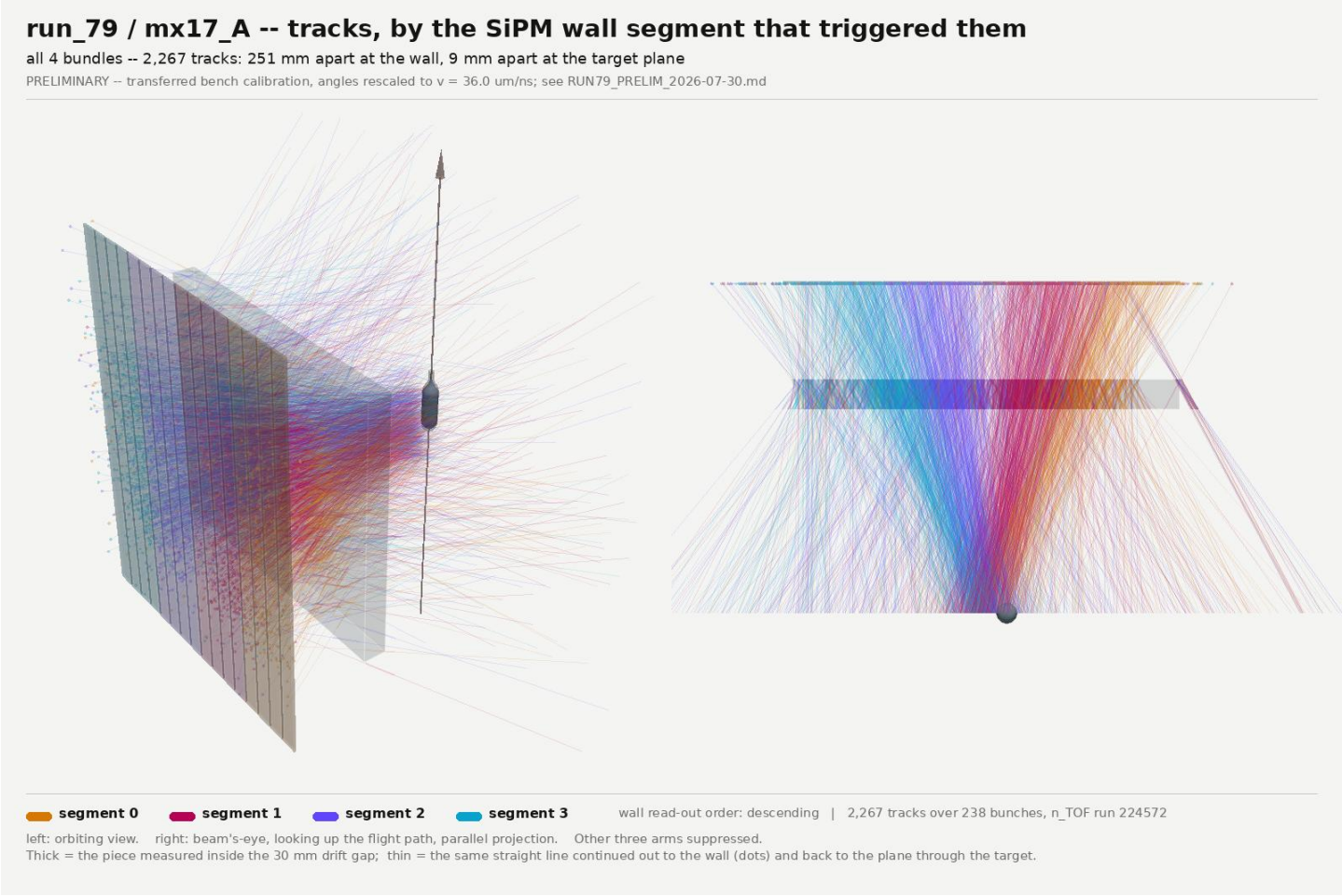
The trigger lives in the thermal window — by design



Recorded trigger rate vs neutron arrival time, against Geant4 — the 5.3 ms thermal peak reproduces; the normalisation runs **2.1×** high (per wall 1.1–2.6×), under investigation

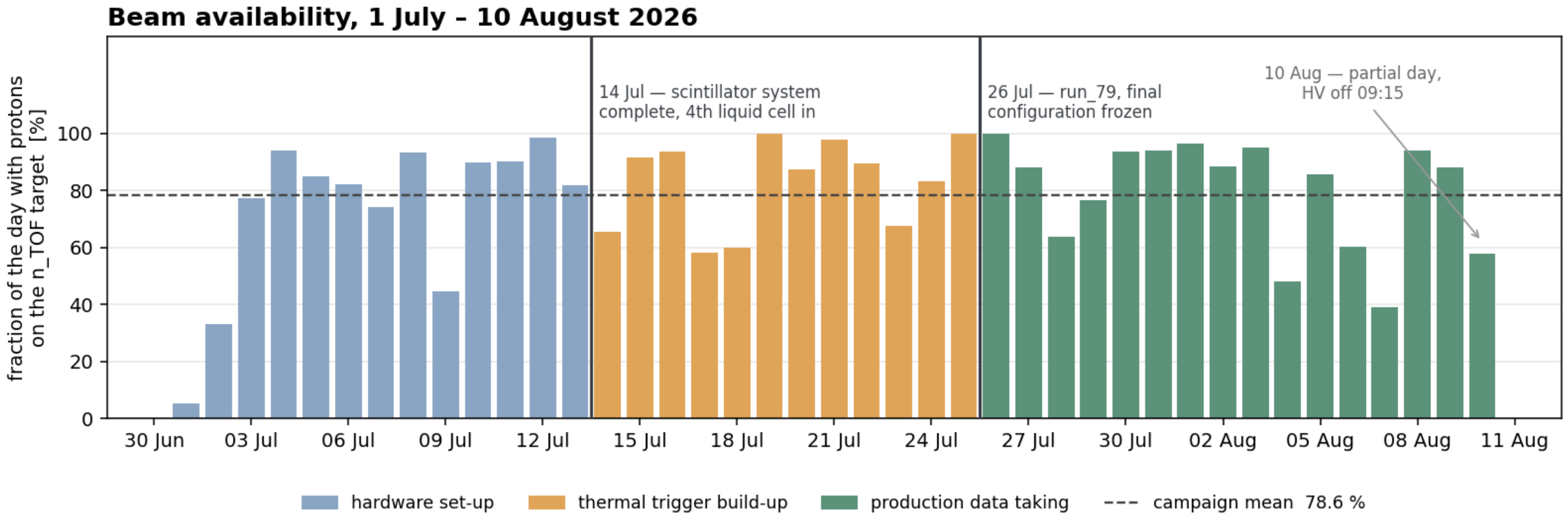
- **Capture loves slow neutrons, and slow neutrons are late.** Over ~20 m of flight path the thermal flux peaks at **5.3 ms** — recorded, and reproduced by Geant4 in shape.
- **The trigger:** per arm, the SiPM wall (any of 4 segments over ½ MIP) **AND** a plastic bar over **0.90 MIP**, within 20 ns — a charged leg crossing the stack, not a γ.
- Simulation says **96 %** of those trigger legs are ^{27}Al-capture γs converting in the capsule wall — the trigger is dominated by the background the analysis has to beat.

The tracks point back at the capsule



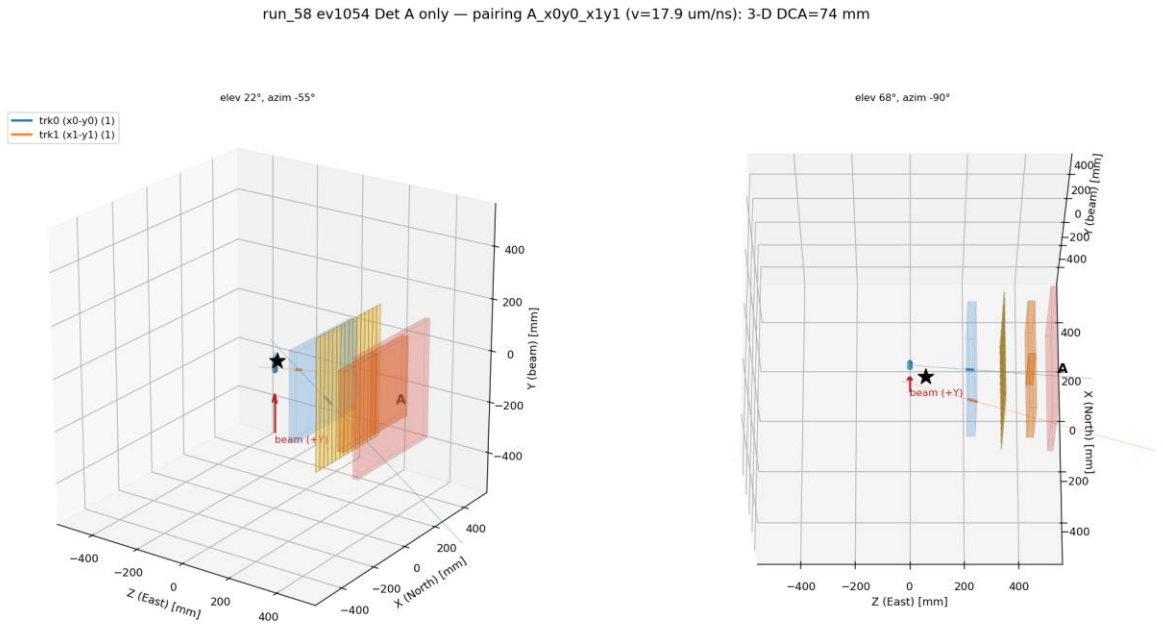
Waveform-first reconstruction on beam data, coloured by **which SiPM wall segment fired** — external position truth on a 100 mm granularity, on a lever arm the chamber knows nothing about. The four bundles are **251 mm apart at the wall and 9 mm apart at the target plane**; shuffling the segment labels collapses that to 28 mm and 3.2 mm. **Ensemble statement** — the waist is a convergence of medians, not a per-track pointing resolution.

Beam availability, day by day



Fraction of each day with protons on the n_TOF target, from the one-minute classification of the last beam-current transformer before the target (FTN.BCT477). Campaign mean **78.6 %**; the 21.4 % without beam is the usual mix of PS supercycle sharing and machine stops — the logger's attribution between the two is not reliable, so only the total is quoted. Colours are the three campaign phases; 10 August is a partial day (HV off 09:15).

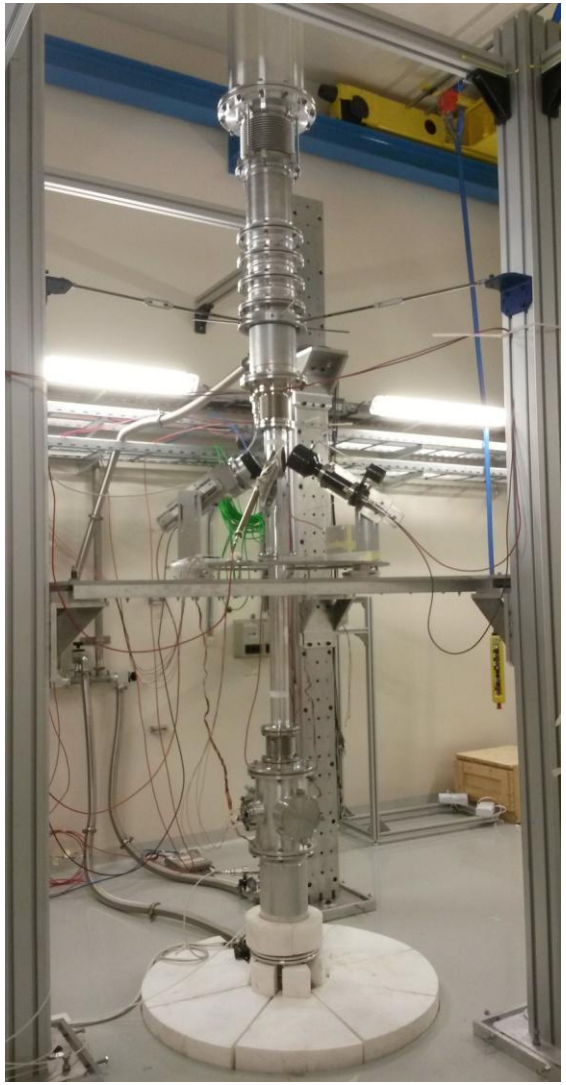
Next: from single tracks to pairs



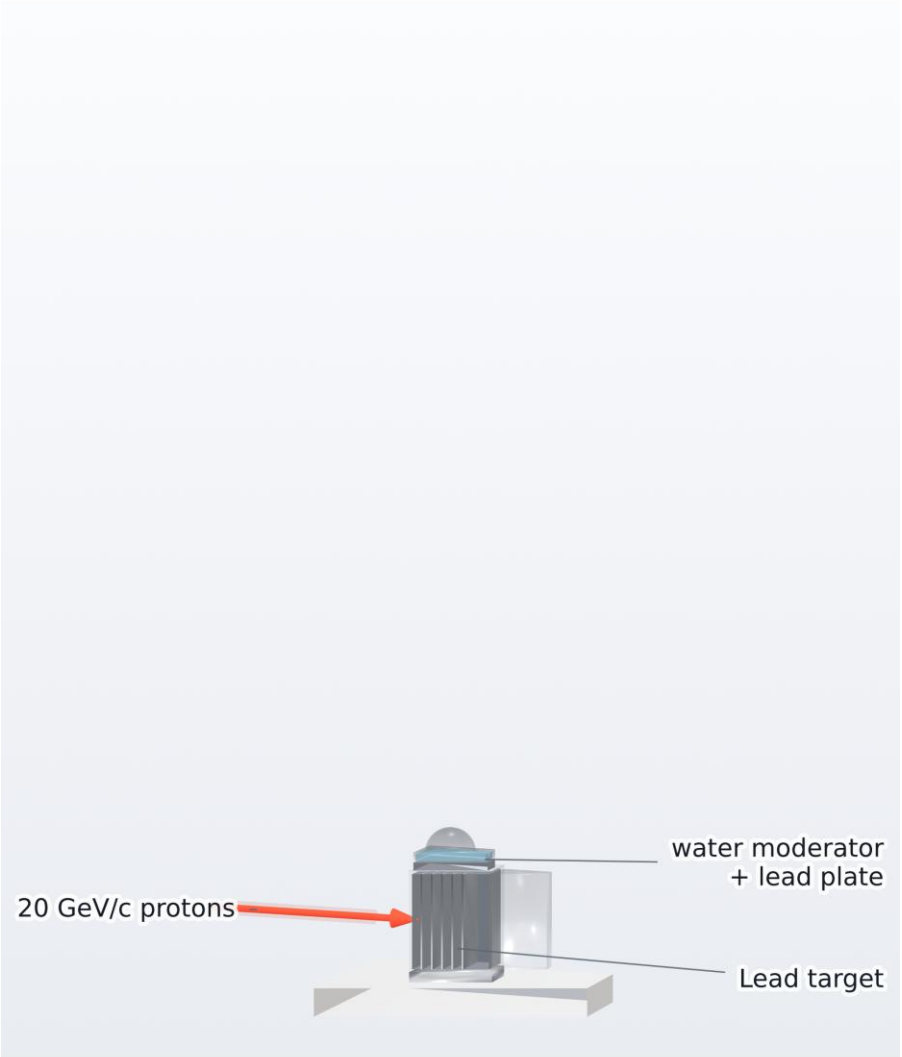
A real two-track event in one chamber (July scan run) — two track-grade lines in **both** planes, paired into 3-D. A demonstration, not a measurement: hits-basis, and the X ↔ Y pairing is two-fold ambiguous

- **What a signal is:** two tracks in *different* arms, opening angle kinematically confined above **109°**, total pair energy 19.6 MeV (slide 4). Everything in this talk exists to make that measurable.
- **The reconstruction is ready for the easy half:** the forward fit now returns *every* candidate track, not just the winner — single-track output bit-identical, candidates shipped alongside. Two tracks merged closer than ~12 mm still fit as one bad-χ^2 compromise: the two-track forward model is the one designed-but-unbuilt piece.
- **The dataset is final and the search is offline:** 41.8 M events on EOS, with the trigger dominated by Al-capture pairs — an enormous calibration sample for exactly this topology.
- **The bar the tracker has to clear:** the Al background separates on *total pair energy*, which asks for **$\sigma(p)/p \lesssim 30$ %** per lepton from scattering or range — the open MPGD question to leave the room with.

n_TOF EAR2: a vertical neutron beam line at CERN



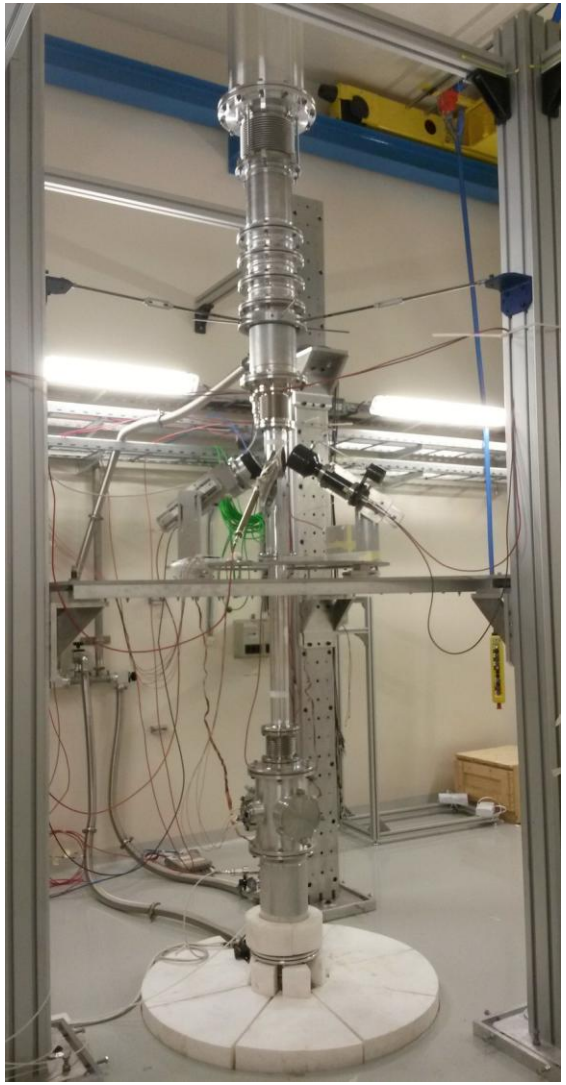
The EAR2 hall — © CERN



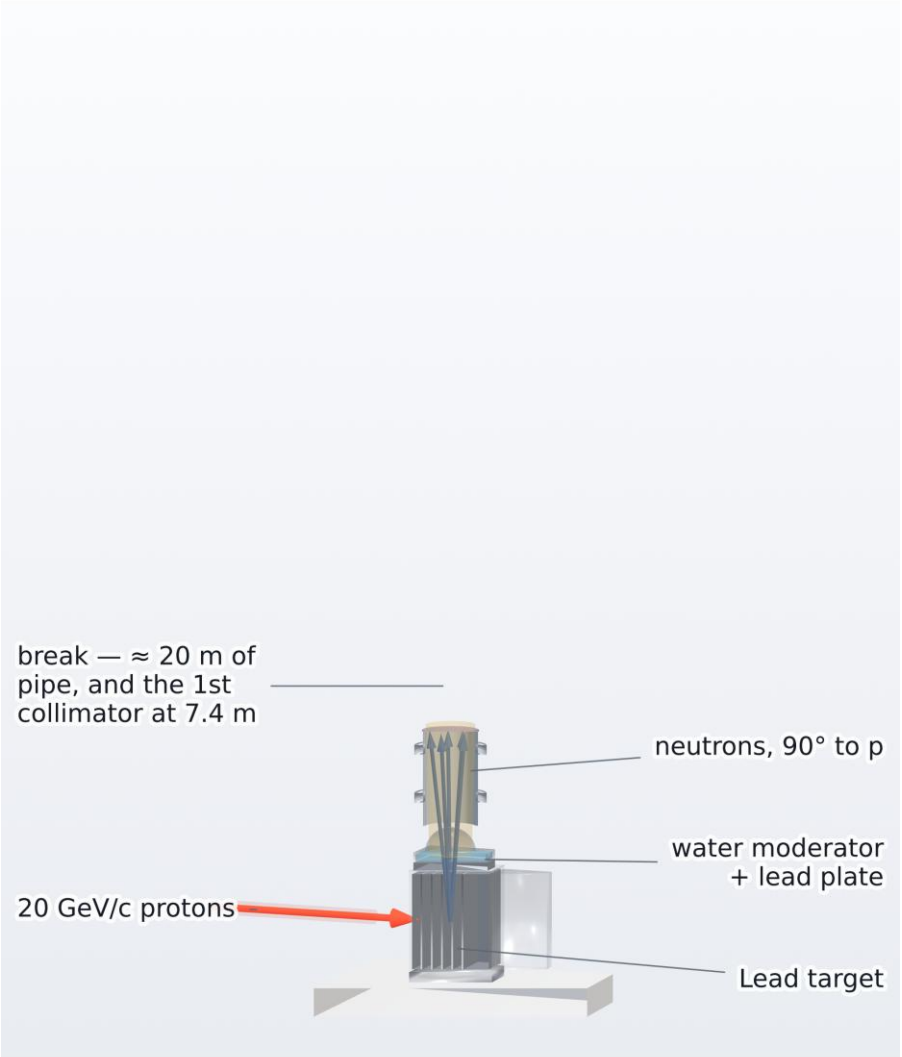
In-house render · Weiß (2015), Pavon-Rodriguez (2025)

- 20 GeV/c PS protons hit a lead spallation target

n_TOF EAR2: a vertical neutron beam line at CERN



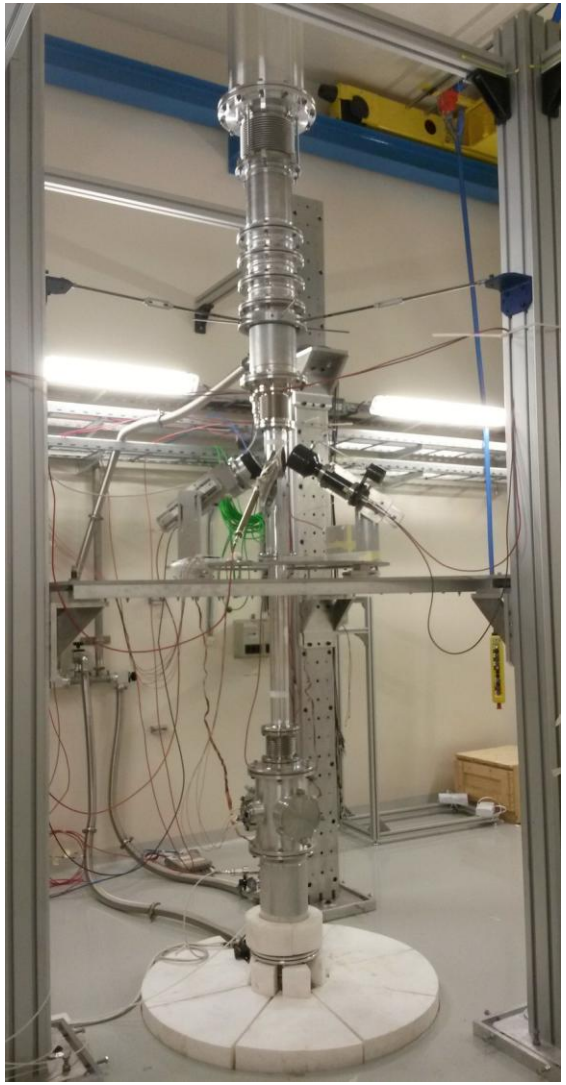
The EAR2 hall — © CERN



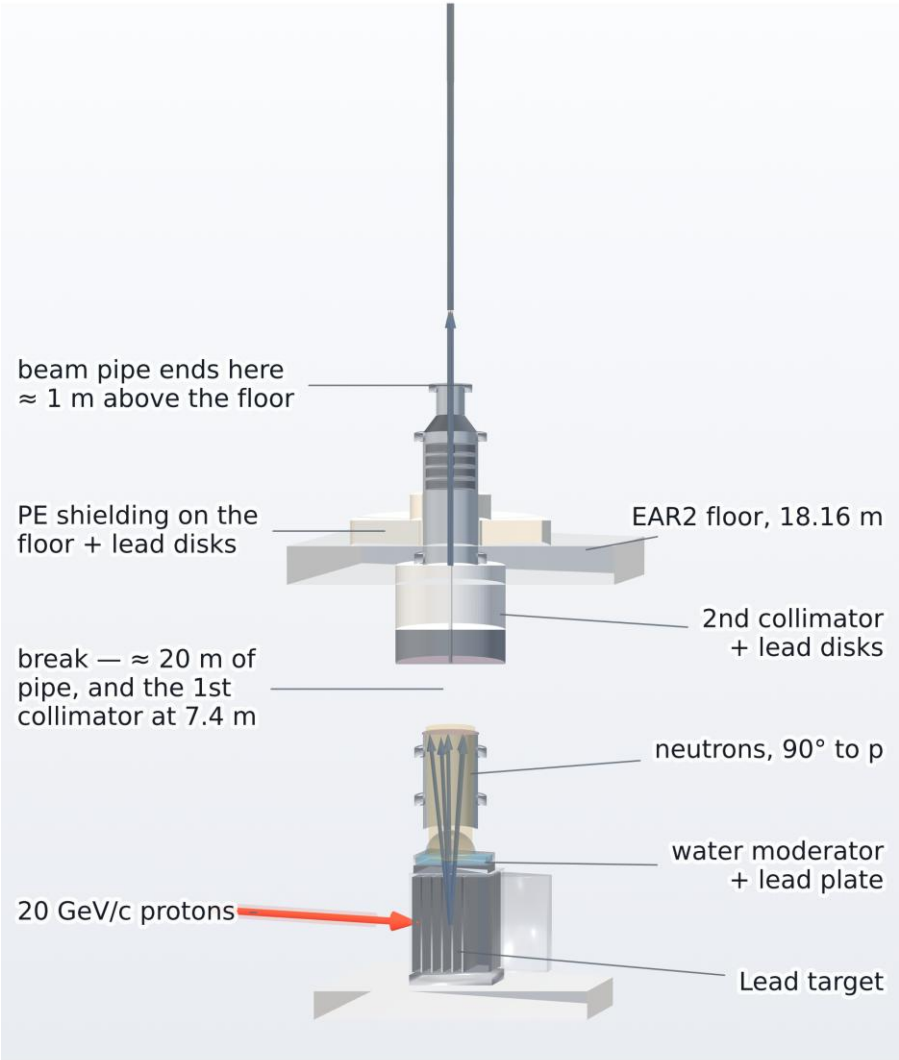
In-house render · Weiß (2015), Pavon-Rodriguez (2025)

- 20 GeV/c PS protons hit a lead spallation target
- Neutrons ejected at **90°** climb **~20 m** to EAR2

n_TOF EAR2: a vertical neutron beam line at CERN



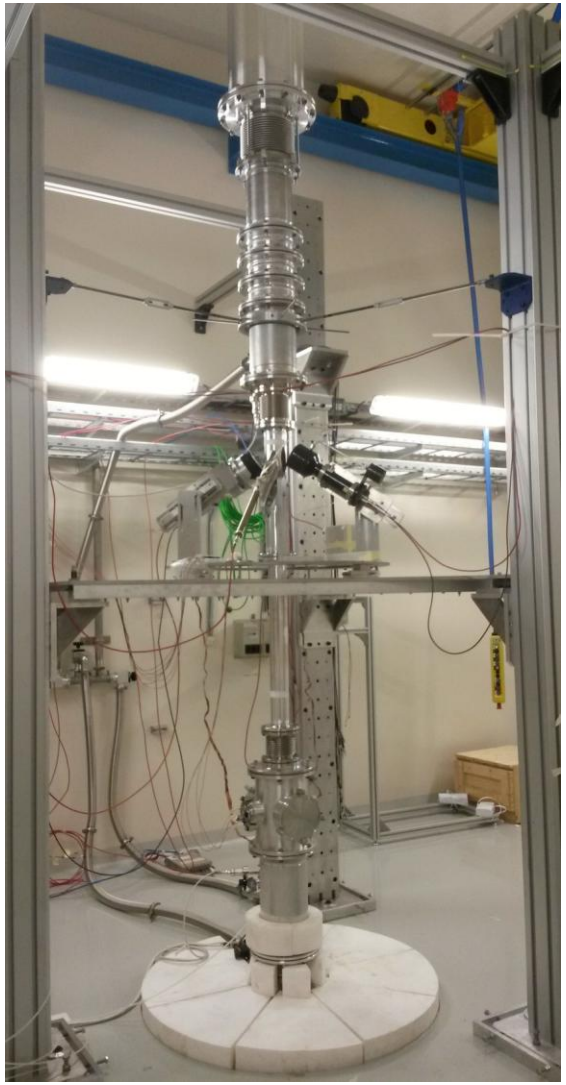
The EAR2 hall — © CERN



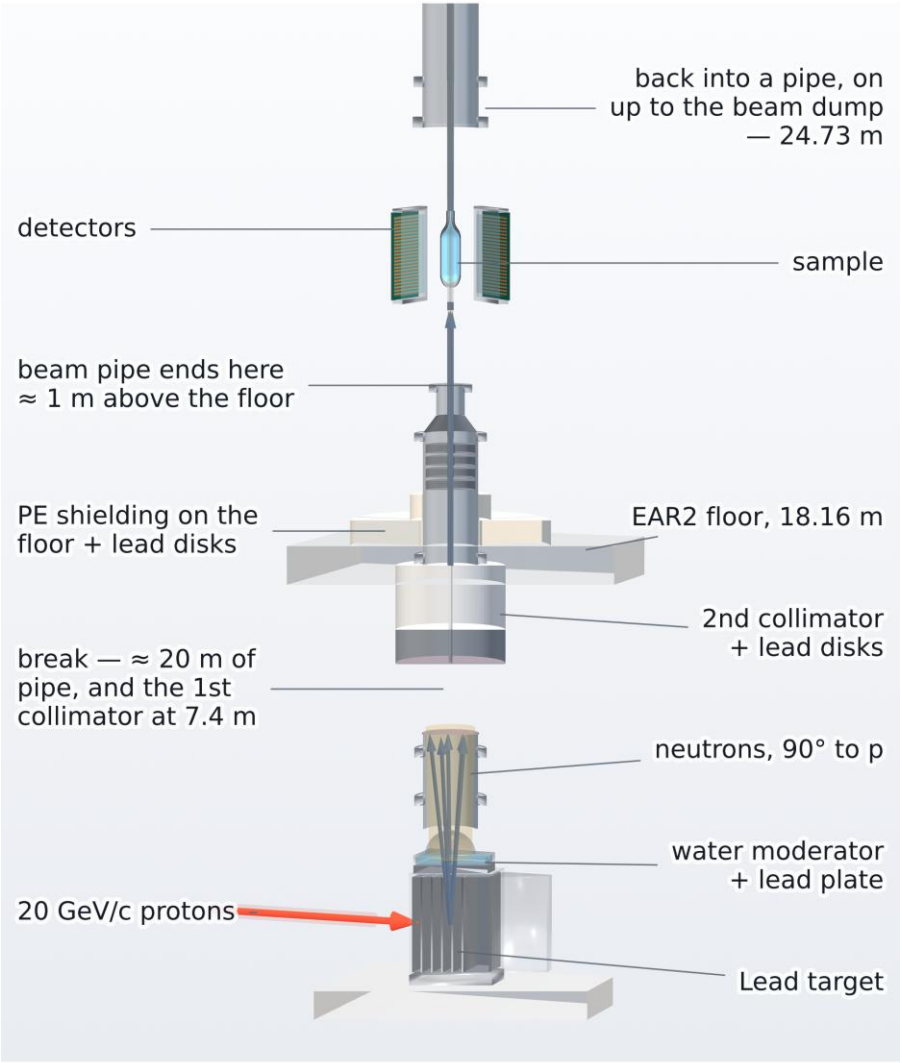
In-house render · Weiß (2015), Pavon-Rodriguez (2025)

- 20 GeV/c PS protons hit a lead spallation target
- Neutrons ejected at 90° climb ~20 m to EAR2
- Collimated to a **≈3 cm** pencil
- The pipe ends: the beam crosses the hall in air

n_TOF EAR2: a vertical neutron beam line at CERN



The EAR2 hall — © CERN



In-house render · Weiß (2015), Pavon-Rodriguez (2025)

- 20 GeV/c PS protons hit a lead spallation target
- Neutrons ejected at 90° climb ~20 m to EAR2
- Collimated to a ≈3 cm pencil
- The pipe ends: the beam crosses the hall in air
- Measurement station sits in that open beam
- The highest instantaneous flux at n_TOF

The Micromegas Compact TPC

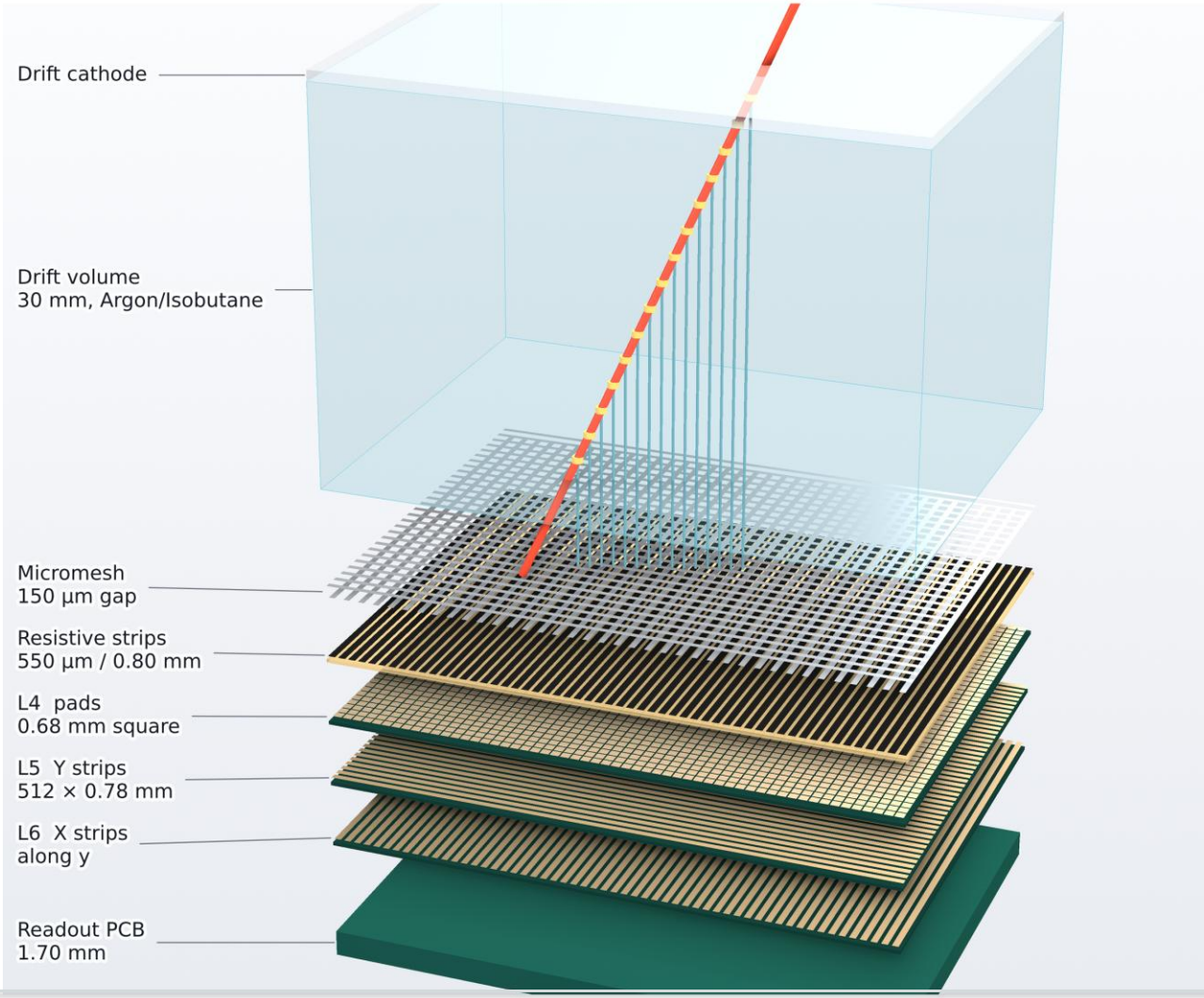
A large-area resistive-strip detector, built to measure the opening angle of the pair.

The Micromegas compact TPC design

DESIGNED & BUILT AT CEA SACLAY BY
Stéphan Aune

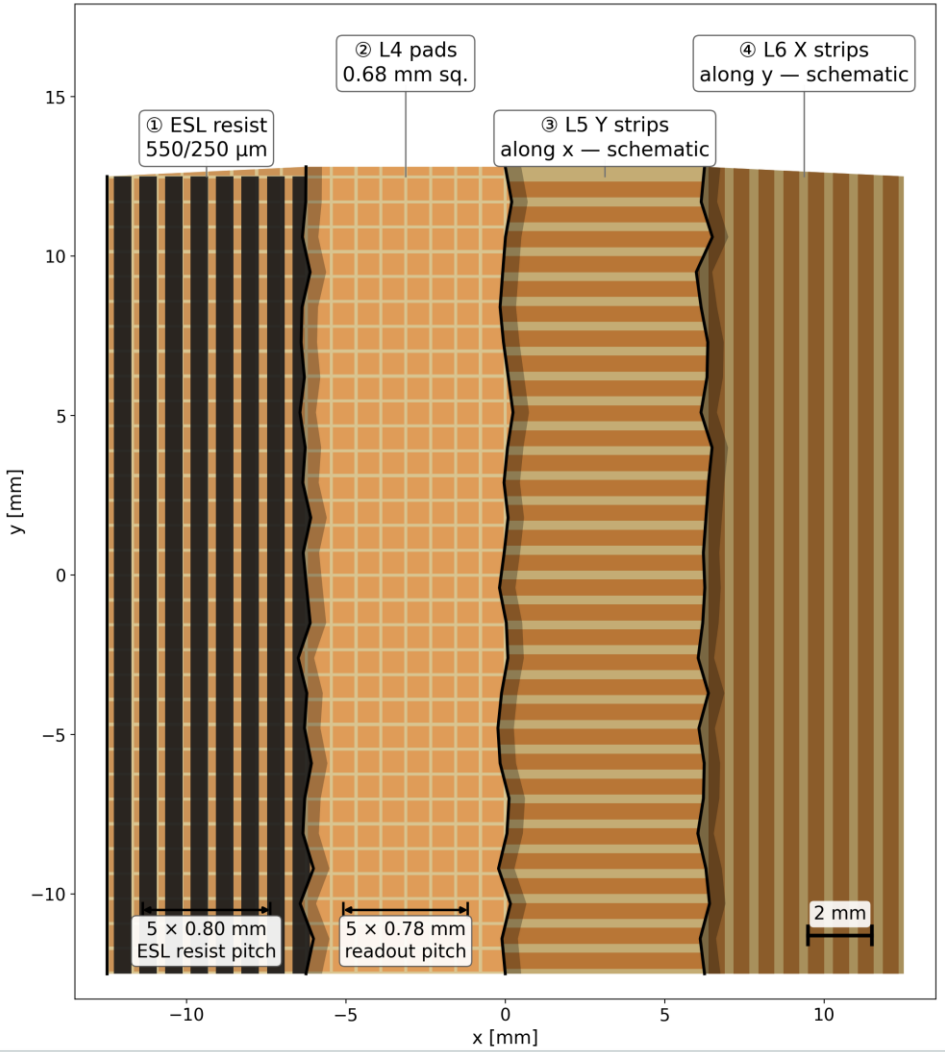
One chamber, exploded along the drift axis

a 44 × 34 mm window at the real pitch



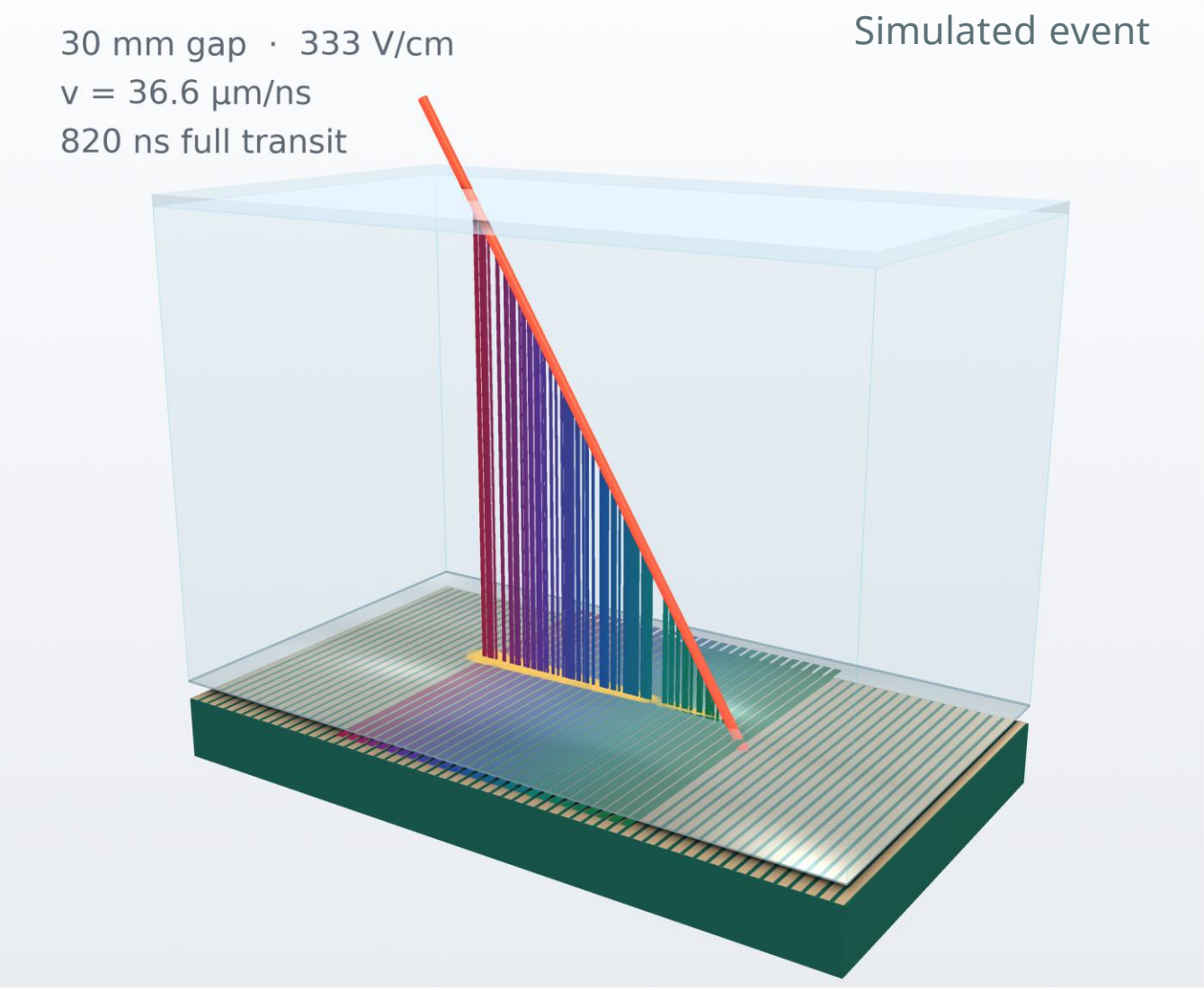
MX17 readout board, four layers peeled back

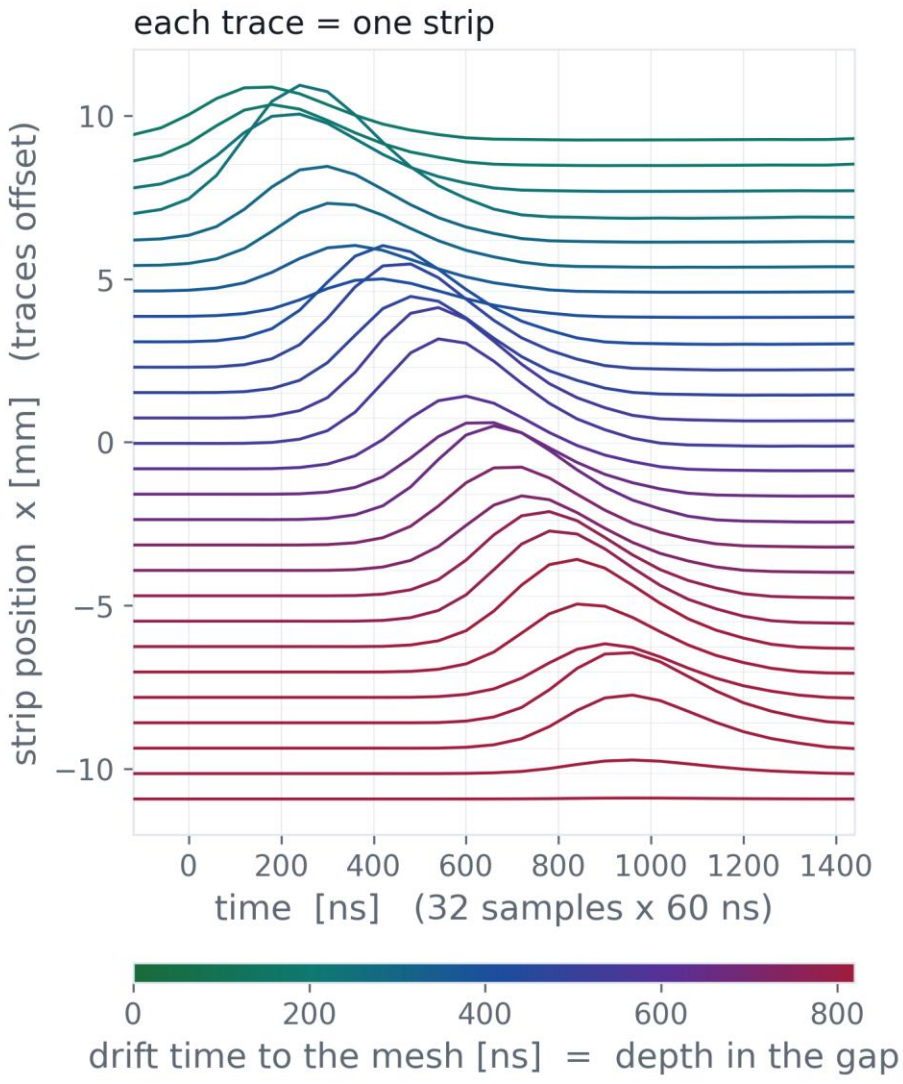
25 × 25 mm of the 470 mm board



DESIGNED & BUILT AT CEA SACLAY BY
Stéphan Aune

TPC operation



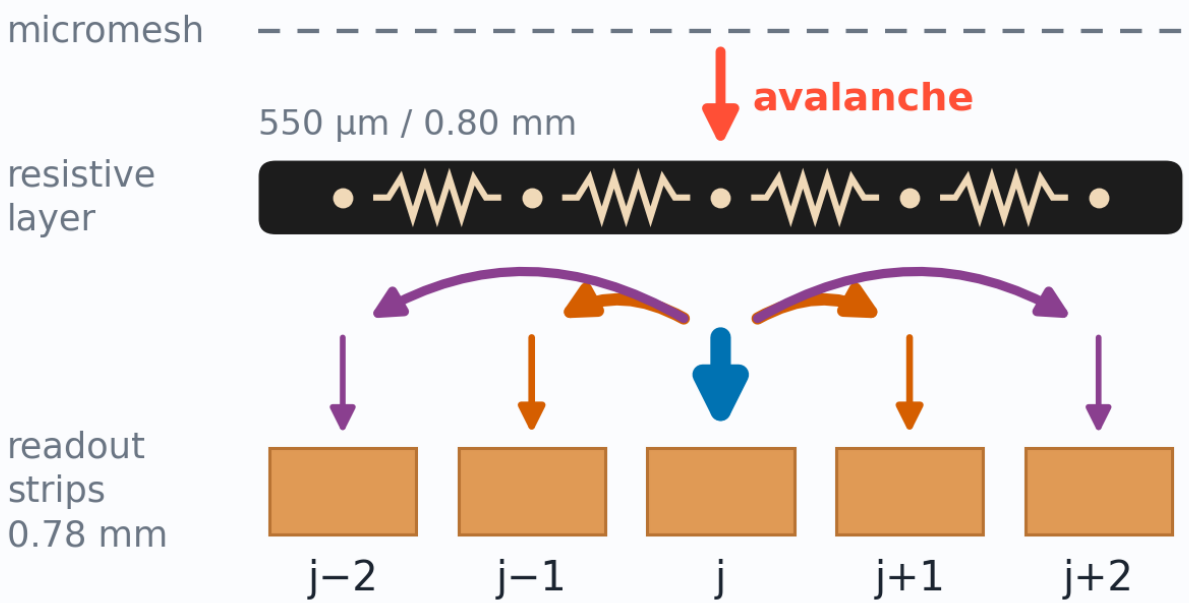


Method inspired by T2K ND280
Attié et al., NIM A 1056 (2023) 168534

WORK IN PROGRESS

Resistive strips spread and delay charge

What happens on the resistive layer



own charge

±1 neighbour — late by 166 ns

±2 — 333 ns

Charge spreads through the resistive strips

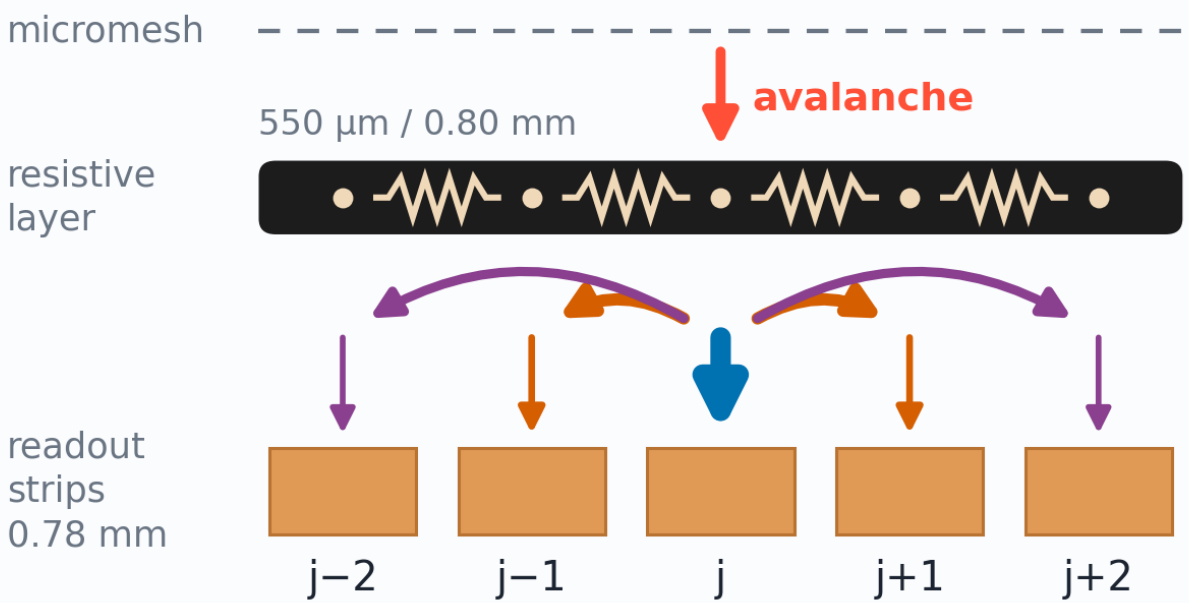
A neighbor’s copy arrives late and dispersed

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WORK IN PROGRESS

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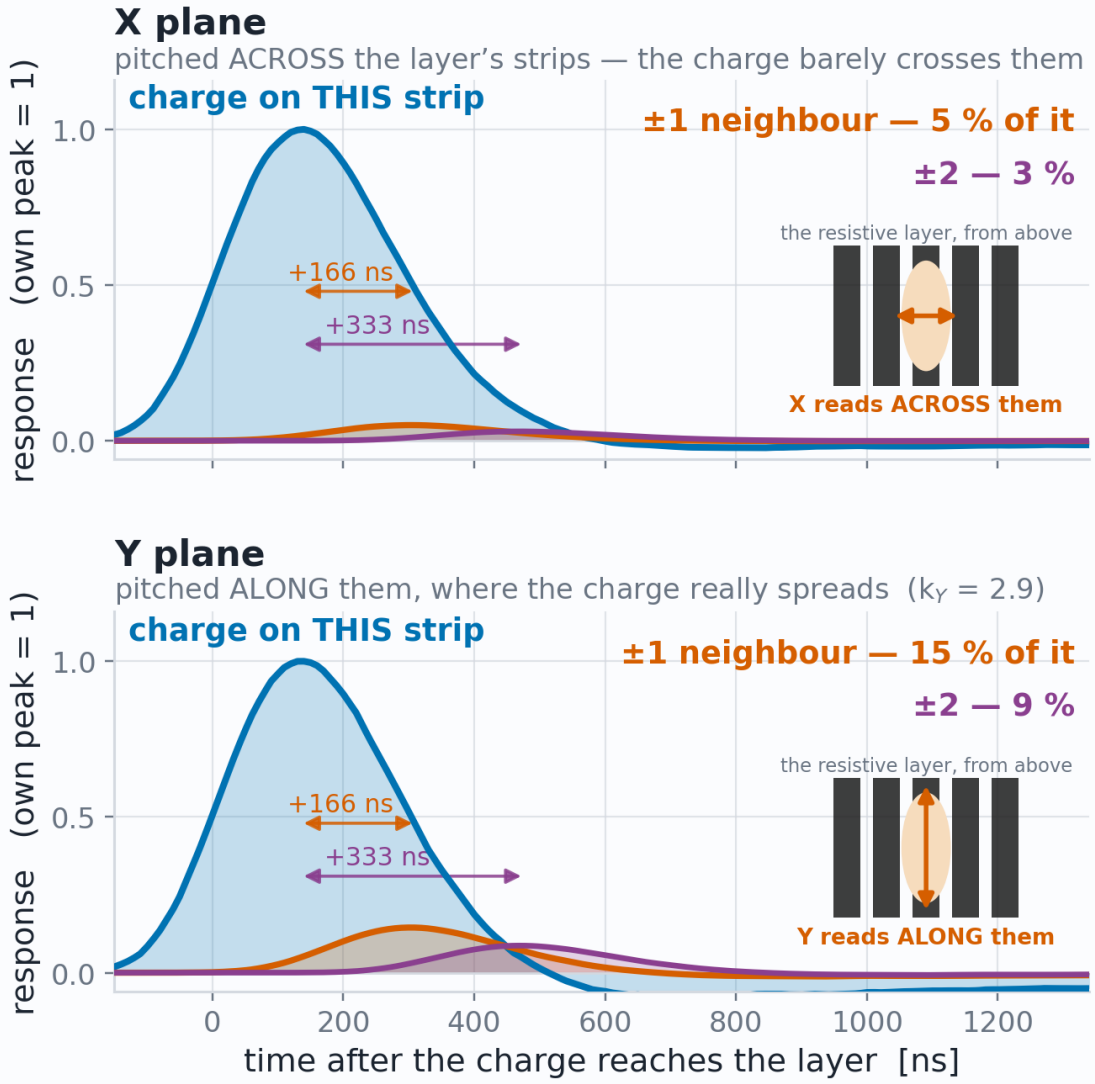
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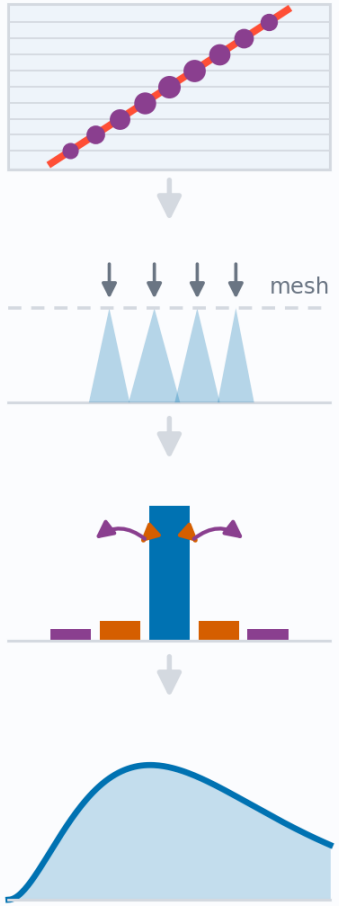
A neighbor’s copy arrives late and dispersed



Method inspired by T2K ND280
Attié et al., NIM A 1056 (2023) 168534

WORK IN PROGRESS

The forward fit: assume a track and predict every waveform



① the track, in 60 ns slices

K = 18 slices of the drift column. Each holds a charge $q_k \geq 0$ at transverse position $p_0 + w \cdot u_k$. The charges are solved, not searched.

② geometric spread

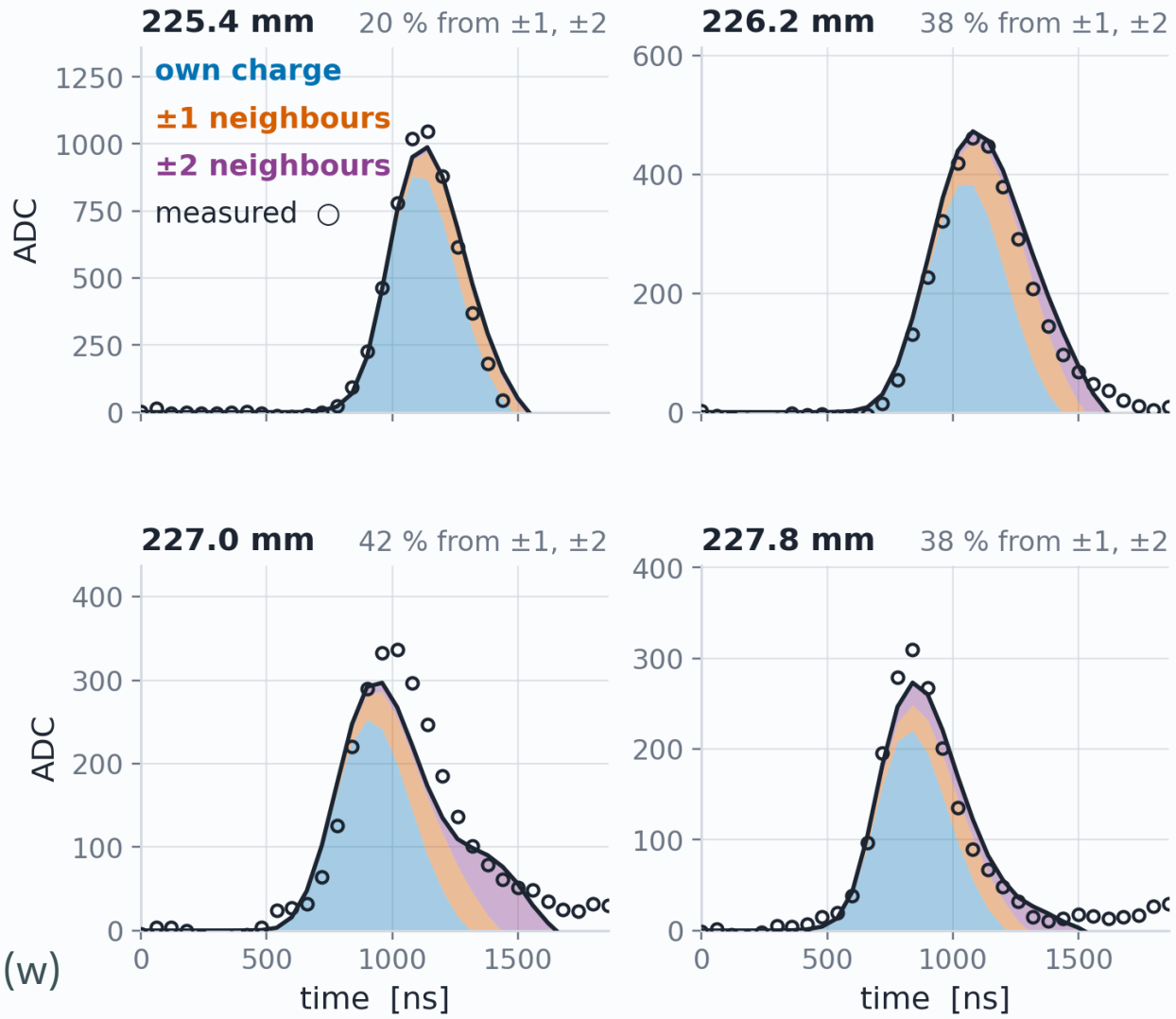
What one slice puts on the layer, integrated over the 0.78 mm pitch. Its width is transverse diffusion plus the amplification width.

③ charge spread

Every strip's charge appears again on ±1 and ±2 — and LATE, by 166 and 333 ns.
Per plane: Y 15 % / 9 %, X 5 % / 3 %.

④ fold with response

The measured single-primary response of this plane, sampled at 32 × 60 ns — one complete predicted (strip × sample) window.



Four consecutive strips of one real muon event

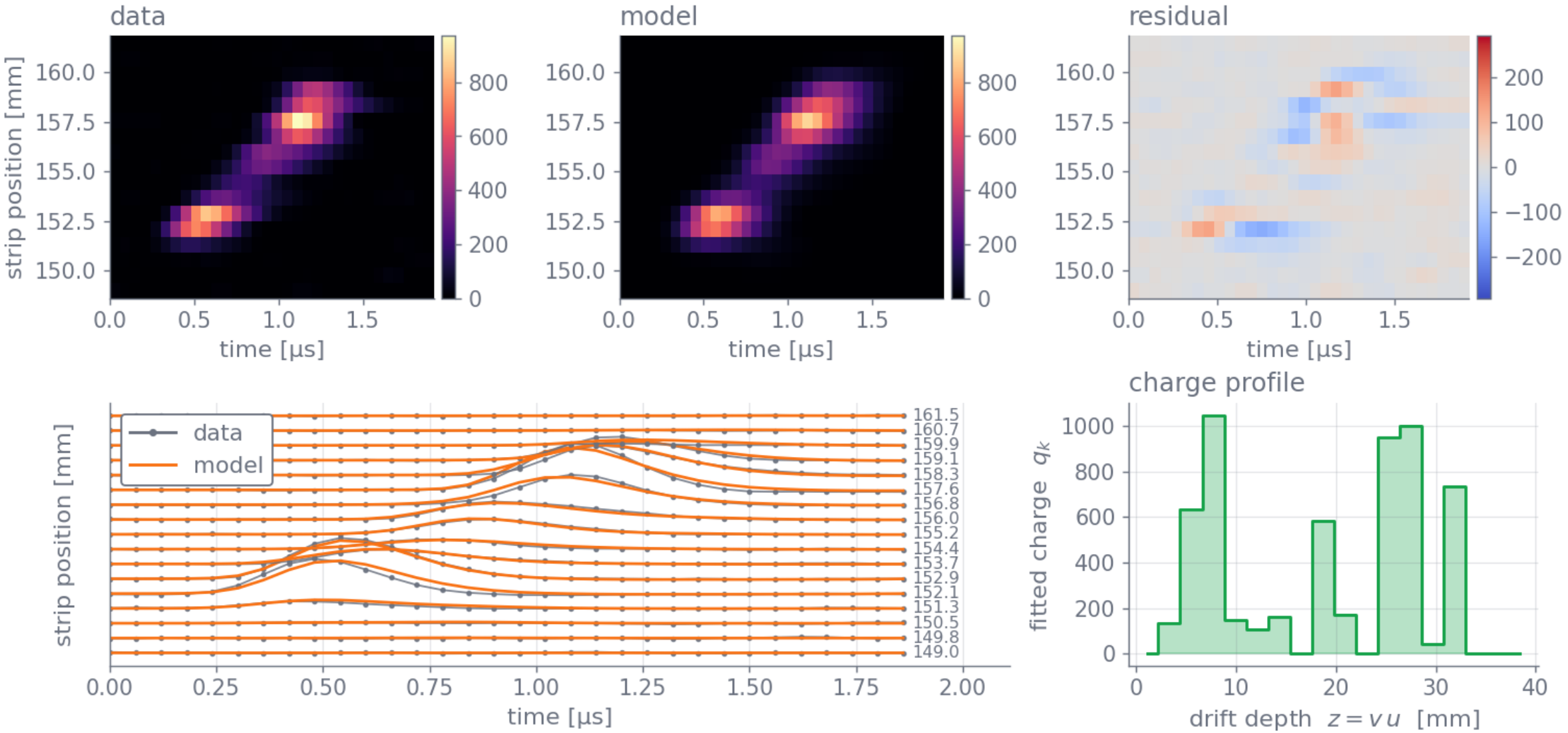
Three numbers are searched:

Mesh arrival position and time (p_0, t_0) and transverse speed (w)

v_{drift} enters once at the end: $\tan\theta = w / v_{drift}$

WORK IN PROGRESS

One muon through the forward fit



Characterization

How well the chamber actually performs,
measured with cosmic muons.

Characterization on the Saclay cosmic bench

A cosmic muon fires the scintillators at the top and bottom of the stack.

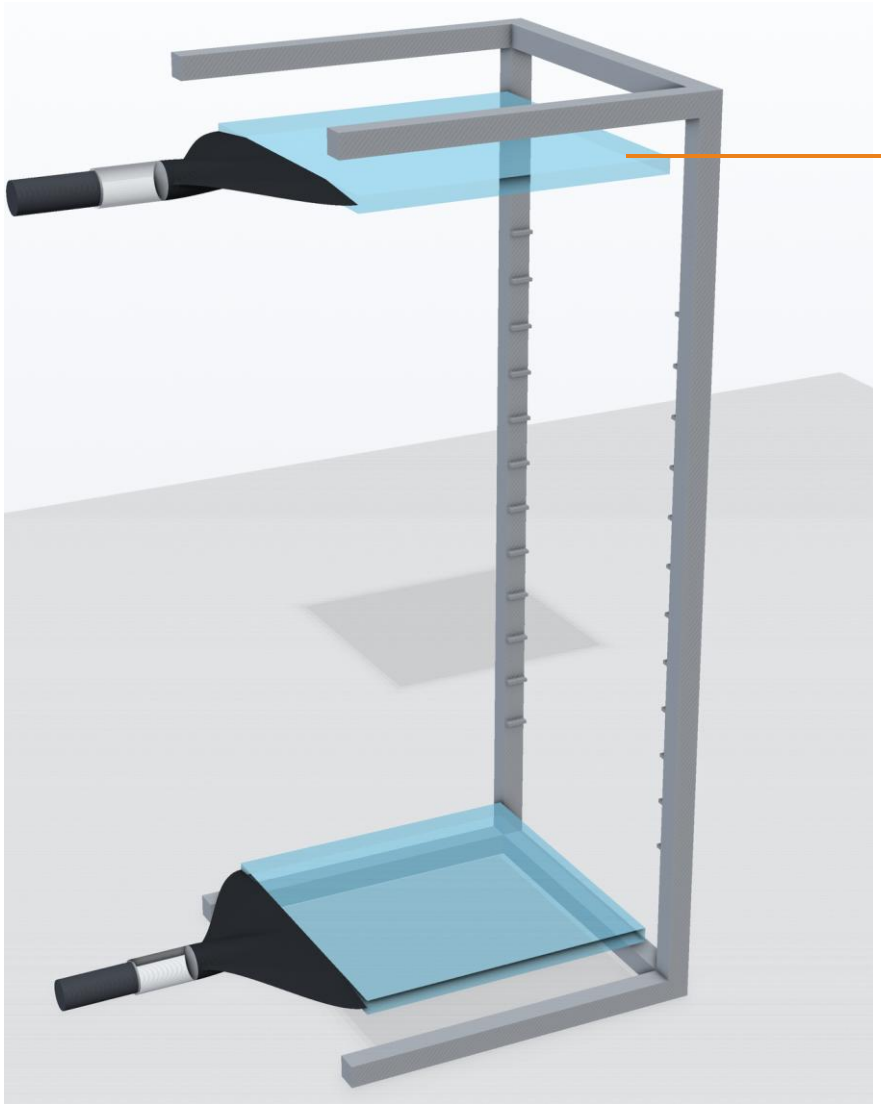
Four reference Micromegas reconstruct its track; the two MX17 detectors under test sit in between.



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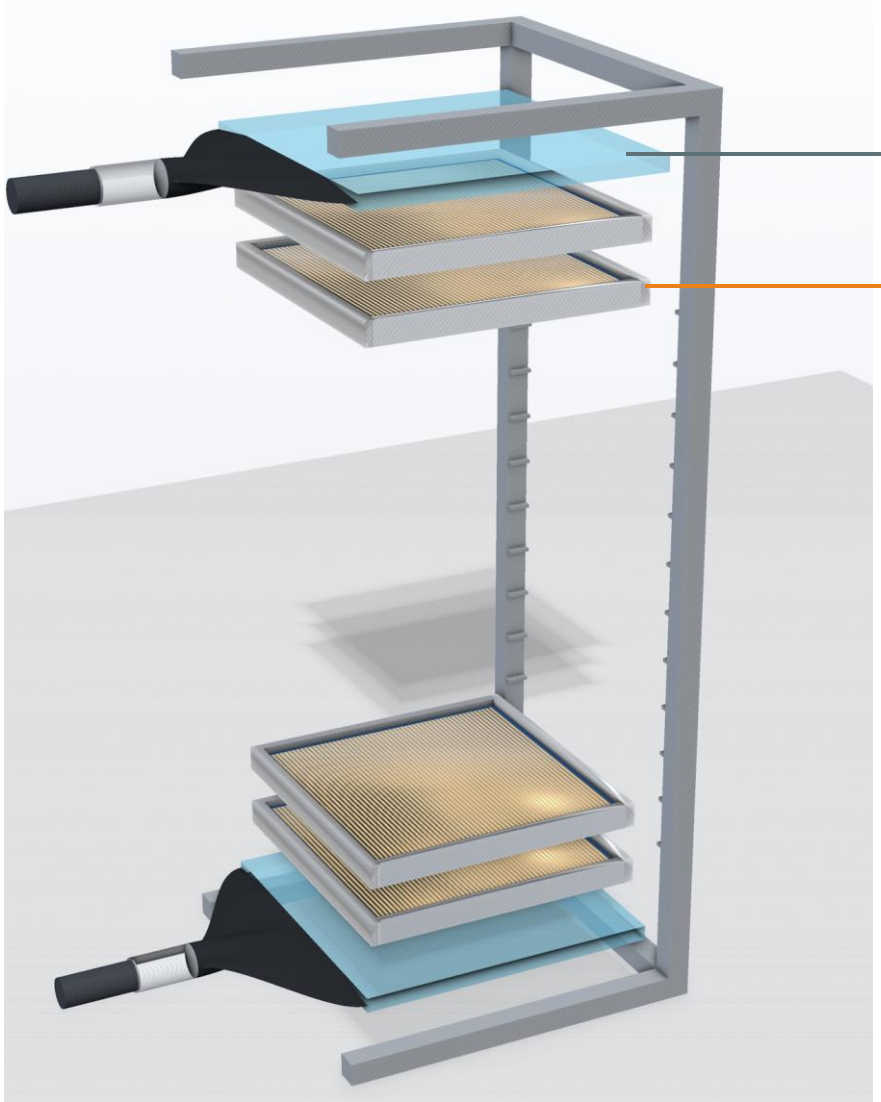


60 × 60 cm trigger scintillators, top and bottom

Characterization on the Saclay cosmic bench

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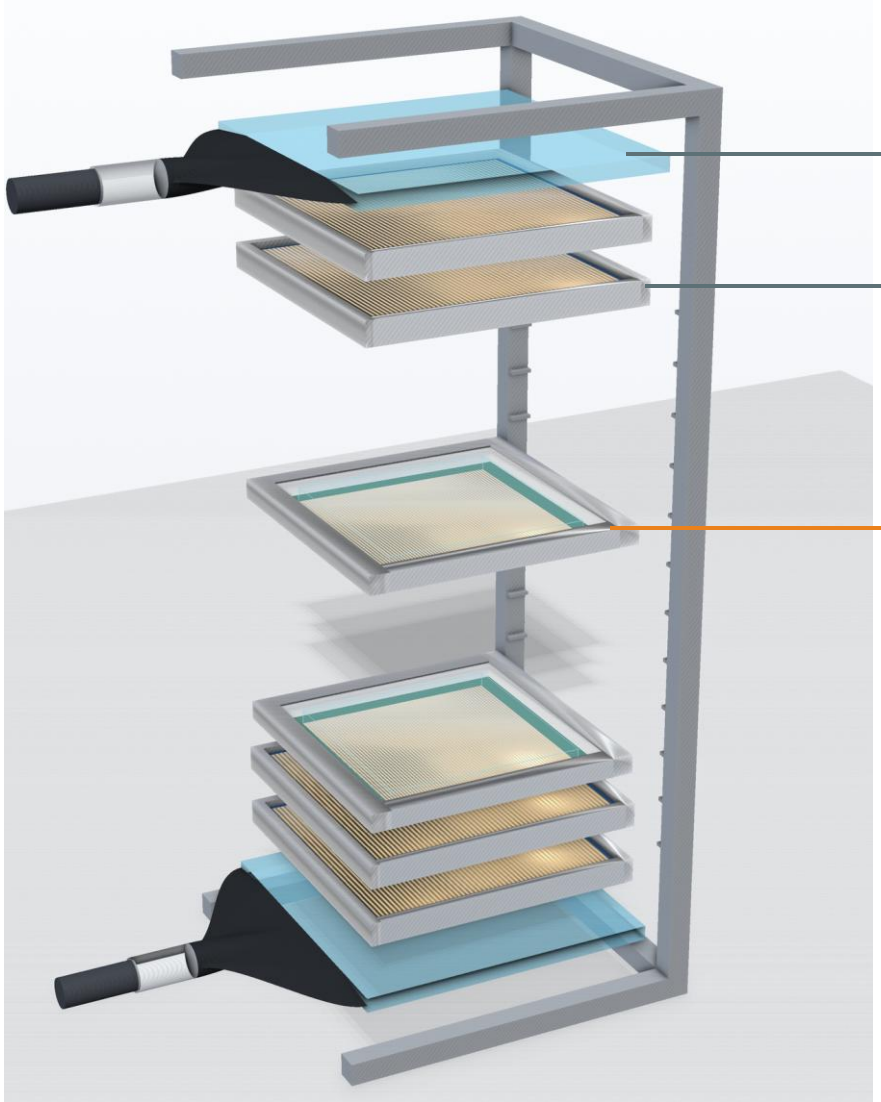
60 × 60 cm trigger scintillators, top and bottom

50 × 50 cm Micromegas reference trackers, four of them

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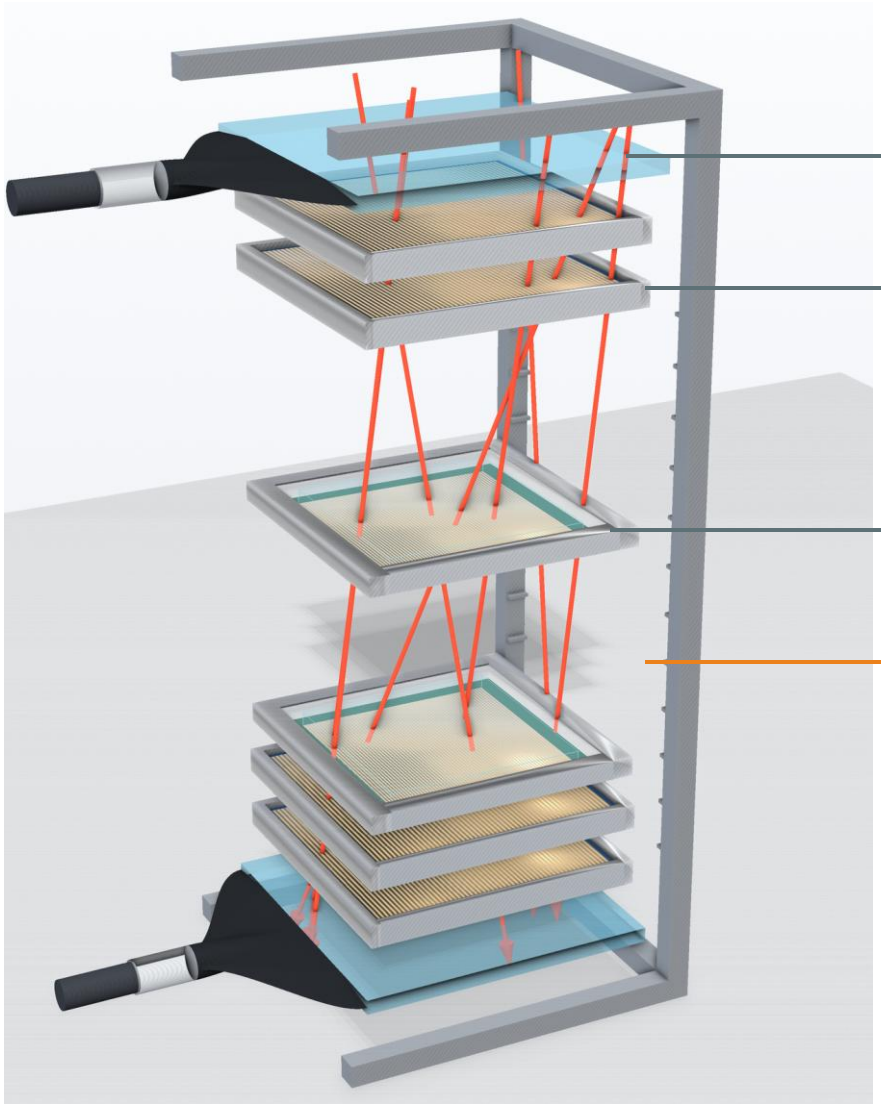
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40 × 40 cm TPC chambers — the ones being characterized

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60 × 60 cm trigger scintillators, top and bottom

50 × 50 cm Micromegas reference trackers, four of them

40 × 40 cm TPC chambers — the ones being characterized

Cosmic muons — two slots measured at once, against the same reference

WORK IN PROGRESS

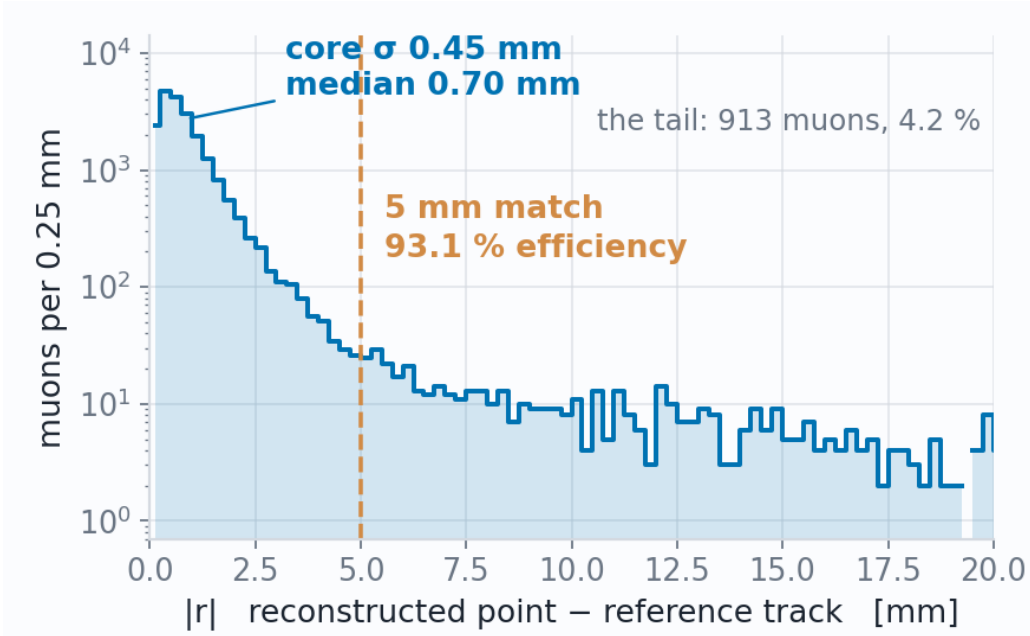
Efficiency: 93% on the best chamber

Where every crossing muon goes

Reconstructed within 5 mm the efficiency	**93.1%** 20,428
Detected, point >5 mm off the near-miss tail†	**4.2%** 913
Discharged during the muon many strips fire uncorrelated with muon	**2.6%** 563
Fired, no valid X+Y point	**0.2%** 46
Silent — no signal at all genuine blindness	**0.01%** 3

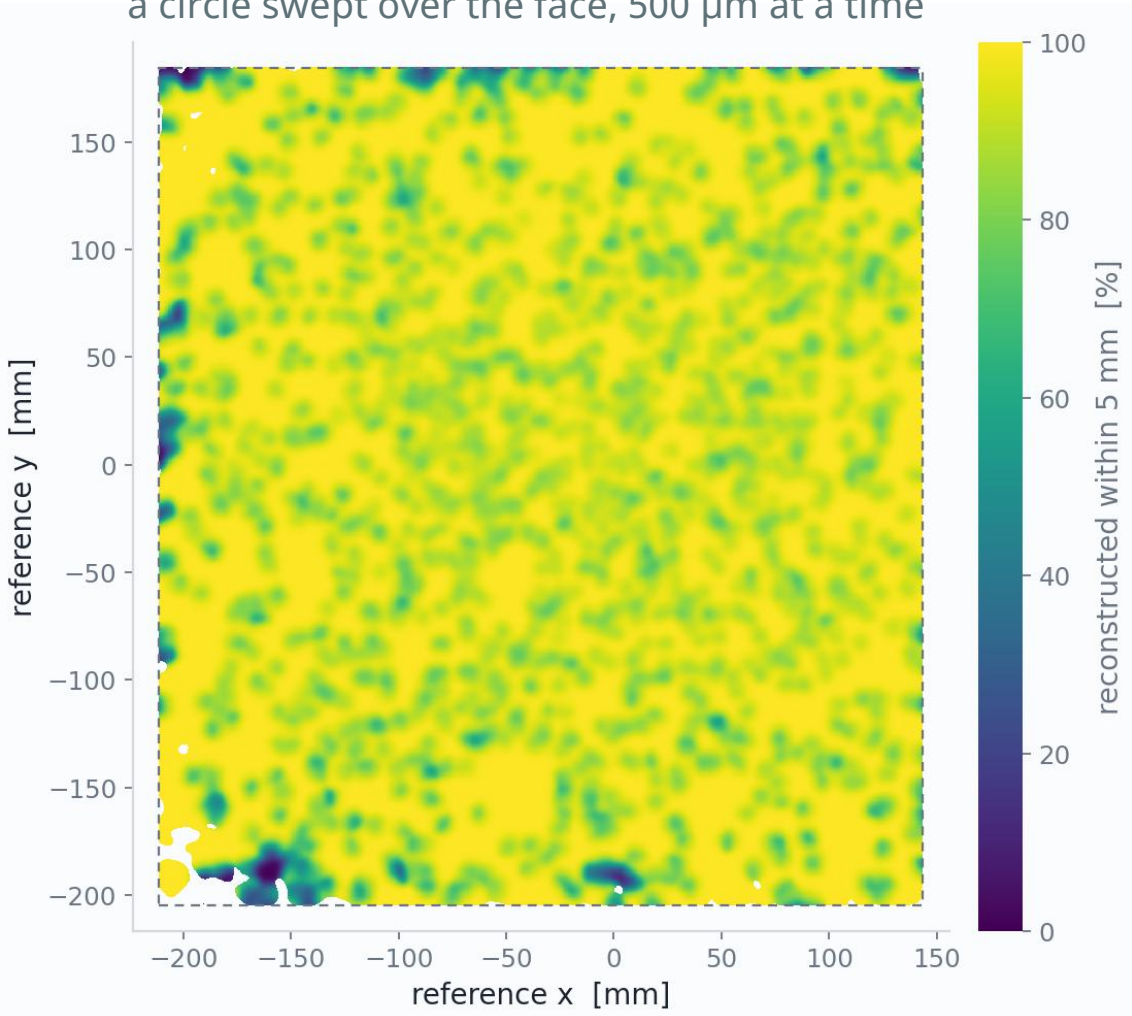
WHY 0.45 mm

Reference trackers resolve to ~500 µm



Flat across the chamber face

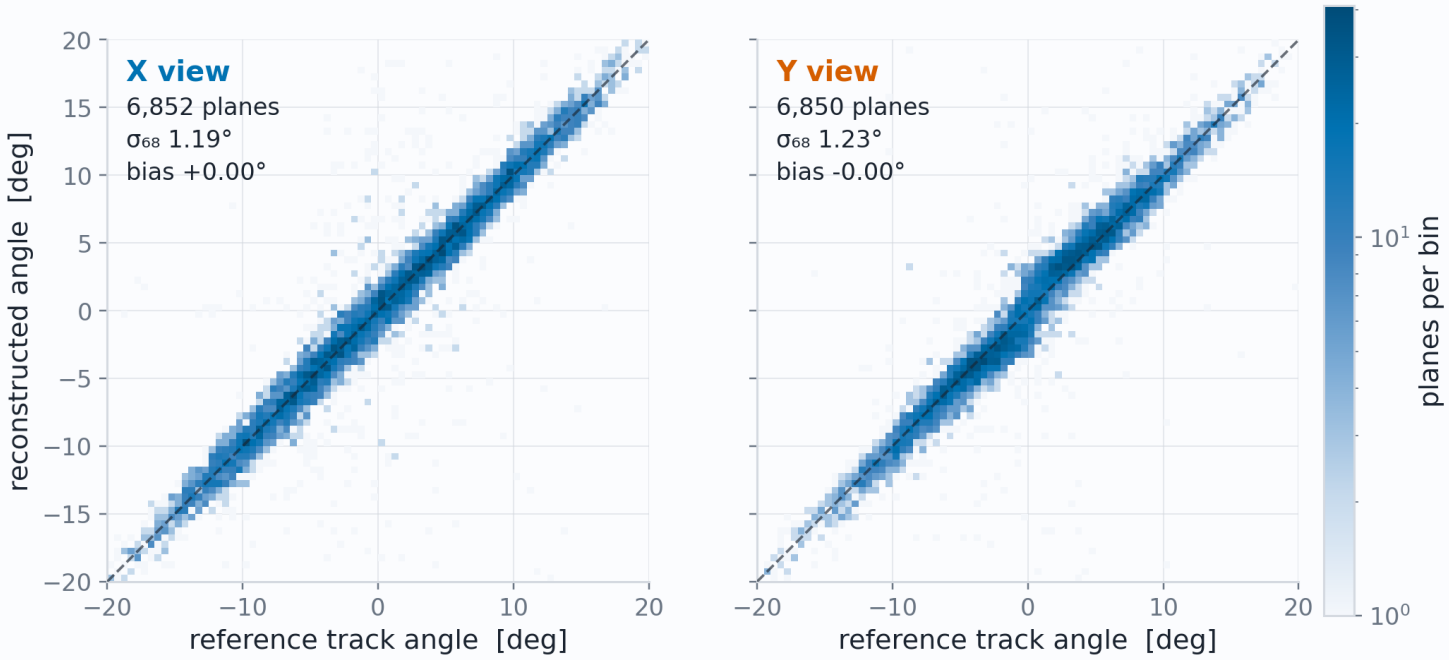
a circle swept over the face, 500 µm at a time



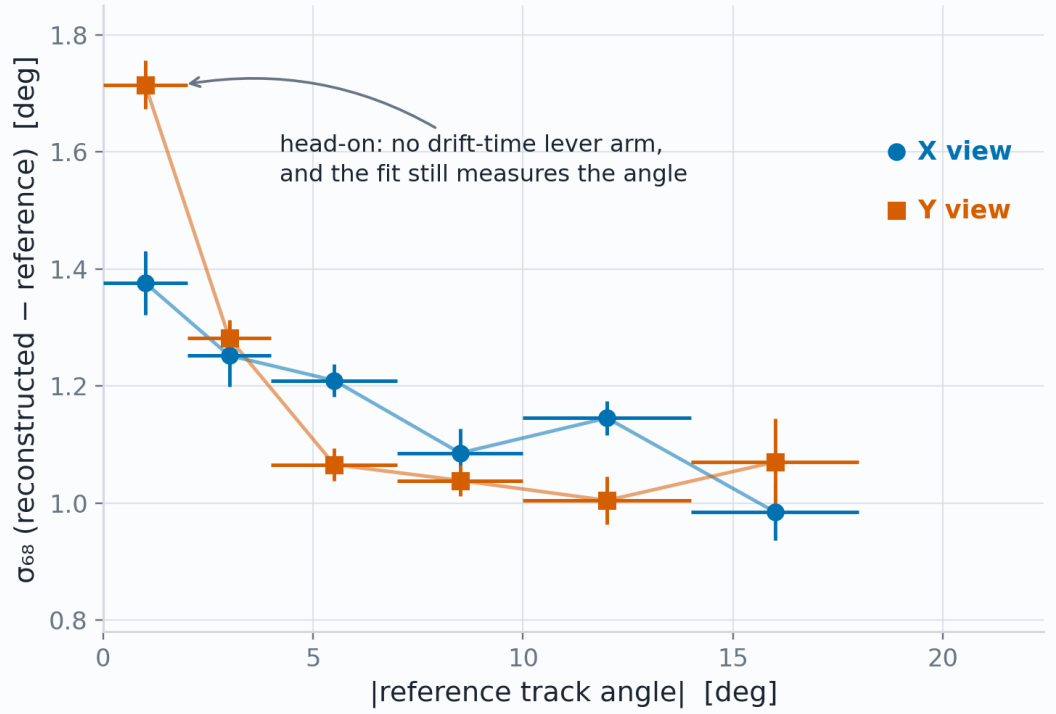
WORK IN PROGRESS

Adequate angular, spatial and temporal resolution

Reconstructed against reference, per plane



Head on tracks reasonable and angled tracks approach 1°

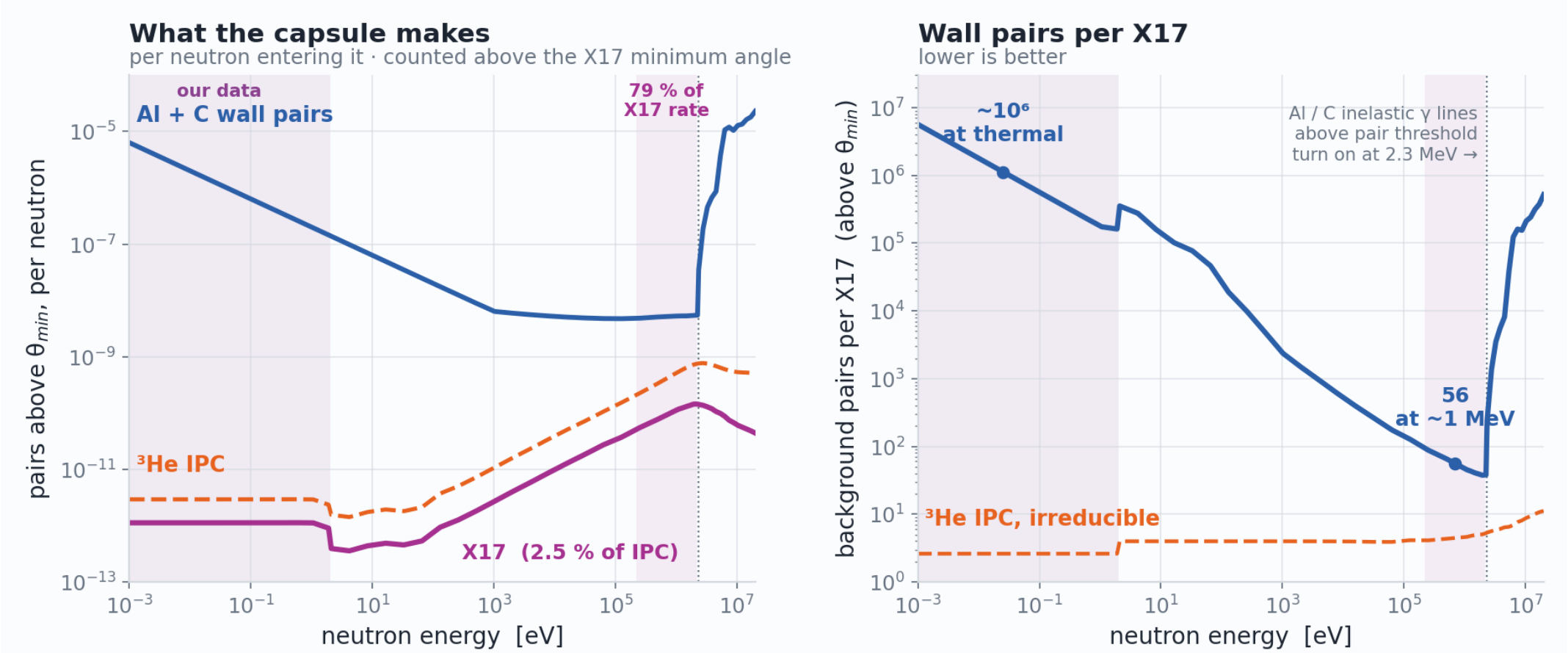


1.19 / 1.23°
angle per plane, X / Y (σ_{68}), bias < 0.1°

180 µm
position at normal incidence, **±0.035 mm** conservatively

33 ns
plane-to-plane timing, ≈ 1.1 mm of drift

Could plan A still work? Wall pairs per X17, by neutron energy



Wall: internal pairs only, no external conversion (adds up to ~×6, mostly at small angles). ^{27}Al (n,γ) below 845 keV bridged by hand (1/v, then log-log to ENDF).
Not included: γ flash, room background, (n,p) two-prongs. X17/IPC = 2.5 % is the rate table's assumption; the ratio scales with it.

The Al + C wall makes ~10^6 pairs per X17 at thermal energies but only ~50 at 0.1–2 MeV, where 79 % of the X17 rate sits — until the Al/C inelastic lines open at 2.3 MeV. X17/IPC = 2.5 % is the rate table's assumption; the ratio scales with it.

Cut for the 15-minute version